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SHOELOTSKEY WEB & ADMIN SUITE: A MULTI-BRANCH MANAGEMENT SYSTEM WITH AI FORECASTING USING PROPHET AI

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APPROVAL SHEET

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CERTIFICATION OF ORIGINALITY

This is to certify that the research work presented in this capstone project, SHOELOTSKEY WEB & ADMIN SUITE: A MULTI-BRANCH MANAGEMENT SYSTEM WITH AI FORECASTING USING PROPHET AI for the degree Bachelor of Science in Information Technology at the Polytechnic University of the Philippines embodies the result of original and scholarly work carried out by the undersigned. This capstone project does not contain words or ideas taken from published sources or written works that have been accepted as basis for the award of a degree from any other higher education institution, except where proper referencing and acknowledgement were made.

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ABSTRACT

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Shoelotskey, a multi-branch shoe care service, faces operational inefficiencies due to reliance on manual processes such as handwritten job orders, Excel-based records, and decentralized communication via Messenger. These practices lead to inconsistent service delivery, delayed updates, poor inventory tracking, and limited scalability. To address these challenges, this study developed the Shoelotskey Web & Admin Suite (SWAS), a centralized web and mobile platform featuring real-time service tracking, automated notifications, role-based access control, inventory management, and AI-driven forecasting using Prophet. The system was evaluated based on ISO/IEC 25010 standards through surveys administered to 109 respondents, including customers, staff, and administrators. Results showed strong agreement across all user groups, with overall mean scores of 4.06 (Customers), 4.58 (Administrators), and 4.70 (Staff), particularly in usability, reliability, and security. Findings confirm that SWAS effectively streamlines multi-branch operations, enhances customer engagement, and provides a scalable digital foundation for business growth.

Keywords: Shoelotskey, Multi-Branch Management System, Real-Time Service Tracking, Automated Notifications, Role-Based Access Control (RBAC), MongoDB, Prophet AI, Web and Mobile Platform, Centralized System.

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Chapter 1

INTRODUCTION

Project Context

It has become increasingly evident across the world today that small or medium-sized enterprises face growing pressure to modernize operations and strengthen customer engagement. Despite this, many service-based businesses continue to rely on traditional workflows. Shoelotskey, a growing shoe care service with three branches and plans to expand, is an example of an enterprise hindered by manual processes that limit scalability, delay service delivery, and disrupt effective communication.

Shoelotskey's current system relies heavily on handwritten job orders, Excel spreadsheets for customer records, and verbal or text-based communication for status updates. These fragmented tools create inconsistencies in service delivery due to varying staff experience, dependence on key personnel, and limited availability during peak periods. The lack of a standardized intake process and structured documentation results in delays, unclear updates, and repeated customer questions. Service tracking is reactive rather than proactive, with admins often unaware of bottlenecks until escalation occurs. Manual inventory handling and unclaimed items further contribute to storage and processing issues. Moreover, communication gaps between staff and customers lead to missed messages, delays in notifications, and disputes, especially when dealing with time-sensitive or demanding client needs.

In view of the above challenges, we, as student developers from the Polytechnic University of the Philippines, propose the Shoelotskey Web & Admin Suite, a centralized web-based platform designed to modernize and streamline the

company's operations. The system will serve Shoelotskey Shoe Cleaning Services and its three branches, namely: SM Grand Central, SM Valenzuela, and SM Baliwag. This project integrates features such as real-time inventory tracking, automated service updates, administrative analytics, and a client-facing portal that allows customers to actively monitor the progress of their service requests. This initiative aligns with the growing demand for digital-first interactions in service industries. According to the GoDaddy Data Observatory (2023), 51% of small businesses in the Philippines have already moved online, with many reporting improved revenues. Additionally, Almtiri and Miah (2020) emphasize the importance of real-time systems and personalized engagement in enhancing customer satisfaction and loyalty. By transitioning to a modern system, Shoelotskey can not only address internal inefficiencies but also position itself for sustainable growth in today's digitally competitive market.

Technical Background

Equipment/Hardware

Shoelotskey's hardware setup is minimal and varies with every branch location. They operate with a small number of essential devices that are distributed between the warehouse and its service branches:

Table 1

Existing Hardware Used by Shoelotskey

Device	Quantity	Location	Purpose/Use
Dell Laptop	1	Warehouse	Used for encoding customer data in Excel and printing receipts
Printer	1	Warehouse	Used for generating customer receipts and documentation

Continuation of **Table 1**

Device	Quantity	Location	Purpose/Use
Mobile Phones	3	One per SM branch	Used for Messenger communication and storing service photos

This hardware is shared among the staff and does not include branch level desktop or tablets. Each SM branch primarily operates using their assigned mobile phone, which acts as both a communication, record-keeping, and image storage tool. The use of a limited hardware setup may not be sufficient for manual workflows.

Software (Client's software)

Shoelotskey's current software infrastructure is composed of basic tools that are no longer efficient for the complexity of its business operations. As per the client interview, the company still relies heavily on:

- **Microsoft Excel** – Used for logging service transactions, managing inventory records, and tracking customer data. This manual entry method is prone to human error and data loss, especially when used across multiple branches without synchronization.
- **Facebook Messenger** – The primary tool for customer communication. It is used for booking appointments, sending order updates, and responding to customer inquiries. However, this method lacks automation, searchability, and centralized tracking.
- **Windows Operating System** – Installed on the warehouse laptop, which is currently the only device used for official data processing.

Shoelotskey does not use any enterprise software, extravagant digital inventory system, or customer management platforms. There is no real-time service tracking or centralized database that results in data fragmentation, inconsistent customer experiences and service delays. These limitations may escalate in the future as the business expands to include more branches to take care of.

Peopleware/Manpower

Shoelotskey's manpower operates with a family-oriented yet professional structure that supports their multi-branch operations. The company is divided into roles to cover management, technical services, customer relations, logistics and marketing. The proposed system is designed to reflect this organizational structure through role-based access control to ensure that each user interacts only with the parts of the system that are relevant to their duties.

Table 2

Organizational Table

Position	**Name(s)**	**Role Description**
Chief Executive Officer (CEO)	Eleazar Dela Cruz Enguero	Oversees overall business direction and long-term strategies. Not involved in daily system use.
Chief Operating Officer (COO)	Suzette Toledo Enguero	Manages day-to-day operations, supervises staff, and ensures service quality. Will have full access to admin modules and analytics dashboards.
Team Leader	Cristina Toledo	Supervises technicians and coordinates service flow. Will access service status dashboards and internal scheduling tools.

Continuation of **Table 2**

Position	Name(s)	Role Description
Shoe Cleaning Technicians	John Lee, Beth, Maricris, Sherwin, Alvin	Perform shoe cleaning and restoration services. Minimal system access, limited to viewing task assignments and service statuses.
Customer Service Representatives (CSRs)	Charlote, Critine, Krishel	Handle customer bookings, provide service updates, and respond to inquiries. Will use the system to update order status, input customer records, and log appointments.
Delivery Staff	Edel Enguero, Sherwin Toledo	Responsible for the pick-up and delivery of shoes. Will use the system to confirm delivery schedules and update delivery logs.
Chief Marketing Consultant (CMC)	Psalms David Toledo Enguero	Advises on promotional strategies and branding. May review sales trends and customer insights through the analytics module.
Chief Marketing Officer (CMO)	Camille Joy Toledo Enguero	Leads the marketing efforts and campaign execution. Will use analytics features for campaign planning.
Marketing Staff	Aliza Cabriana	Executes marketing plans and supports social media promotions. May use limited analytics tools within the system.

Network Infrastructure/Architecture

Shoelotskey's network infrastructure is simple and decentralized. Each branch location uses its own independent internet connection, with no centralized or branch-connected network architecture in place.

Warehouse Setup:

- **Connectivity Type:** Wi-Fi

- **Device Connection:** 1 laptop and 1 printer are connected via a standard wireless router.
- **Usage:** The laptop and printer are used to encode the customer records using Excel, print receipts, and perform limited data handling tasks.

SM Branches Setup (3 Locations)

- **Connectivity Type:** Mobile Data (SIM-based)
- **Device Connection:** Each branch uses 1 smartphone with mobile internet access.
- **Usage:** Phones are used for customer communication via Messenger, taking photos of items, and storing job-related media.

Network Architecture Summary:

- **No LAN or shared intranet** setup across branches.
- **No centralized server or file sharing** between devices or branches.
- **No internal communication or VPN setup** for secure data transmission between locations.

Storage, Backup, and Recovery Procedure

Currently, Shoelotskey stores service documentation and transaction references in a decentralized and informal manner. Most service-related media, such as before-and-after photos, are saved in Messenger photo albums linked to their official Facebook page. Additionally, service logs and customer records are stored locally on mobile phones assigned to each of the three SM branch locations. While this method

offers quick access for staff, it poses a high risk of data inconsistency, device failure, or accidental loss due to the lack of centralization and standardized backup procedures.

Security Procedures

Shoelotskey currently implements basic security protocols to safeguard service-related information across its three branches. Staff devices used for managing customer records and service images are password-protected, ensuring only authorized personnel can access them. The client also maintains antivirus software and firewalls on computers where applicable, as part of their standard protective measures. Access to mobile phones and online accounts used in operations is restricted to designated staff, preventing unauthorized viewing or modification of data. While these practices help maintain a basic level of data protection, there is no centralized system in place to monitor user activity or enforce security protocols uniformly across all branches.

Policies and Procedures

Shoelotskey follows internal policies and procedures to manage its service operations, staff responsibilities, record-keeping, and customer communication. These protocols help ensure consistent handling of customer requests and service documentation across its multiple branches. The table below outlines the current practices observed by the client:

Table 3

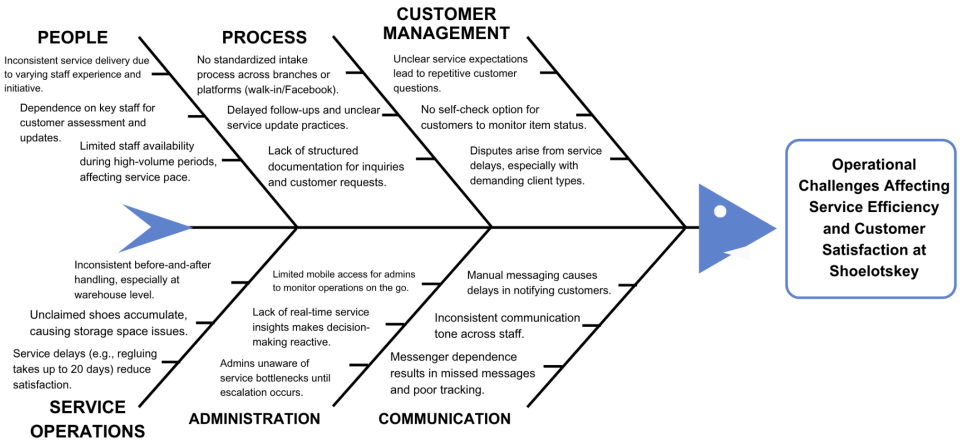
Existing Policies and Procedures of Shoelotskey

Policy / Procedure Area	Current Practice
Service Handling	Services are assessed in-store by a sales associate before proceeding. Customers cannot directly choose a service online.
Staff Access and Responsibility	Each branch has designated staff handling customer interactions and service documentation. Device access is limited to assigned personnel only.
Record-Keeping	Service records (photos, logs) are saved on branch-assigned mobile phones and Messenger albums.
Customer Updates	Customers are updated manually via Messenger or text, usually by the same staff who handled the service.
Service Confirmation	Customers confirm receipt of services or resolved concerns verbally or via chat. No formal digital receipt is used.
Branch Autonomy	Each branch follows the same basic process but manages data independently.

Problem Analysis

Fishbone Diagram

Figure 1. Fishbone Diagram



Problem and Solution Statement

Shoelotskey, a growing shoe care service provider, currently faces several operational challenges that hinder efficiency and customer satisfaction. Across its three branches, inconsistencies in staff handling, service intake, and customer communication contribute to delays and confusion. Service quality varies depending on personnel initiative, and there is no unified process for handling customer inquiries or follow-ups. Customers rely heavily on Facebook, Messenger or walk-ins for updates, often resulting in missed messages and miscommunication. Unclaimed shoes create storage issues, and the lack of real-time service transparency leads to disputes, especially with high-demand clients. Additionally, administrators struggle to gain immediate insight into branch operations, making oversight reactive instead of proactive. These issues collectively limit Shoelotskey's ability to scale efficiently, maintain high-quality service, and deliver a seamless customer experience.

To address these operational challenges, a web and mobile-based centralized management system, Shoelotskey Web & Admin Suite (SWAS), will be developed. The system will feature real-time service tracking, centralized inventory management, and role-based access control to streamline branch coordination. Additionally, AI-assisted forecasting (using Prophet), automated notifications, and a customer mobile app will enhance service transparency and user engagement. By digitizing core business functions, the proposed solution will improve accuracy, reduce human error, enable data-driven decision-making, and support the company's expansion into a scalable, efficient, and customer-oriented service provider.

Problem-Requirements Matrix

Table 4 identifies and categorizes key issues and their associated requirements for effective resolution:

Table 4

Problem-Requirements Matrix

Identified Problem	System Requirement
Inconsistent handling of customer inquiries and service intake across branches.	The system shall provide a centralized, real-time dashboard for managing service requests and customer records across all branches.
No standardized communication process results in delays and miscommunication.	The system shall give notification via the app for every updated movement from the inventory for timely and consistent customer updates.
Delays in manual follow-ups and service updates cause bottlenecks in customer service.	The system shall allow batch updates for service statuses to reduce admin workload and speed up customer notifications.
Unclaimed shoes accumulate due to lack of timely reminders and follow-ups.	The system shall notify customers of pending pickups and automatically flag overdue items with customizable scheduling.
Storage and retrieval of customer and service data is not scalable.	The system shall utilize MongoDB for scalable, flexible storage with support for real-time queries and export functions.
Service tracking and customer inquiries are dependent on manual or verbal updates.	The system shall include a customer-facing mobile app with real-time service tracking and history access.

Continuation of **Table 4**

Identified Problem	System Requirement
Varying access levels for staff leads to inefficiencies and security concerns.	The system shall integrate Role-Based Access Control (RBAC) to manage user permissions based on roles (admin, staff, warehouse, etc.).
Inventory is updated manually, leading to restocking delays and data mismatches.	The system shall feature an inventory module with real-time visibility, audit trails, and low-stock alerts.
Admins have limited visibility on service flow and must check statuses manually.	The system shall provide a visual service pipeline or dashboard showing current job progress per branch.
No insight into service demand trends or peak periods.	The system shall integrate AI forecasting using Prophet to analyze trends and predict future demand.

Purpose and Description

The prime objective of SWAS or Shoelotskey Web & Admin Suite is to provide better workflow prevailing in the business operations that happens because of outdated manual processes. The digitization of operations aims to turn a business into a centralized, scalable, and secure real-time service tracking, efficient inventory management, and improved customer interaction is what helps a client.

The proposed system has been tailored to fit Shoelotskey's multi-branch system as it automates workflows and reduces operational mistakes while improving their data accuracy drastically. It will enable the business to harness analytics to make more informed decisions as well as provide an improved customer experience and prepare the business for scalable growth in a highly competitive digital market.

Specific Objectives

1. **To Implement a Centralized Branch Management System**

   Enable branch-wide process synchronization and real-time visibility into service requests and sales through a unified admin panel.

2. **To Introduce an Automated Order Updates and Notification System**

   Improve customer communication by sending notifications for every updated movement of their item from the inventory**.**

3. **To Create a Batch Update Management Tool for Administrators**

   Allow bulk status updates and administrative actions across multiple branches for consistency and time efficiency.

4. **To Integrate a Role-Based Access Control System for Secure Data Management**

   Enhance security and organization by assigning specific permissions based on user roles (e.g., branch staff, super admin).

5. **To Implement a Scalable MongoDB Data Management Solution**

   Facilitate future-proof data handling and ensure fast access to historical records, analytics, and customer history.

6. **To provide real-time service tracking and visual dashboards for transparency**

   Enable customers and staff to monitor service progress through an interactive interface, reducing inquiry volume and improving trust.

7. **To integrate a customer-facing mobile application for engagement and convenience**

   Allow customers to book appointments, check status updates, and view past transactions on their Android devices.

8. **To automate administrative notifications and reminders for efficient service flow**

   Ensure that critical events such as overdue pickups, appointment confirmations, or stagnating records trigger system-generated alerts to relevant users.

9. **To enable AI-powered business forecasting using Prophet**

   Provide predictive insights into service demand and sales trends to aid in strategic planning and resource allocation.

Scope and Limitations

Scope

- The system will serve Shoelotskey Shoe Cleaning Services and its branches. Namely: SM Valenzuela, SM Grand Central, and SM Balintawak.
- Core modules include: service tracking, inventory management, customer record integration, admin tools, and notification (manual and templated).
- The web-based platform will be accessible through modern browsers and fully responsive on desktop, tablet, and smartphone devices.
- A customer mobile app (Android only) will allow users to:
    - Book appointment dates
    - View the status of their service
    - Access past transaction history
- The AI component (via Prophet) will provide forecasting insights for admin use.

Limitations

- The system does not support third-party payment gateways (e.g., GCash, PayPal, PayMaya).
- Automated SMS notifications are not implemented due to cost constraints; only manual messaging by staff is supported.
- Appointment bookings made via the mobile app are for scheduling purposes only—actual service types are determined in person by a sales associate.
- The mobile app will be available on Android only; iOS support is excluded due to development constraints and a limited timeline.
- AI-generated notifications are limited to predefined templates.
- Offline functionality is not supported; the system requires an internet connection.
- The scope excludes third-party courier integration for logistics or delivery tracking.

Definition of Terms

- **SWAS (Shoelotskey Web & Admin Suite):** A web-based platform designed to digitalize and centralize Shoelotskey's business operations including order/service tracking, inventory management, customer communication, and administrative tools.
- **Admin Panel:** A secure backend interface where administrators and super admins manage users, view analytics, monitor operations, and perform updates across branches.

- **Centralized System:** A system architecture in which data and management tools are hosted and controlled from a single location, enabling consistent and coordinated operations across multiple branches.
- **Responsive Web Design:** A design approach that ensures the system's interface adapts smoothly to different devices and screen sizes (desktop, tablet, mobile).
- **Inventory Management:** A software module that enables the tracking and updating of item stock levels, service materials, and product availability across all branches.
- **Real-Time Tracking:** The ability to monitor and update the status of customer service requests instantly across all branches through a centralized digital dashboard.
- **Batch Update:** A feature that allows system administrators to make bulk changes (e.g., update service requests statuses or inventory counts) to multiple records simultaneously, improving efficiency.
- **Order Status:** The stage or progress of a customer's service request (e.g., Received, In Process, Completed, Ready for Pickup) which is visible to both staff and customers.
- **Open-Source Technology:** Software whose source code is freely available for use and modification, such as Svelte and MongoDB, reducing licensing costs and increasing customization flexibility.
- **Dashboard Analytics:** Visual summaries and performance metrics (e.g., order volume, top services, sales per branch) used by business owners and admins for data-driven decision-making.

- **RBAC (Role-Based Access Control):** A security mechanism that restricts system access based on assigned user roles (e.g., branch admin, super admin) to ensure proper authorization and data protection.
- **User Role:** A predefined access level granted to users (e.g., super admin, branch staff) that determines their permissions and capabilities within the system.
- **MongoDB:** A NoSQL database management system used for storing unstructured and flexible data formats, ideal for rapidly growing and dynamic information like customer records and service logs.
- **Data Export:** A feature that allows users to extract stored data into external formats such as CSV or Excel, enabling offline access, reporting, and backups.

Chapter 2

REVIEW OF LITERATURE, STUDIES, AND SYSTEMS

Digital Transformation in MSMEs and Service-Based Businesses

The related literature conducted by Ranieses (2024) discusses the rapid shift of Philippine micro, small, and medium enterprises (MSMEs) toward digitalization during the COVID-19 pandemic, noting that despite initial resistance, businesses that embraced digital tools became more resilient and competitive in the post-pandemic market. Similarly, Bruce et al. (2023) emphasize that digital marketing tools, particularly those leveraging social media, allowed small businesses in developing nations to scale operations and improve market visibility. The Department of Trade and Industry (2024) formalized this transformation by embedding it in the MSME Development Plan 2023–2028, citing digitalization as a critical driver for economic inclusion and competitiveness. Hurey (2022) provided real-world examples from local enterprises that implemented cloud-based point-of-sale systems and online order tracking to streamline customer service and reporting. Internationally, Mahmood et al. (2021) examined Indonesian and Malaysian SMEs and found that digital readiness, measured through e-commerce adoption and cloud service usage, was positively correlated with revenue growth, customer engagement, and long-term sustainability.

Meanwhile, PwC Philippines (2020) noted that despite the promising benefits, digital transformation among Filipino MSMEs remains limited due to infrastructure gaps, digital skills shortage, and financial constraints. To address this, ASA Philippines Foundation pioneered cloud-based financial reporting systems across its branches, leading to improved data transparency and operational efficiency (ASA, 2023). APEC (2020) also reported that MSMEs across the Asia-Pacific face common issues in digital

uptake but benefit greatly from government-led infrastructure support and public-private training initiatives. In a similar vein, Guillen and Lim (2023) explored how Facebook advertising and growth hacking strategies enabled MSMEs in the Philippines to attract niche markets without the need for large marketing budgets. From a global standpoint, Ramadani et al. (2021) analyzed digital transformation models across Europe and concluded that agile adaptation and modular systems were most successful among service-based businesses with limited technical staff. Finally, the United Nations Economic and Social Commission for Asia and the Pacific (UNESCAP, 2022) underscored that integrating digital platforms into small enterprise operations is no longer optional but imperative, especially in environments affected by economic uncertainty and public health disruptions. These local and foreign studies collectively justify the design and implementation of Shoelotskey's Web & Admin Suite (SWAS), which aligns with proven strategies for digital growth, automation, and resilience in service-oriented MSMEs.

Centralized Multi-Branch Management Systems

Operating multiple business branches without a centralized system often leads to fragmented processes, duplicated tasks, and inconsistent service quality. This is particularly true for service-based Micro, Small, and Medium Enterprises (MSMEs), where real-time monitoring, coordination, and data access are crucial for maintaining efficiency. A study by SWEEDU (2024) highlights that centralized management systems streamline operations by integrating various administrative functions, allowing institutions to monitor schedules, process fees, and allocate resources efficiently across branches. In the Philippine context, the Department of Science and Technology (2025) emphasized that while programs like SETUP have significantly contributed to

MSME growth, businesses still need improved administrative and monitoring systems to maximize their technological upgrades. Megaventory (2025) further supports the implementation of centralized inventory management, explaining how this system improves accuracy and scalability across business locations. Similarly, RSI Concepts (2023) explains that centralized queue management across branches leads to more consistent customer service experiences and faster service turnaround.

Government and development agencies also stress the need for centralized digital systems. The World Bank (2023) noted that the Philippine government's own move toward digital transformation relies heavily on unified systems for improving service delivery, providing a model for MSMEs to follow. In the education and service sectors, SWEEDU (2024) reported that centralized dashboards improved multi-branch coordination through real-time data access and resource control. On a broader scale, APEC (2020) and UNESCAP (2022) recognized that centralized platforms enhance internal market access and streamline operations for SMEs trying to scale. Similarly, the Department of Finance (2024) highlighted digital integration as essential for national resilience, including for risk management. In the private sector, The Business Research Company (2025) projected major growth in Computerized Maintenance Management Systems (CMMS), noting that centralized management software is key to optimizing multi-site maintenance operations. These findings collectively support the development of Shoelotskey's Web & Admin Suite, reinforcing the necessity of centralization for operational efficiency, visibility, and scalability across branches.

Real-Time Service Tracking and Customer Engagement

Real-time service tracking has emerged as a critical tool in enhancing customer satisfaction and business efficiency, particularly among service-based MSMEs.

According to Salcedo (2023), small businesses in the Philippines that adopted real-time service dashboards improved their client retention by 34%, owing to greater transparency and communication. The Department of Trade and Industry (DTI, 2024) reinforces this view in its MSME e-commerce enablement program, where they emphasize automated updates and tracking systems as key to digital competitiveness. Reyes and Mariano (2023) conducted a case study on Metro Manila-based courier services and found that mobile-based order tracking helped reduce customer inquiries by 48%, showing how proactive updates reduce friction in the service journey. A similar study in Vietnam by Le et al. (2022) revealed that integrating real-time service updates into customer experience strategy resulted in higher customer trust and 21% more repeat transactions for online cleaning services. This is supported by McKinsey & Company (2022), which states that 70% of customers now expect businesses to provide real-time service updates, especially for services involving logistics, repair, or appointments.

Moreover, interactive tracking tools have proven to improve perceived service quality, especially when coupled with mobile notifications and visual progress bars. Villanueva et al. (2024) discuss the implementation of in-app service update notifications in small appliance repair shops in CALABARZON, which resulted in more organized pick-up schedules and increased user engagement. Likewise, the World Bank (2023) highlights the Philippines' push for improved digital infrastructure in support of mobile-based business models, where timely customer communication is central to quality service delivery. Internationally, Shopify (2021) reported that e-commerce and service businesses with live order and status updates experience 23% fewer customer support tickets and higher review ratings. PwC (2020) emphasized that businesses that integrated customer tracking features such as

estimated delivery times, live service status, or chatbot follow-ups reported increased customer retention, especially among millennials and Gen Z. Lastly, UNESCAP (2022) concluded that small businesses adopting real-time communication tools experienced significantly better post-sale engagement and fewer disputes, underlining the necessity of real-time service updates as a core digital capability for MSMEs. These studies strongly support Shoelotskey's plan to integrate real-time status tracking, service dashboards, and automated messaging to boost customer trust and streamline operations.

MongoDB and NoSQL Databases for Scalable Business Applications

The shift from traditional relational databases to NoSQL systems like MongoDB has revolutionized how businesses, particularly MSMEs, manage, scale, and access data. MongoDB, a document-oriented NoSQL database, offers flexibility in data modeling and enables rapid development through dynamic schema design. According to Chodorow (2019), MongoDB's JSON-like document structure allows developers to store complex data objects with ease, making it ideal for applications with evolving data requirements. The official MongoDB Atlas Cloud Report (2023) highlights that startups and SMEs worldwide prefer MongoDB due to its auto-scaling capability, horizontal partitioning (sharding), and simplified deployment across cloud platforms. In the Philippine context, a study by Arcenas and Beltran (2021) found that local businesses using MongoDB in point-of-sale systems significantly reduced system downtimes and experienced faster reporting speeds compared to legacy MySQL setups. Meanwhile, Abu and Ogbuokiri (2020) concluded in a comparative analysis that MongoDB outperformed relational databases in handling high-write operations typical of e-commerce platforms and retail inventory systems.

In terms of scalability and responsiveness, NoSQL databases are especially useful in real-time applications. A case study by IBM (2022) shows how small logistics companies scaled their web applications using MongoDB to support thousands of concurrent users with minimal latency. Likewise, Rani and Sharma (2020) emphasized that MongoDB's indexing and replication strategies allow SMEs to run distributed systems with high availability, making it suitable for mobile applications and dashboard systems. According to Google Cloud (2023), MongoDB is widely adopted in customer-facing apps due to its ability to manage unstructured and semi-structured data efficiently, enhancing real-time data synchronization. In Southeast Asia, the Digital Innovation Council (2022) documented that NoSQL adoption among startups increased by 41% in 3 years, with MongoDB being the most popular due to its ease of integration with JavaScript frameworks like Node.js and React. Pineda and Santos (2024) showed how a shoe repair and cleaning startup in Manila used MongoDB to manage orders, update records in real time, and reduce spreadsheet use. These findings support the Shoelotskey Web & Admin Suite's choice of MongoDB for scalable, real-time, and secure backend operations aligned with SME digital transformation best practices.

Role-Based Access Control (RBAC) in Business Systems

Role-Based Access Control (RBAC) is a widely adopted access control strategy used in business systems to assign permissions based on user roles, helping organizations maintain data confidentiality, integrity, and accountability. According to Sandhu and Coyne (2019), RBAC simplifies security management by ensuring users can only access the data and functions relevant to their role, minimizing risk and reducing administrative overhead. In a business setting, RBAC is especially beneficial in multi-user systems where tasks must be clearly divided among employees.

Concentric AI (2025) highlights how RBAC is central to modern data governance, especially in platforms that process sensitive customer and transaction data. A study by Cruz and Vergara (2022) on small enterprises in Quezon City found that RBAC implementation reduced unauthorized system changes by 63% and improved employee accountability. Moreover, research by Choi et al. (2021) in South Korea emphasized that RBAC helped retail businesses maintain tighter controls over their POS and admin operations across branches, leading to more reliable transaction records and improved trust among staff.

RBAC is not just about access limitation; it also enhances system usability by tailoring interfaces and functions to the user's needs. According to Google Cloud (2023), integrating RBAC in cloud-based services streamlines authentication, improves user experience, and reduces system overload by eliminating irrelevant options for each role. In the local context, a case study by Javier and Monteverde (2023) evaluated the use of RBAC in a Philippine-based appointment scheduling platform, which helped distinguish privileges between admins, staff, and clients, reducing confusion and data leaks. IBM (2022) also discusses how RBAC boosts compliance with data privacy regulations such as the Philippine Data Privacy Act by enforcing “least privilege” principles. Additionally, Wang et al. (2021) presented a hybrid RBAC model that adapts dynamically to user behavior, particularly useful in businesses with seasonal or contract-based workers. Linx Global (2023), an enterprise tech provider in Southeast Asia, reported that businesses adopting RBAC alongside audit logging and MFA (multi-factor authentication) reduced internal data breaches by 45% across industries. These findings reinforce Shoelotskey’s use of RBAC to manage permissions across super admin, branch admins, and staff levels that enhance security, user clarity, and control in a multi-branch environment.

AI-Assisted Notifications and Analytics in Customer Service

Artificial Intelligence (AI) has increasingly become integral to enhancing customer service through personalized notifications and predictive analytics. As explained by Accenture (2022), AI systems enable businesses to send targeted, real-time updates based on user behavior, improving the customer experience and operational efficiency. In the Philippine context, Dizon and Mallari (2023) observed that SMEs implementing AI-driven messaging systems saw a 35% increase in repeat clients, largely due to improved service transparency and follow-up engagement. Google Cloud (2024) supports this by showing how AI-powered alerts, using natural language processing and machine learning, automate customer interactions and improve satisfaction rates. Meanwhile, Mercado and Javier (2023) conducted a case study on local food delivery services, showing that AI-generated delivery notifications led to a 22% reduction in support tickets. Globally, Salesforce (2023) reported that 73% of customers expect companies to understand their needs and provide proactive communication, something AI systems now support through automated, behavior-based triggers.

Aside from communication, AI's role in analytics is equally vital in shaping customer service strategies. IBM (2021) notes that AI analytics systems can detect patterns in customer complaints, helping service teams identify weaknesses before they escalate. In Southeast Asia, Digital ASEAN (2022) stated that predictive analytics in small businesses helped preempt order delays and allocate resources more efficiently. In a Philippine study, Reyes and Cruz (2024) highlighted how AI-driven dashboards helped MSMEs monitor customer behavior and adapt marketing strategies in real time. Microsoft Azure (2023) explains that AI-assisted analytics tools now come

with integrated sentiment analysis, allowing businesses to assess customer satisfaction beyond star ratings or text reviews. Finally, Sharma and Lim (2020) stress that AI's role in real-time data processing improves operational responsiveness, especially in service industries where timing and accuracy of communication directly affect customer retention. These findings support Shoelotskey's planned integration of AI-generated email notifications and dashboard analytics as key features that aim to improve service delivery, customer engagement, and long-term loyalty.

Synthesis of the Study

The review of related literature presents a comprehensive understanding of how modern digital platforms are transforming service-based Micro, Small, and Medium Enterprises (MSMEs) by enhancing user experience, improving operational workflows, and expanding market reach. This synthesis supports the foundation of the proposed system, Shoelotskey Web & Admin Suite (SWAS). A centralized, scalable platform designed to streamline business management for a growing shoe cleaning and restoration service with multiple branches. As emphasized by Ranieses (2024) and DTI (2024), digital transformation is no longer optional but necessary for MSMEs to remain competitive. The reviewed works collectively underscore the need for seamless service visibility, automation, and customer engagement, particularly in a post-pandemic environment where clients demand faster, transparent, and personalized experiences.

The trend observed across the literature reveals a shift from manual processes to centralized, real-time service platforms. Studies on MongoDB and NoSQL databases highlight the significance of using scalable backend solutions to accommodate the growing needs of service businesses (Chodorow, 2019; Arcenas & Beltran, 2021). These technologies ensure responsive data handling and are especially suitable for

dashboard systems with user-facing service logs and before-and-after photos. Furthermore, the integration of Role-Based Access Control (RBAC) allows differentiated access and workflow security across user types from super admins to branch-level staff (Cruz & Vergara, 2022; IBM, 2022). This proves essential in managing service requests, verifying customer data, and minimizing administrative errors. Additionally, works by McKinsey (2022), Google Cloud (2024), and Villanueva et al. (2024) affirm that real-time tracking, customer notifications, and visual service confirmation features significantly increase customer satisfaction and trust.

Complementing the user-facing functionality are service review systems with Facebook API integration. As demonstrated in literature on social-driven platforms, users are more likely to trust and share feedback when systems are connected to familiar channels like Facebook (Salesforce, 2023). This integration not only enhances marketing reach but also reinforces brand credibility through organic user engagement. On the backend, the use of filterable subtabs, advanced sorting, and database-intensive views aligns with findings by Megaventory (2025) and SWEEDU (2024) that emphasized the role of centralized dashboards in operational monitoring and staff productivity. These are further supported by secure and scalable backend infrastructure, with MongoDB offering flexibility in data structure while maintaining performance across web and mobile environments.

In conclusion, the studies reviewed collectively highlight the growing demand for intelligent, role-aware, and highly interactive systems in MSME digital ecosystems. The insights gained from both local and international literature justify the development of the Shoelotskey Web & Admin Suite. By integrating features such as real-time customer tracking, automated notifications, and robust role-based access, the system provides a scalable solution tailored for service-based businesses navigating digital

transformation. This synthesis reinforces the importance of transitioning from fragmented manual workflows to centralized, customer-focused digital platforms that empower both the business and its clients.

Chapter 3

METHODOLOGY

Requirements Analysis

Requirements – Features Matrix

Table 5

Requirements – Features Matrix

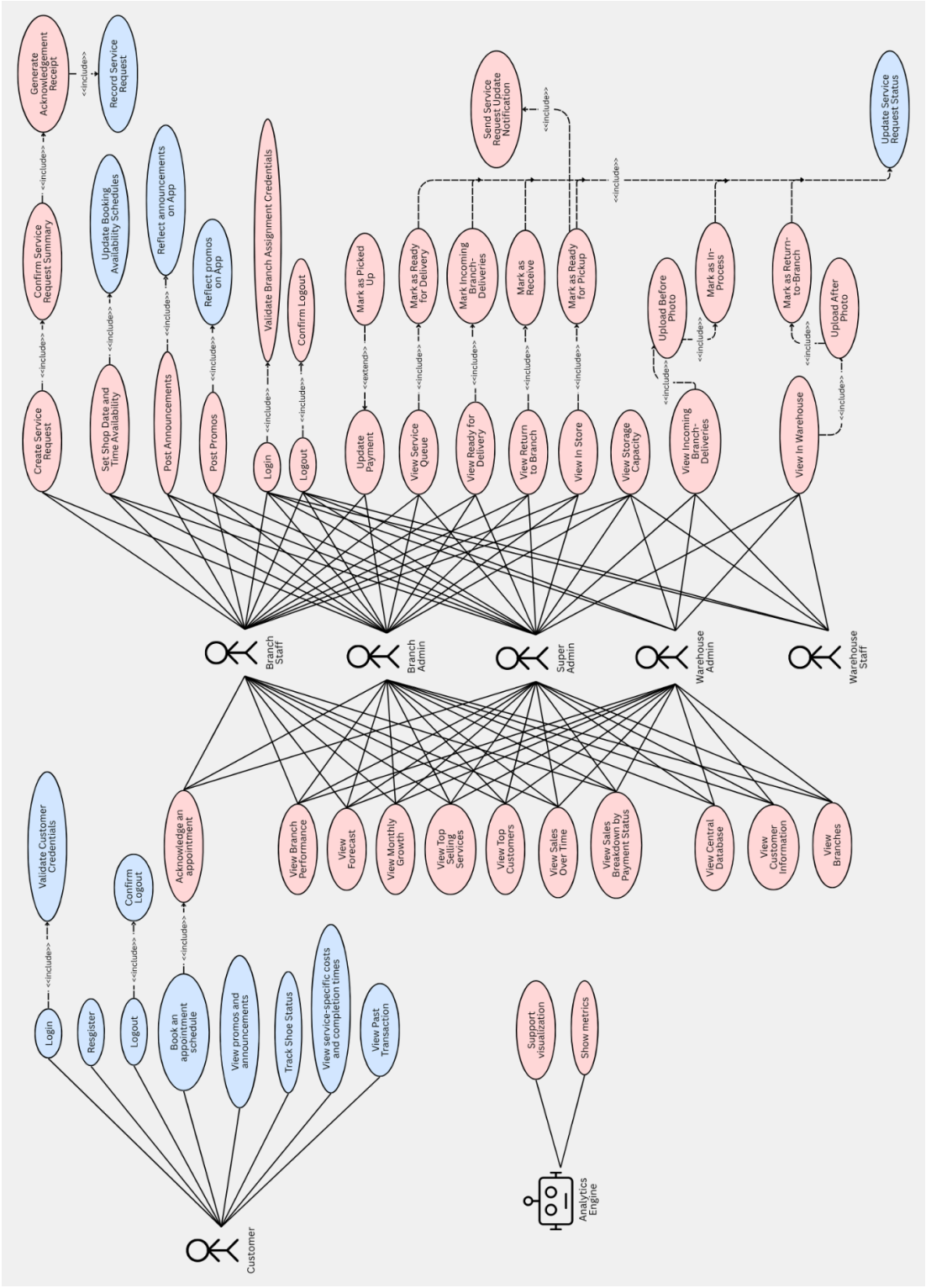
Requirement Description	Corresponding System Feature	Priority Level	Status
Centralized control of multiple branches and processes.	Admin dashboard with centralized branch monitoring and real-time process tracking.	High	Developed
Real-time updates on service requests and appointments for customers.	Notification system for service request updates and appointment confirmations.	High	Developed
Reduction of manual administrative tasks across branches.	Batch update tool for multi-record status changes.	High	Developed
Secure data access based on staff roles.	Role-Based Access Control (RBAC) for admins, staff, and super admins.	High	Developed
Scalable data handling for growing operations and transaction records.	MongoDB-based data management and export functionality (CSV, Excel).	High	Developed
Unified inventory control across all service branches.	Branch-synced inventory management module.	Medium	Developed

Continuation of **Table 5**

Requirement Description	Corresponding System Feature	Priority Level	Status
Easy access to historical customer data.	Searchable customer records and service request history feature.	Medium	Developed
Automated generation of acknowledgement receipts for inquiries and payments.	Generate an acknowledgment receipt for both the initial inquiry and the final payment.	Medium	Developed
Tracking of inactive items to identify overlooked or delayed transactions.	Color-coded highlighting (yellow, orange, red) that visually indicates the number of days an item has remained inactive.	Medium	Developed
One-month-ahead sales forecasting with detailed breakdown and monthly growth tracking.	Sales analytics module with forecast projection, category-based breakdown, and growth trend chart.	Medium	Developed
Branch-aware report generation with export options in PDF and Excel formats.	Report generation module with branch-specific filtering and export to PDF and Excel.	Medium	Developed
Centralized announcements and promo visibility for mobile users.	Dedicated announcement page displaying updates and promos, with integrated data tracking for analytics regression.	Medium	Developed

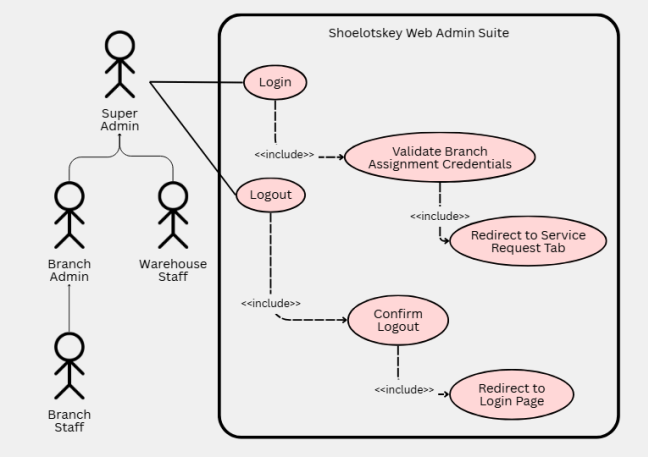
Use Case Diagram

Figure 2. General Use Case Diagram



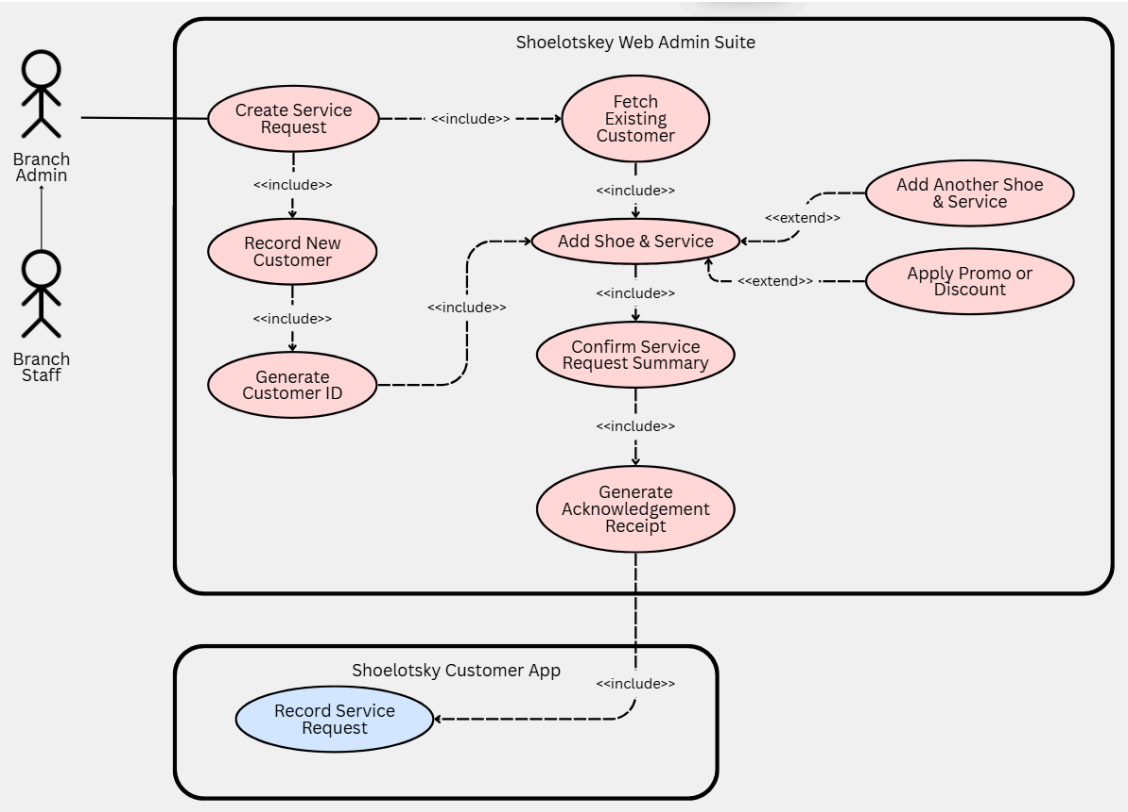
This use case diagram illustrates the core functionalities of the Shoelotskey system, which is divided into two main platforms: the Shoelotskey Web Admin Suite (in red) and the Shoelotskey Customer App (in blue). The Customer App allows users to manage their profiles, book appointments, track service status, view promotions, and calculate service costs. Meanwhile, the Web Admin Suite supports different roles, including Branch Staff, Branch Admin, Super Admin, and Warehouse Staff, in managing service workflows such as service requests, bookings, logistics, analytics, customer information, and internal communications. An automated email system and analytics engine further enhance operational efficiency by handling email updates and performance metrics.

Figure 3. Manage User Authentication Use Case Diagram



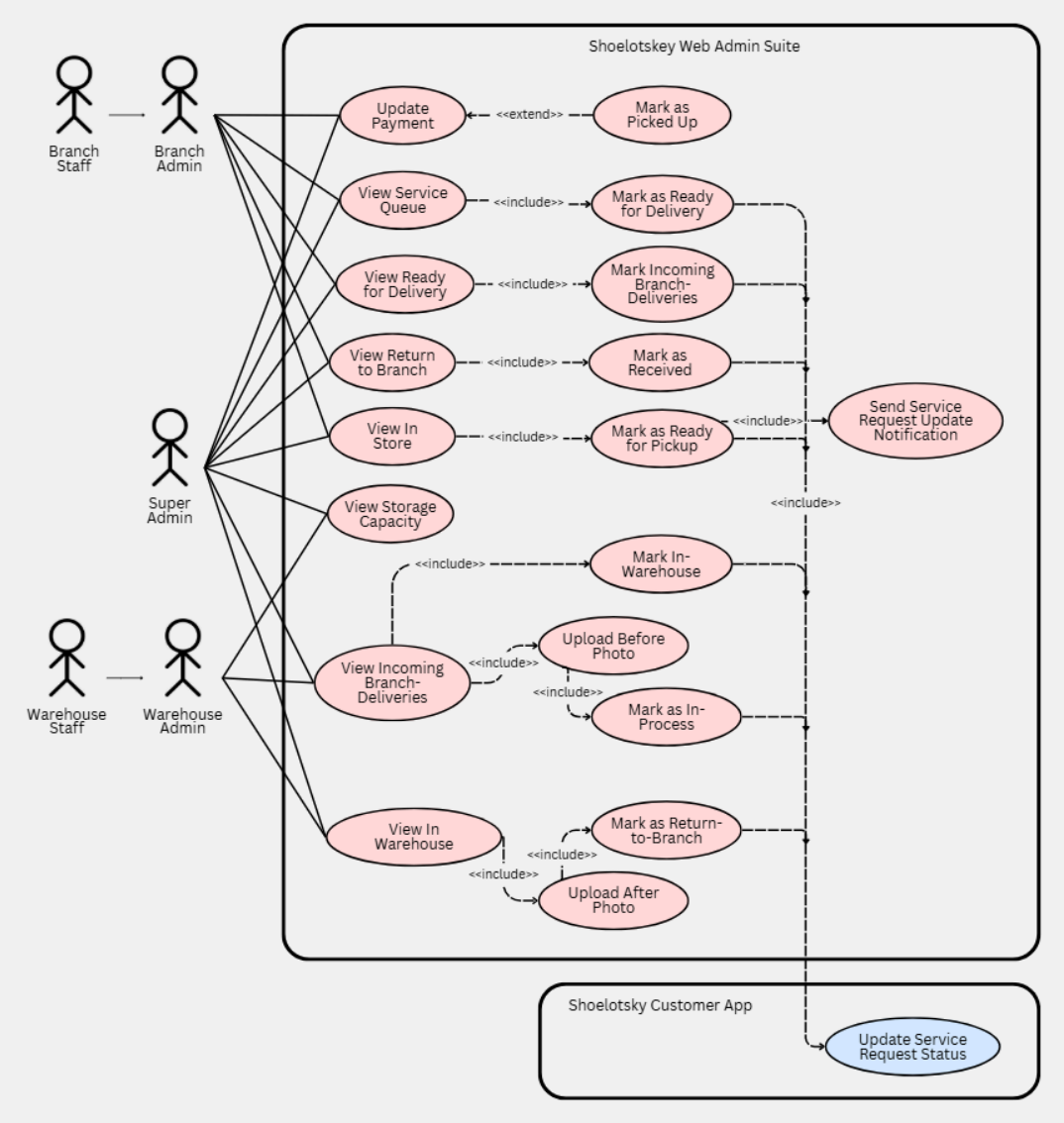
This use case diagram illustrates the authentication mechanisms in both the Shoelotskey Web Admin Suite and Customer App. It involves customers and staff members logging in using validated credentials and logging out with confirmation. The purpose is to ensure secure access to features and services. Each user type interacts only with their respective authentication flow. The separation of concerns helps maintain a secure and role-specific environment.

Figure 4. Handle Service Request Use Case Diagram



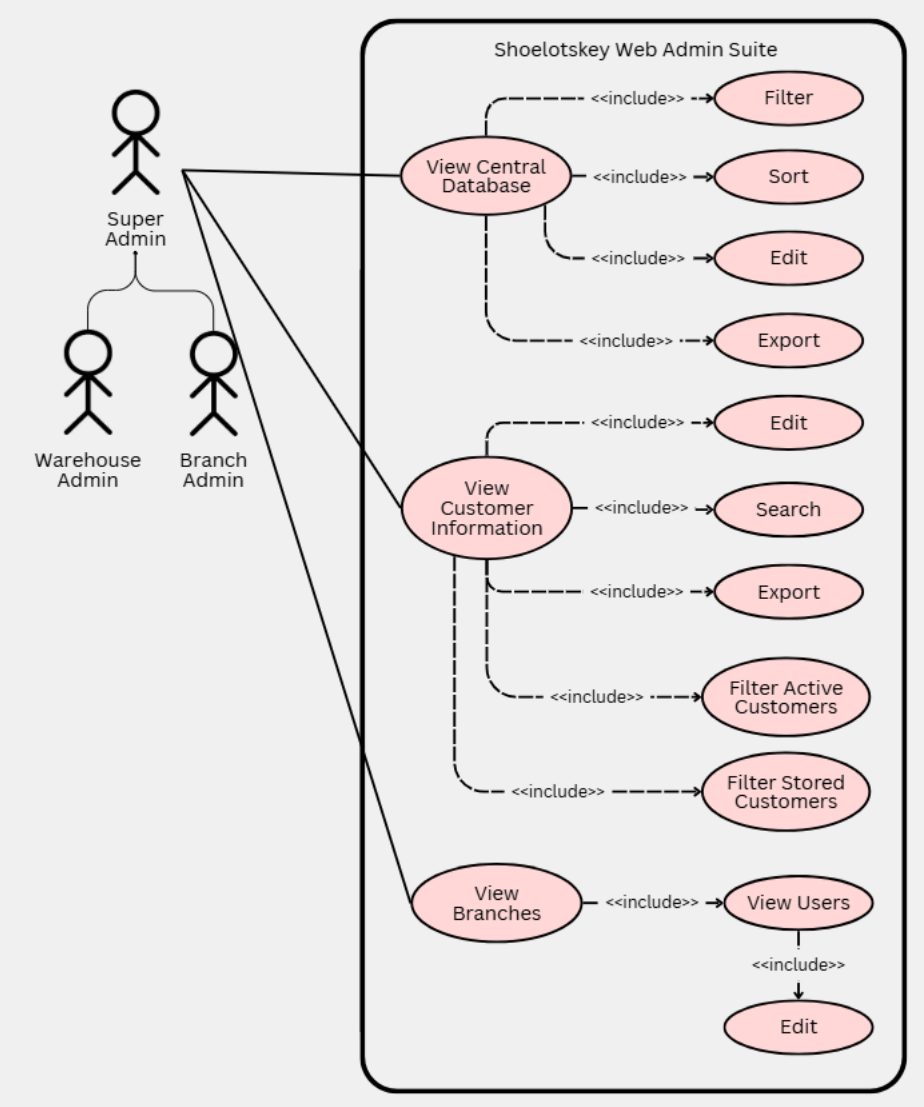
The Handle Service Request use case diagram allows branch staff to process walk-in customer service requests efficiently. When a customer arrives, the staff opens the service request form, identifies if the customer is new or existing, and enters all necessary personal and service-related details. The system ensures no duplicate accounts are created and generates an acknowledgment receipt with the estimated cost and turnaround time. If applicable, rush options, discounts, or promos are applied before finalizing the request.

Figure 5. Coordinate Branch and Warehouse Operations Use Case Diagram



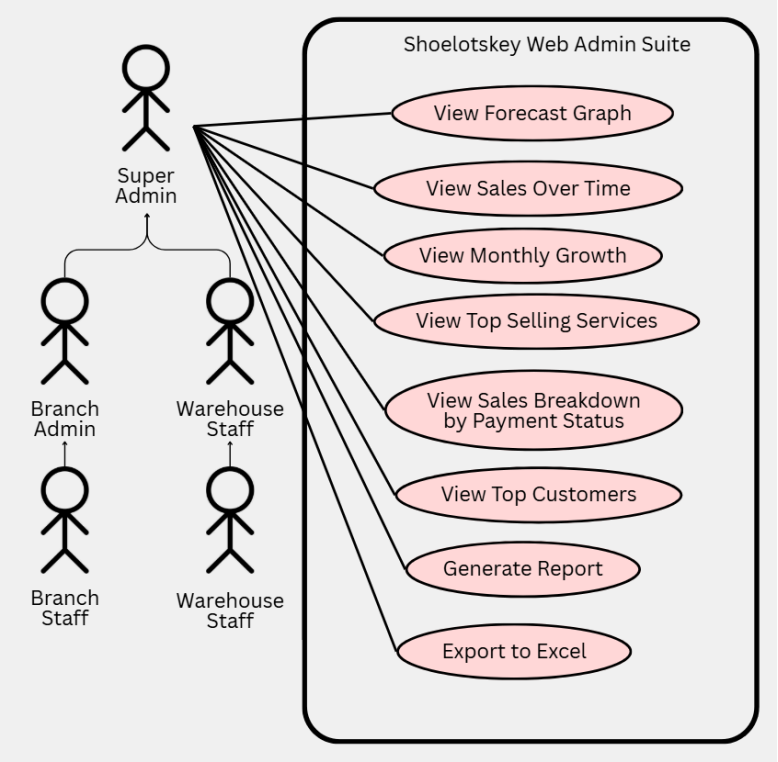
This use case diagram allows branch and warehouse staff to coordinate item movement by updating service statuses like “Ready for Delivery” and “In-Process” through the Shoelotskey Web Admin Suite. Handoffs are acknowledged, photos are uploaded when needed, and each status shows the “last date modified” to help track delays. For warehouse staff, remaining days are also displayed based on the estimated turnaround time, helping them stay aware of upcoming due dates. This ensures clear communication and real-time item tracking.

Figure 6. Access and Manage Database Records Use Case Diagram



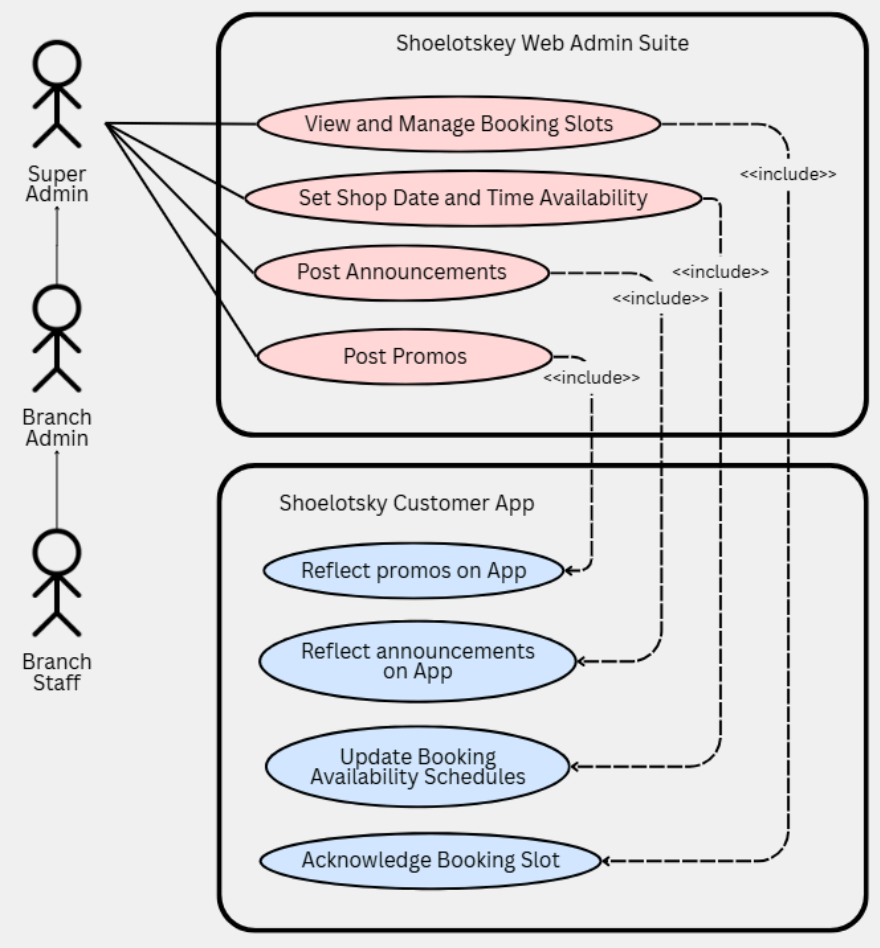
This diagram outlines how administrative roles interact with the system's central data. It covers actions such as filtering, sorting, editing, and exporting data. They can also view customer information, perform searches, export customer data, and apply filters to view active or stored customers. Additionally, both roles can view and manage branches, which includes the ability to view users under each branch and perform edit operations as needed.

Figure 7. Generate and View Business Analytics Use Case Diagram



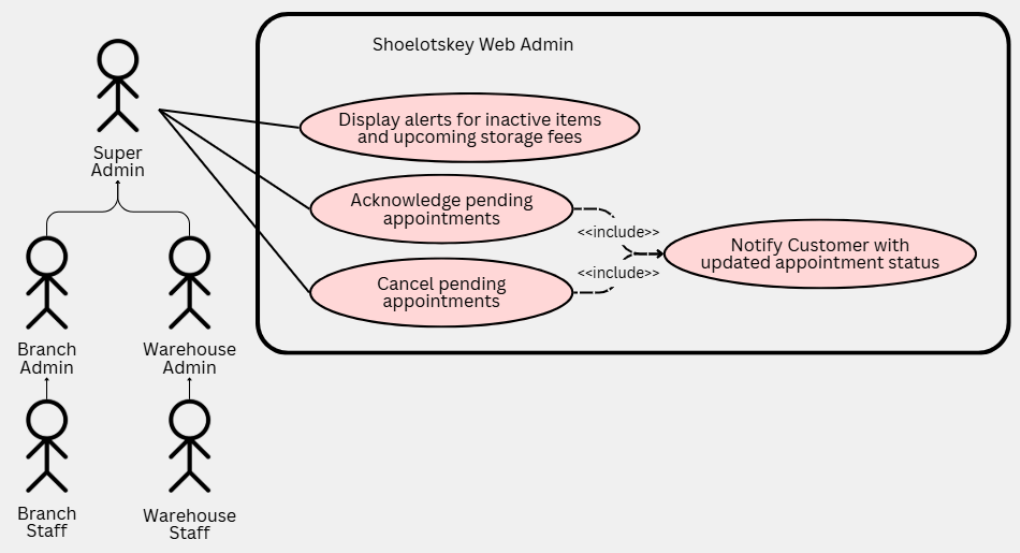
The analytics use case diagram includes clearly defined labels to make it easier to understand. It showcases how branch and super admins gain insights from collected data. The diagram supports viewing labeled metrics such as average order value, monthly growth, daily trends, and sales breakdowns. A narrative accompanies the daily trend to provide context and explain patterns, helping users make informed decisions. These insights are essential for performance monitoring and strategic planning. It ensures the Shoelotskey system is data-driven and adaptable to business needs.

Figure 8. Manage Users and Schedules Use Case Diagram



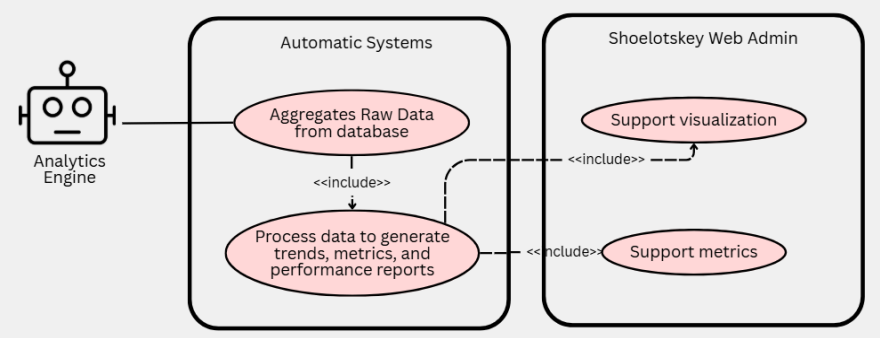
This diagram focuses on managing user schedules and appointment availability. Branch staff can manage booking slots, set availability, and post announcements. These updates are reflected in the customer app for transparency. The system supports real-time schedule updates and synchronization between staff and customer views. This ensures efficient service scheduling and communication.

Figure 9. Send Customer Notification Use Case Diagram



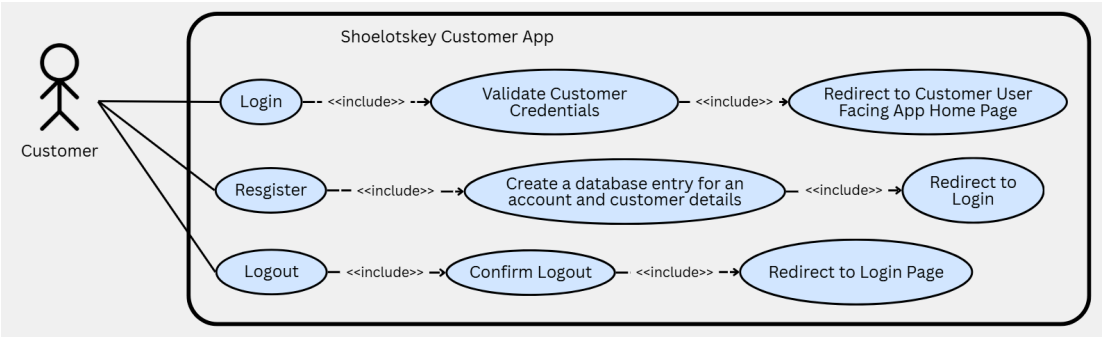
The Admin Side Notification use case ensures that administrators and staff are promptly informed whenever a customer books a scheduled appointment. Once a valid appointment date is submitted by the customer, the system detects the event and automatically sends a notification to the relevant admin. Additionally, the system notifies staff and admins if a transaction record has not been updated for a predefined number of days, serving as a reminder to check for possible delays. For warehouse staff, a separate notification is triggered when the remaining days in the estimated turnaround time are running low, helping them manage service progress proactively.

Figure 10. Automate System Processes Use Case Diagram



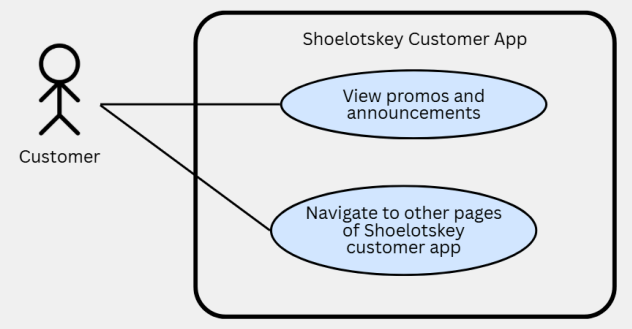
This diagram shows how automation supports both staff and customer processes. The automated system handles email communications triggered by service events like booking acknowledgements and delivery readiness. It also collaborates with the analytics engine to provide business insights. Automation reduces manual workload and ensures consistent communication. It is a key component in scaling Shoelotskey's operations efficiently.

Figure 11. Register and Log In Customer Accounts Use Case Diagram



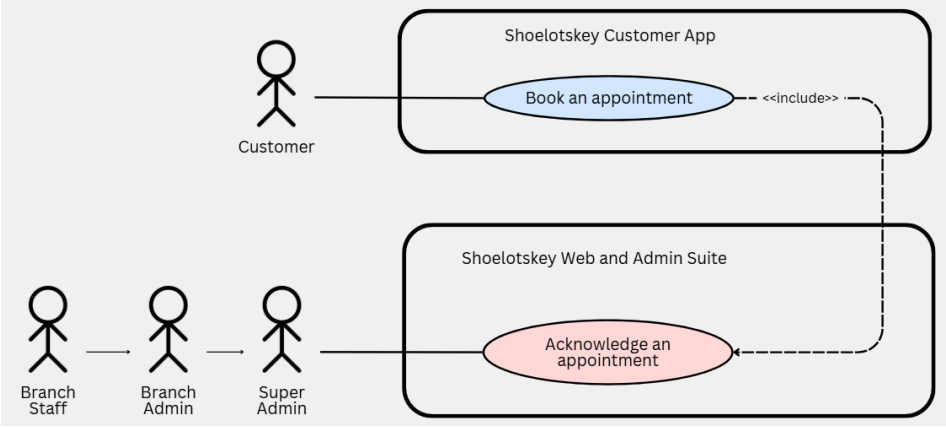
This use case describes the customer's onboarding and authentication in the Shoelotskey Customer App. Customers can register a new account or log in to an existing one. The system validates credentials and manages successful or failed login attempts. This foundation enables access to personalized services and features. It ensures a smooth and secure user experience.

Figure 12. Access Customer Dashboard Use Case Diagram



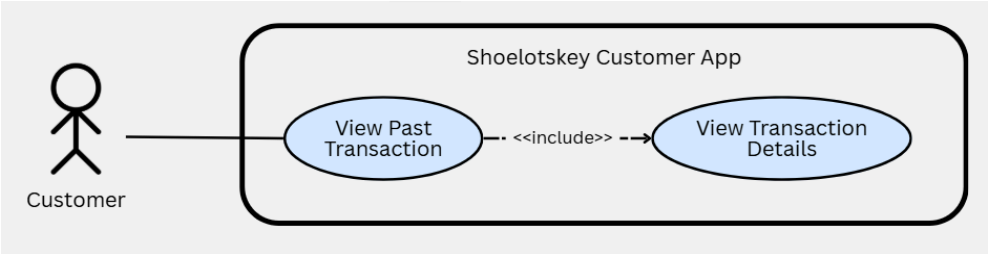
The dashboard use case shows the core functions accessible after login. Customers can view promos, announcements, track service status, and view past transactions. It acts as the dashboard for navigating the app. The system ensures the home page reflects real-time updates from branch staff. It's a central hub for customer interaction and engagement.

Figure 13. Book Appointment Slots Use Case Diagram



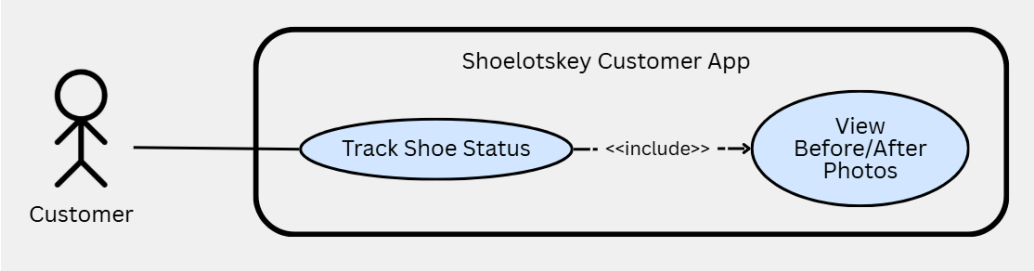
This diagram outlines the process of booking an appointment. Customers select from available slots managed by branch staff. Once booked, the staff acknowledges the appointment and updates the schedule accordingly. This ensures a structured workflow from booking to acknowledgment. It promotes organized service delivery and customer satisfaction.

Figure 14. View Transaction History Use Case Diagram



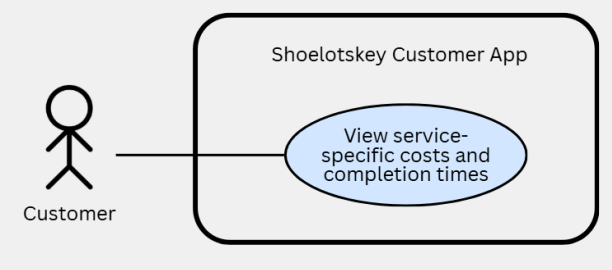
This use case lets customers view past transactions through their app. It displays previous services, dates, and payments for reference. This fosters accountability and easy access to service records. The feature supports both customer reviews and potential follow-ups. It enhances the overall user experience by keeping everything in one place.

Figure 15. Track Service Status Use Case Diagram



This use case focuses on how customers track the progress of their shoe service. Statuses like “In-Warehouse”, “Ready for Delivery”, or “Picked Up” are visible to the customer. The updates are managed by staff and reflected in real time. This transparency improves customer trust. It also reduces inquiry loads for staff.

Figure 16. Estimate Service Cost and Turnaround Time Use Case Diagram



This diagram illustrates the customer's ability to estimate both the cost and expected turnaround time for services before booking. By selecting the service type and related details, the app shows the price as well as how many days it will take before the shoes are ready for pickup. This feature allows customers to plan, both financially and logistically. It encourages transparency by providing upfront information. Overall, the tool enhances customer confidence and convenience before committing to a service request.

Table 6

Use Case Actors

Actors	**Role Description**
Customer	- Registering for an account - Logging in and logging out - Booking appointment slots - Viewing promos and announcements - Tracking service status - Estimating service cost and pickup time - Viewing past transactions
Branch Staff	- Creating and updating service requests - Managing payments and booking slots - Acknowledging appointments and bookings - Posting announcements - Marking service status (e.g., Ready for Delivery, Received, Ready for Pickup)

Continuation of **Table 6**

Actors	Role Description
Branch Admin	- Performing all Branch Staff actions - Viewing analytics (e.g., average order value, top customers) - Viewing and managing database records (customers, transactions, branches, etc.)
Super Admin	- Accessing full system data across all branches - Viewing the central database, user information, services, and transactions - Viewing business analytics and reports
Warehouse Staff	- Managing warehouse backlog and storage capacity - Uploading service images (before/after) - Marking service items as in-process, return-to-branch, etc.
Warehouse Admin	- Performing all Warehouse Staff actions - Viewing and managing warehouse operation records (inventory, transfers, service logs, etc.) - Coordinating inter-branch item movements
Analytics Engine	- Supporting metric generation and visualization - Providing trend data for admin analysis

Table 7

Use Cases

Use Case ID	Use Case Name	Description
UC-01	Manage User Authentication	Allows customers, staff, and admins to log in and log out securely using validated credentials. Ensures access is restricted based on user roles.
UC-02	Handle Service Requests	When a customer arrives, the staff records their details, ensures no duplicate accounts, and inputs service details (shoe model, services, applicable promos, payment). Then the system generates a receipt.

Continuation of **Table 7**

Use Case ID	Use Case Name	Description
UC-03	Coordinate Branch and Warehouse Operations with Customer Updates	Facilitates seamless coordination between branch and warehouse operations by enabling staff to manage item statuses throughout the service process, provide customer transparency through before-and-after photos of serviced items, update the mobile app's customer service tracker, and send notifications when items are ready for pickup.
UC-04	Access and Manage Database Records	Permits admins to view and manage data such as customer profiles, branches, services offered, and past transactions.
UC-05	View Business Analytics	Allows admins to view key business metrics like monthly growth, sales breakdown, and revenue forecast.
UC-06	Manage Users and Schedules	Enables staff to set booking availability, manage customer appointments, and post announcements and promos to be reflected in the customer app.
UC-07	Send Customer Notifications	Notifies customers of appointment booking status.
UC-08	Automate System Processes	Automates routine back-end operations, including scheduled analytics updates, to improve efficiency.
UC-09	Register and Log In Customer Accounts	Enables new users to create an account by submitting required information. Also includes the login process.
UC-10	Access Customer Dashboard	Allows customers to view current service status, ongoing promos, announcements, appointment booking and quick access to service descriptions.
UC-11	Book Appointment Slots	Customers select and book available appointment times for drop-offs which are later confirmed by branch staff.

Continuation of **Table 7**

UC-12	Track Service Status	Provides customers with a view of real-time updates on the status of their service requests, from warehouse to branch pick-up.
UC-13	Display estimated service cost and turnaround time for customer reference	Service information module showing standard pricing and expected completion time per service
UC-14	View Transaction History	Allows customers to view all past service transactions including service details, dates, and payment summaries.

Use Case Reports

Table 8

Manage User Authentication

Use case name:	**MANAGE USER AUTHENTICATION**
Scenario:	A user wants to securely access or exit the system by logging in or logging out using their credentials.
Triggering event:	The user opens the web portal
Brief description:	This use case allows customers, staff, and admin users to authenticate themselves in the system. The system validates the provided credentials and either grants access or displays an error. On logout, the user is securely exited from the session. If incorrect credentials are entered, an error message is displayed.
Actors:	Branch Staff, Branch Admin, Super Admin, Warehouse Staff
Include Use Case:	‘Validate Credentials’, ‘Confirm Logout’
Extend Use Case:	None
Stakeholders:	All registered administrators

Continuation of **Table 8**

Preconditions:	The user must have an existing registered account in the system.	
Postconditions:	The user is either successfully logged in with session access based on their role and branch, or logged out and returned to the login/home page.	
Flow of activities:	**Actor**	**System**
	1. Selects “Login” or “Logout” option	1. Shows the login page or confirms the logout action
	2. Enters login credentials	2. Validates credentials
	3. Clicks the login or logout button	3a. If login is successful, redirects to the dashboard 3b. If the login fails, it shows an error 3c. If logged out, ends the session and returns to the login
Exception conditions:	If the system is down, an error message is displayed. The system fails to retrieve user credentials, and an error message is displayed.	

Table 9

Handle Service Requests

Use case name:	**HANDLE SERVICE REQUESTS**
Scenario:	A branch staff member manually receives a service request from a walk-in customer and records it in the system.
Triggering event:	The customer walks into a branch and requests a shoe service. The branch staff initiates the request form.

Continuation of **Table 9**

Brief description:	This use case enables branch staff to handle in-person service requests by recording customer and item details into the system. Staff may choose whether the customer is new or existing. If new, an account is created to avoid duplication. Shoe and service information are entered, along with optional rush processing. The system automatically generates an acknowledgement receipt based on the input. Discounts and promos can be applied, and the estimated turnaround time and cost are calculated.	
Actors:	Branch Staff	
Include Use Case:	Record Customer Details, Record Shoe & Service Info, Generate Acknowledgement Receipt, Calculate Estimated Time & Cost	
Extend Use Case:	Add Another Shoe Service, Apply Discounts or Promos	
Stakeholders:	Branch Staff, Customers	
Preconditions:	The customer must be physically present. Branch staff must be logged in. If the customer is new, their account must be created.	
Flow of activities:	**Actor**	**System**
	1. Opens the service request form	1. Displays fields for input
	2. Selects "Old" or "New" customer	2. Retrieves or creates a customer profile to prevent duplication
	3. Input customer details (name, birthdate, contact, address, etc.)	3. Stores the customer information
	4. Inputs the shoe brand and model, , and selects the requested service(s)	4. Stores the shoe and service information

Continuation of **Table 9**

Flow of activities:	**Actor**	**System**
	5. Applies rush request or discounts/promos if applicable	5. Updates total pricing and shows a preview of the acknowledgement receipt.
	6. Submits the form	6. Calculates estimated cost and turnaround time. Generates a digital acknowledgement receipt with full details.
Exception conditions:	System errors during saving or generating a receipt, or duplicate customer detection.	

Table 10

Coordinate Branch and Warehouse Operations

Use case name:	**COORDINATE BRANCH AND WAREHOUSE OPERATIONS**
Scenario:	Branch staff and warehouse staff collaborate to track item progress and delivery coordination.
Triggering event:	A staff member updates the item status via the admin suite.
Brief description:	This use case allows branch and warehouse staff to manage item movement and status updates. They coordinate delivery logistics, acknowledge item handoffs and update statuses (e.g., "for delivery," "in branch"). This keeps service requests moving and visible to the customer-facing dashboard.
Actors:	Branch Staff, Warehouse Staff
Include Use Case:	Update Item Status, Log Delivery Details, Acknowledge Item Transfers
Extend Use Case:	Send Service Request Update via Email
Stakeholders:	Branch Staff, Warehouse Staff, Customers
Preconditions:	Items must be logged in the system through service request and tagged for processing.

Continuation of **Table 10**

Postconditions:	Items are updated with new statuses and any delivery information is logged.	
Flow of activities:	**Actor**	**System**
	1. Views the service queue	1. Display the service queue panel
	2. Mark as ready for delivery	2a. Update service request on customer app. 2b. Transfer the process to the Ready for Delivery Panel
	3. Views Ready for Delivery Panel	3. Displays Ready for Delivery Panel
	4. Mark as Incoming Branch-Deliveries	4a. Update service request on customer app. 4b. Transfer the process to the Incoming Branch Deliveries Panel
	5. Views Incoming Branch Delivery	5. Displays Incoming Branch-Deliveries Panel
	6a. Mark as In-Process 6b. Upload Before Picture	6a. Opens update module - Update service request on customer app. - Reflect Before Picture in Customer App 6b. Transfer the process to the In-warehouse panel
	7. View In-Warehouse Panel	7. Displays In-Warehouse Panel
	8. Marks as Return-to-Branch 8b. Upload After Picture	8a. Opens update module - Update service request on customer app. - Reflect AfterPicture in Customer App 8b. Transfer the process to Return-to-Branch

Continuation of **Table 10**

	9. Views Return to Branch Panel	9. Displays Return-to-Branch
	10. Mark as Received	10a. Update service request on customer app. 10b. Transfer the process to the Ready for Pickup Panel
	11. Views In Store	11. Displays Ready for Pickup Panel
	12. Mark as Ready for Pickup	12a. Update service request on customer app. 12b. Notify Customer that the shoe is Ready for Pickup
	13. N/A	13.10 days after pickup notice, each day the shoes are not picked up, the system will charge a daily storage fee.
Exception conditions:	Status update fails if the system connection is lost. Upload fails if the image format or size is invalid.	

Table 11

Access and Manage Database Records

Use case name:	**ACCESS AND MANAGE DATABASE RECORDS**
Scenario:	An admin wants to view, update, or delete records related to customers, branches, or services.
Triggering event:	The admin navigates to the database section in the Shoelotskey Web Admin Suite.
Brief description:	This use case allows admins to manage the underlying records of the system. They can view customer profiles, branch data, service lists, and historical transactions. Admins can also update or remove outdated information. This ensures accurate, secure, and clean records across the system.
Actors:	Admin

Continuation of **Table 11**

Include Use Case:	View Records, Edit Records, Delete Records	
Extend Use Case:	Export Records, Clear Records	
Stakeholders:	Admin	
Preconditions:	Admin must be logged in and have access rights.	
Postconditions:	Records are updated, deleted, or viewed. All changes are logged.	
Flow of activities:	**Actor**	**System**
	1. Clicks on a data module	1. Shows a table of entries
	2. Views or selects a record	2. Opens details
	3. Edits or deletes entry	3. Saves or removes data
	4. (If exporting)	4. Backup records
Exception conditions:	Unauthorized access or failed edits trigger error prompts. Deletion errors prompt retry or cancel.	

Table 12

Generate and View Business Analytics

Use case name:	**GENERATE AND VIEW BUSINESS ANALYTICS**
Scenario:	The admin wants to see metrics like income, service trends, and performance charts.
Triggering event:	The admin clicks on the analytics module in the Web Admin Suite.
Brief description:	This use case lets admins visualize data trends through charts and summaries. It includes insights like forecasting and AI analytics(forecasting daily revenue trend(with narrative), monthly income, most-requested services, peak service periods, and branch performance). This helps in evaluating business performance and making informed decisions.
Actors:	Admin
Include Use Case:	View Reports, Filter by Date or Branch

Continuation of **Table 12**

Extend Use Case:	None	
Stakeholders:	Admin	
Preconditions:	Valid admin session; data must be available.	
Postconditions:	Admin can view reports.	
Flow of activities:	**Actor**	**System**
	1. Opens analytics section	1. Loads dashboards
	2. Views chart breakdown	2. Displays visual reports
	3.Clicks Generate Report/ Export Excel	3. Generates PDF report/ Exports Excel
Exception conditions:	Data load errors prompt retry.	

Table 13

Manage Users and Schedules

Use case name:	**MANAGE USERS AND SCHEDULES**
Scenario:	A staff member needs to post announcements, manage user access, or update available appointment slots.
Triggering event:	Staff logs into the admin suite and navigates to the users and schedule module.
Brief description:	This use case allows staff to manage appointment availability and public-facing announcements. It includes setting available drop-off times and ensuring schedule conflicts are avoided. Announcements posted by staff are automatically synced to the customer app.
Actors:	Branch Staff
Include Use Case:	Set Appointment Availability, Post Announcement, Edit Schedule
Extend Use Case:	None
Stakeholders:	Branch Staff, Customer

Continuation of **Table 13**

Preconditions:	Staff must be authenticated with scheduling privileges.	
Postconditions:	Updated schedules and announcements are saved and reflected to customers.	
Flow of activities:	**Actor**	**System**
	1. Navigates to the scheduling section	1. Displays current availability
	2. Edits schedule or posts announcements	2. Updates the database
	3. Saves changes	3. Syncs announcement to app
Exception conditions:	Invalid inputs or schedule overlap trigger an error.	

Table 14

Admin Side Notifications

Use case name:	**ADMIN SIDE NOTIFICATIONS**
Scenario:	The system needs to inform admins whenever a customer books a scheduled appointment.
Triggering event:	A customer booked an appointment schedule.
Brief description:	This use case handles the automated sending of notifications to admins whenever a customer booked an appointment.
Actors:	System
Include Use Case:	Send Appointment Notification
Extend Use Case:	None
Stakeholders:	Admins and Customer
Preconditions:	Customer has inputted a valid date of appointment
Postconditions:	Notifications are sent successfully or failed attempts are logged.

Continuation of **Table 14**

Flow of activities:	**Actor**	**System**
	1. Opens Notification sheet	1. Display alerts for inactive items and upcoming storage fees, and pending appointments
Exception conditions:	If notification fails to send, log the failure and try again once.	

Table 15

Automate System Processes

Use case name:	**AUTOMATE SYSTEM PROCESSES**
Scenario:	Background system processes automatically execute analytics updates and data synchronization tasks without manual input.
Triggering event:	A scheduled cron job runs or a system event (e.g., service status change) triggers an analytics update.
Brief description:	This use case covers all automated background processes performed by the analytics system. These include periodic data aggregation, report generation, and dashboard synchronization in response to service request or status updates.
Actors:	System
Include Use Case:	Update Analytics Dashboard
Extend Use Case:	None
Stakeholders:	Admin, Branch Staff, Customers
Preconditions:	A scheduled cron time, Data sources are available and accessible.
Postconditions:	Analytics dashboards and reports are updated successfully.

Continuation of **Table 15**

Flow of activities:	**Actor**	**System**
		1. Detects cron schedule
		2. Executes data extraction from an operational database.
		3. Processes and aggregates new data for reporting.
		4. Updates analytics dashboard with refreshed data.
Exception conditions:	Cron job fails to execute due to server downtime or scheduling error. Dashboard update fails due to analytics service timeout.	

Table 16

Register and Log In Customer Accounts

Use case name:	**REGISTER AND LOG IN CUSTOMER ACCOUNTS**
Scenario:	A new customer wants to create an account and log in to access the features of the customer app.
Triggering event:	The customer either selects the “Register” or “Login” button on the mobile app and automatically logs in using the newly created credentials.
Brief description:	A customer either logs in or registers by filling in personal details and agreeing to the terms of service. After submitting the form, their account is created. Once registered, the customer logs in using the credentials they created. This provides them access to various services like booking appointments, tracking items, and viewing announcements and promotions.
Actors:	Customer
Include Use Case:	‘Validate Customer Credentials’, ‘Confirm Logout’, ‘Create a database entry for an account and customer details’

Continuation of **Table 16**

Extend Use Case:	None	
Stakeholders:	New and returning customers	
Preconditions:	The customer must not already have an existing account (for registration), and must have valid credentials (for login).	
Postconditions:	A new customer account is created upon successful registration. The customer gains access to their dashboard after login.	
Flow of activities:	**Actor**	**System**
	1. Selects the “Register” button	1. Displays registration form
	2. Inputs name, email, password, and other required info	2. Validates and submits registration data
	3. Clicks “Submit”	3. Creates account and confirms registration
	4. Returns to login page	4. Prompts for login
	5. Enters credentials and logs in	5a. If credentials are valid, redirects to dashboard 5b. If invalid, displays error message
Exception conditions:	Registration fails due to missing or invalid fields; system prompts for correction. Login fails due to incorrect credentials; system prompts retry.	

Table 17

Access Customer Dashboard

Use case name:	**ACCESS CUSTOMER DASHBOARD**
Scenario:	A customer opens the app to view their service and profile information.
Triggering event:	Customer logs into their account.

Continuation of **Table 17**

Brief description:	The dashboard gives customers a summarized view of ongoing services, status updates, current promos, notifications, and quick links to service tools like booking or tracking.	
Actors:	Customer	
Include Use Case:	View Announcements, View Active Service Requests, View Promos	
Extend Use Case:	None	
Stakeholders:	Customers	
Preconditions:	Customer must be logged in to the mobile application.	
Postconditions:	Customer can interact with the dashboard elements.	
Flow of activities:	**Actor**	**System**
	1. Opens the customer app	1. Verifies session and authentication
	2. Accesses dashboard	2. Displays service status, current promos, announcements
	3. Taps dashboard features	3. Redirects to corresponding tools or detail views
Exception conditions:	If data fails to load, an error message is shown with the retry option. If session expired, user is redirected to login.	

Table 18

Book Appointment Slots

Use Case Name	**BOOK APPOINTMENT SLOTS**
Scenario	A customer wants to reserve a drop-off slot at a branch.
Triggering Event	The customer selects “Book Appointment” from the dashboard or service tools.

Continuation of **Table 18**

Brief Description	Customers can view available time slots posted by staff. The appointment is saved and acknowledged by the staff later.	
Actors	Customer	
Include Use Case	View Available Slots, Submit Appointment Request	
Extend Use Case	Send Appointment Acknowledgement via Email	
Stakeholders	Customers, Branch Staff	
Preconditions	The customer must be logged in. Appointment slots must have been published by the staff.	
Postconditions	A pending appointment request is submitted and queued for confirmation by branch staff.	
Flow of activities:	**Actor**	**System**
	1. Selects “Book Appointment”	1. Displays list of available slots based on branch and day
	2. Chooses a slot	2. Verifies availability
	3. Confirms appointment	3. Saves appointment with status “Pending”
		4. Notifies staff for review
Exception Conditions	If a slot becomes unavailable before confirmation, the system notifies the user and prompts to pick another.	

Table 19

Track Service Status

Use Case Name	**TRACK SERVICE STATUS**
Scenario	A customer wants to know the current status of their service request.
Triggering Event	The customer opens the dashboard or selects a specific shoe currently booked.

Continuation of **Table 19**

Brief Description	The system displays real-time updates of service progress, such as “In Progress,” “Ready for Pickup,” or “In Transit.”	
Actors	Customer	
Include Use Case	View Dashboard, Fetch Service Progress	
Extend Use Case	None	
Stakeholders	Customers, Warehouse Staff, Branch Staff	
Preconditions	Customer must be logged in and have active service requests in the system.	
Postconditions	The customer sees the most recent status of their items.	
Flow of activities:	**Actor**	**System**
	1. Navigates to “Track Service”	1. Loads current active service request(s)
	2. Selects a request	2. Displays status timeline, key timestamps, and before and after photo of shoes
Exception Conditions	If there is no active service, the system shows a message: “No active requests found.” If the system fails to fetch status, it prompts the customer to retry or refresh.	

Table 20

Estimate Cost and Turnaround Time

Use Case Name	**ESTIMATE SERVICE COST AND TURNAROUND TIME**
Scenario	A customer wants to check the estimated cost and number of days for a service before booking.
Triggering Event	The customer selects the estimator tool.

Continuation of **Table 20**

Brief Description	The system provides a tool for customers to estimate the total cost and service duration based on selected services. The result is a projected estimate.	
Actors	Customer	
Include Use Case	None	
Extend Use Case	None	
Stakeholders	Customers	
Preconditions	The customer must be logged into their account.	
Postconditions	An estimate is shown with calculated service cost and approximate turnaround time.	
Flow of activities:	**Actor**	**System**
	1. Selects a service	1. Shows service cost and estimated completion time
Exception Conditions	If an error occurs, an error message is displayed suggesting a retry.	

Table 21

View Transaction History

Use Case Name	**VIEW TRANSACTION HISTORY**
Scenario	A customer wants to review all previous service transactions.
Triggering Event	The customer selects “Transaction History” from their dashboard.
Brief Description	The system displays a chronological list of the customer’s past service records, including service details, total payments, and service dates.
Actors	Customer
Include Use Case	View Acknowledgement Receipt
Extend Use Case	None

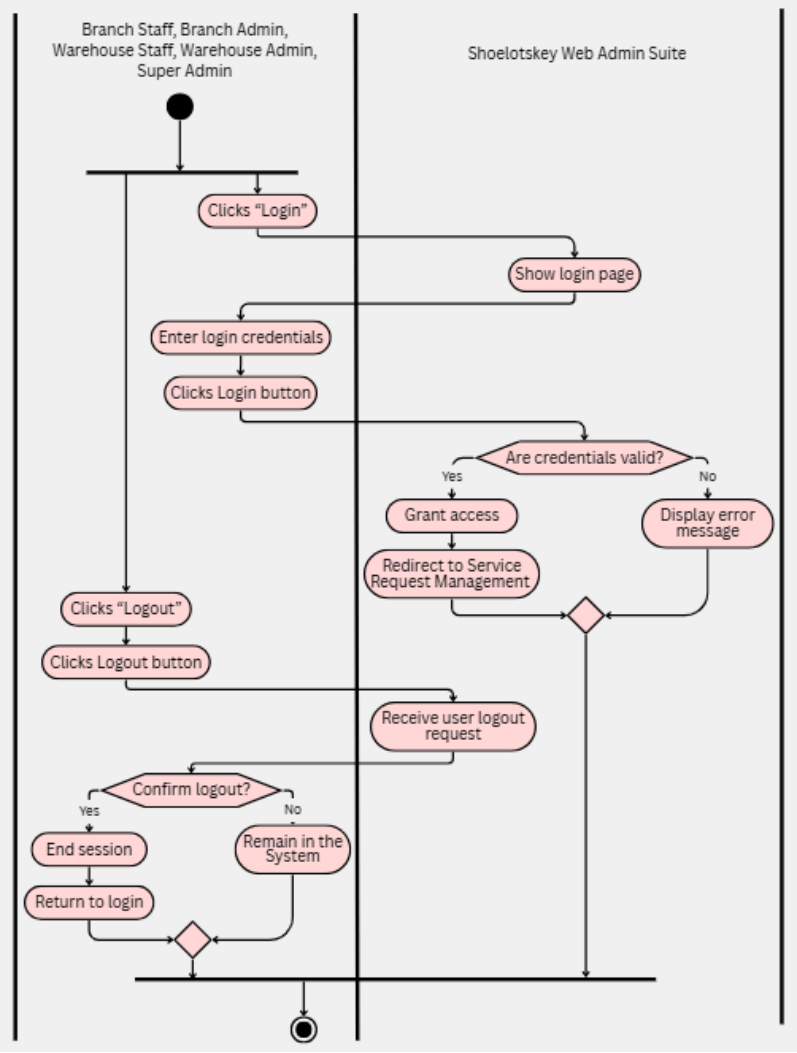
Continuation of **Table 21**

Stakeholders	Customers	
Preconditions	The customer must be logged into their account.	
Postconditions	The transaction history is displayed in detail, and the customer can review individual receipts.	
Flow of activities:	**Actor**	**System**
	1. Selects “Transaction History”	1. Fetches and displays the list of past transactions
	2. Selects a specific record	2. Shows full details including shoe serviced, services rendered, amount paid, receipt ID, and service date
Exception Conditions	If no transactions exist, the system shows a message: “No transactions found.” If loading fails, an error message is displayed suggesting a retry.	

Design Specifications

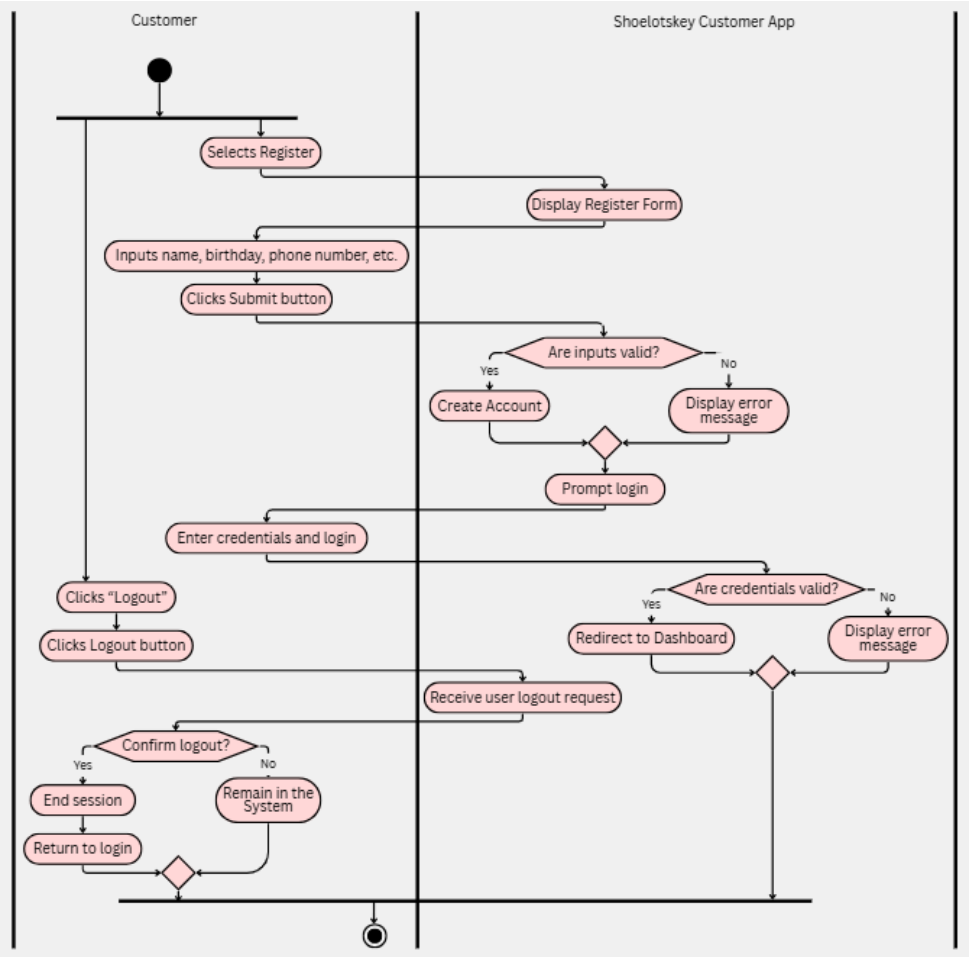
Activity Diagram

Figure 17. Activity Diagram for Manage User Authentication



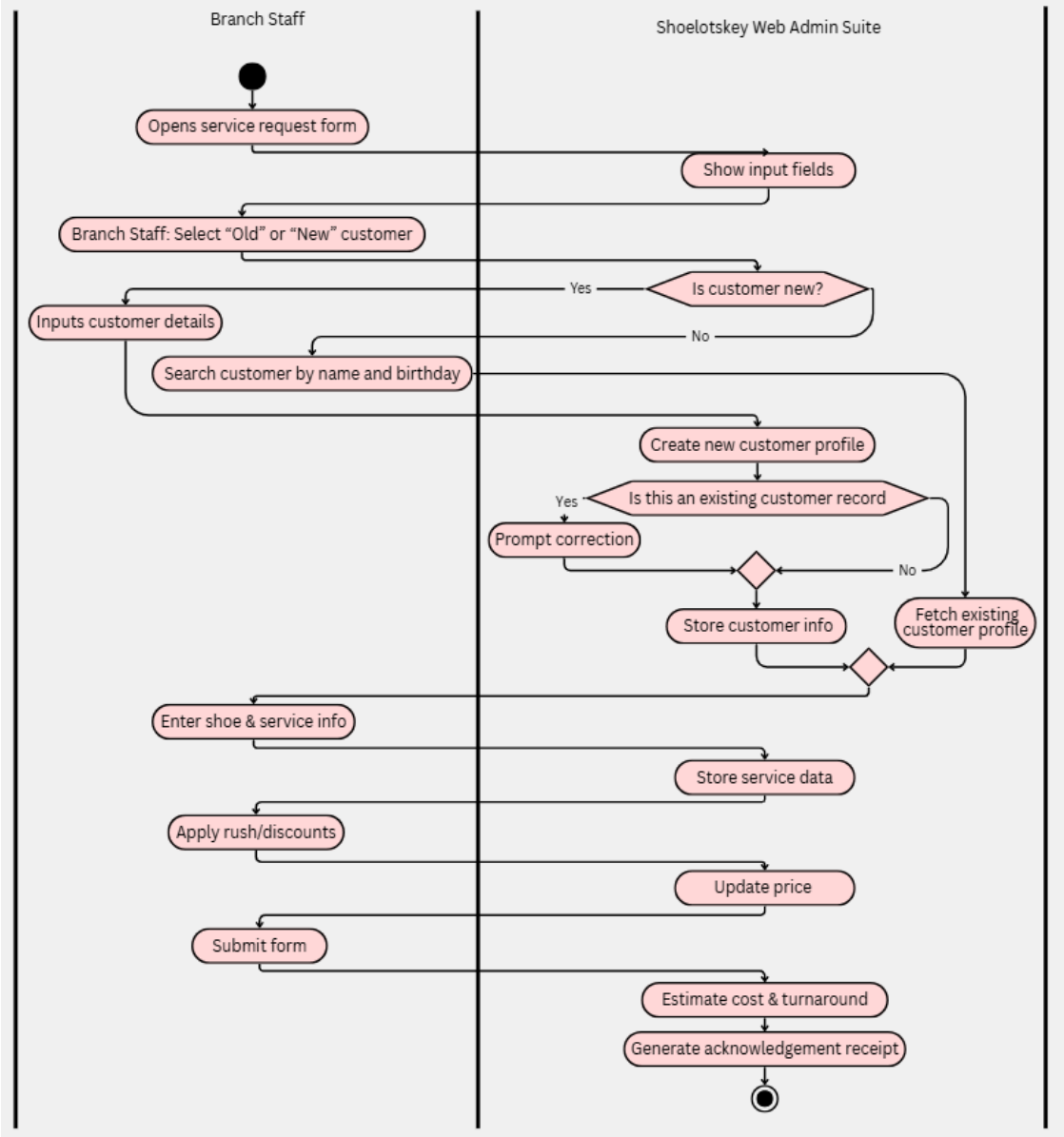
This activity diagram illustrates the process for system login and logout by internal users (staff/admin). The user begins by choosing a login or logout action. Based on the input, the system either validates credentials to grant access or terminates the session securely. Invalid credentials prompt an error, maintaining security.

Figure 18. Activity Diagram for Register and Log In Customer Accounts



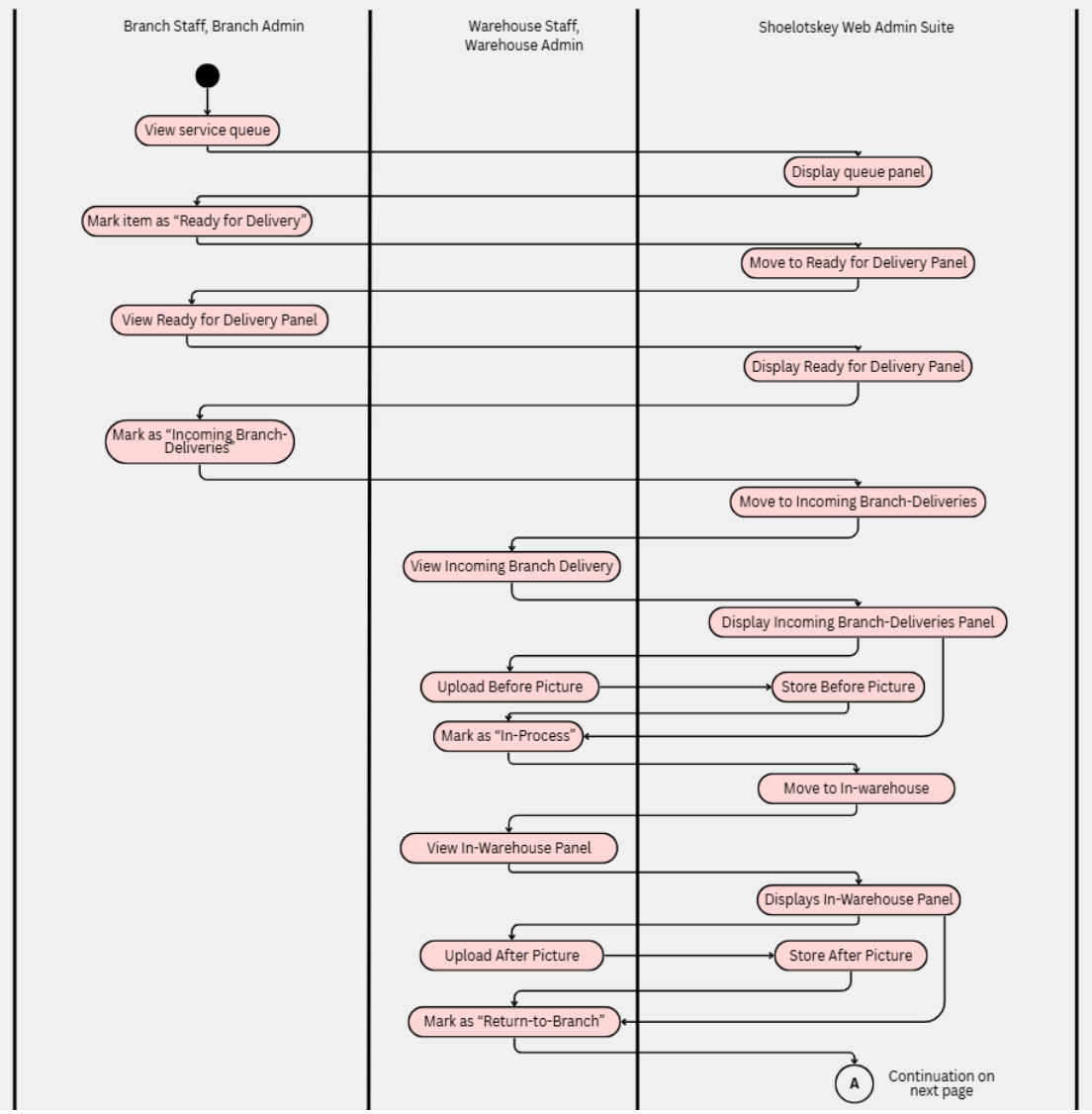
This diagram outlines the customer account registration and login workflow. A customer can either register a new account or log in with existing credentials. The system verifies input, creates the account if valid, and allows access to services. Error handling ensures user feedback for invalid inputs.

Figure 19. Activity Diagram for Handle Service Request

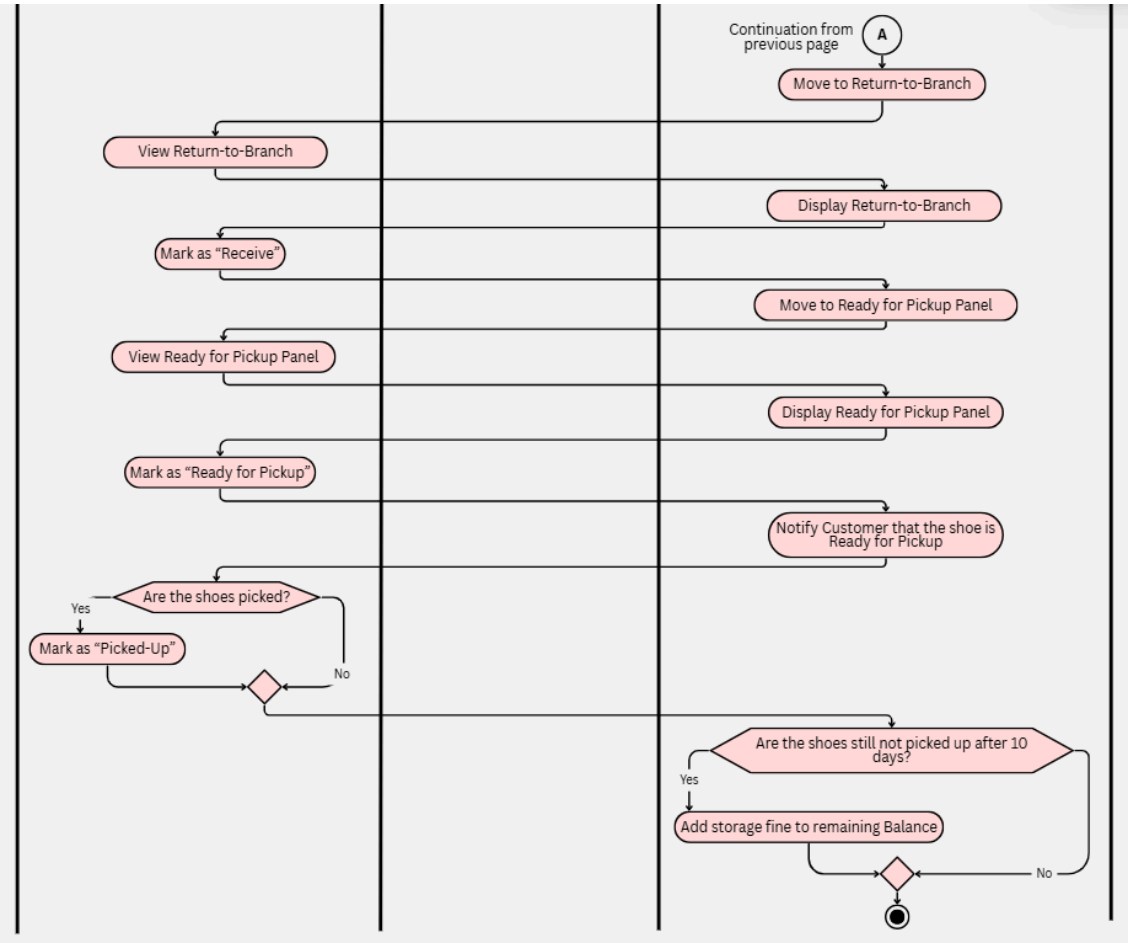


Here, the process for in-branch service intake is shown. Branch staff either create or retrieve a customer profile, input service details, apply optional settings (like rush or discounts), and submit the request. The system then generates an acknowledgment receipt. This ensures structured service recording.

Figure 20. Activity Diagram for Coordinate Branch and Warehouse Operations

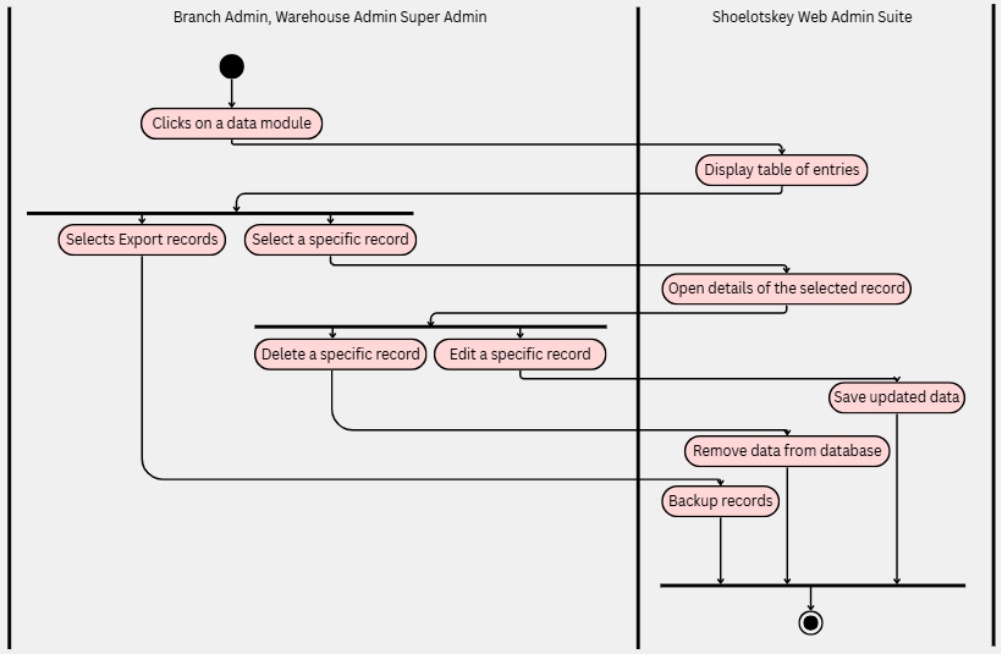


Continuation of **Figure 20**



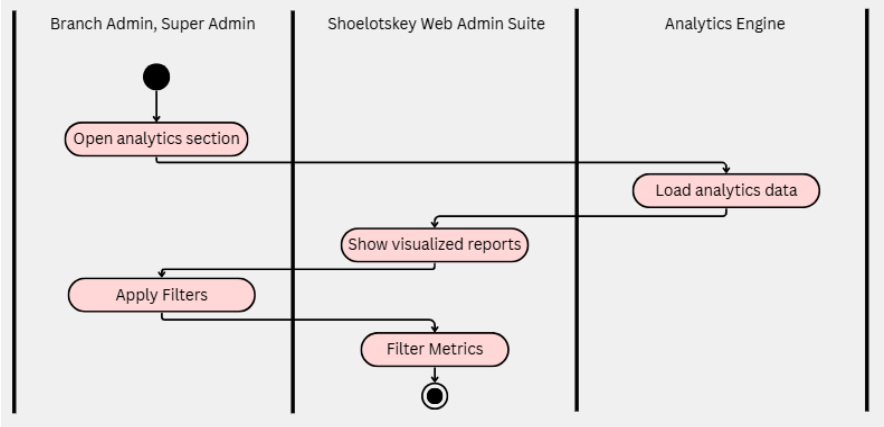
This diagram shows the coordination between branch and warehouse staff during item processing. Status updates and photo uploads guide the shoe's journey from intake to pickup. Updates are reflected in the customer app. Automated reminders are issued if pickups are delayed.

Figure 21. Activity Diagram for Access and Manage Database Record



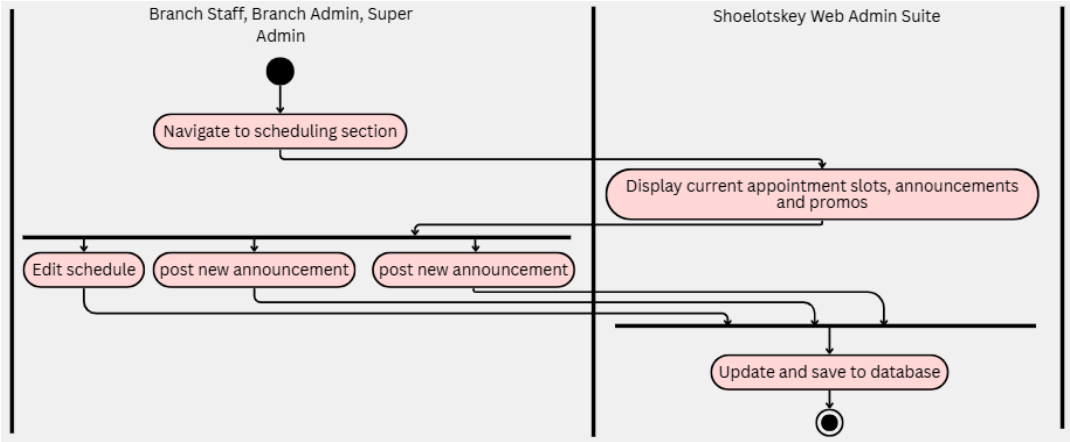
This process allows admin users to manage customer, service, or branch records. Admins can view, edit, delete, or export data modules. The system ensures these actions are secure and logged. Error handling is included for unauthorized access or deletion failures.

Figure 22. Activity Diagram for Generating and Viewing Business Analytics



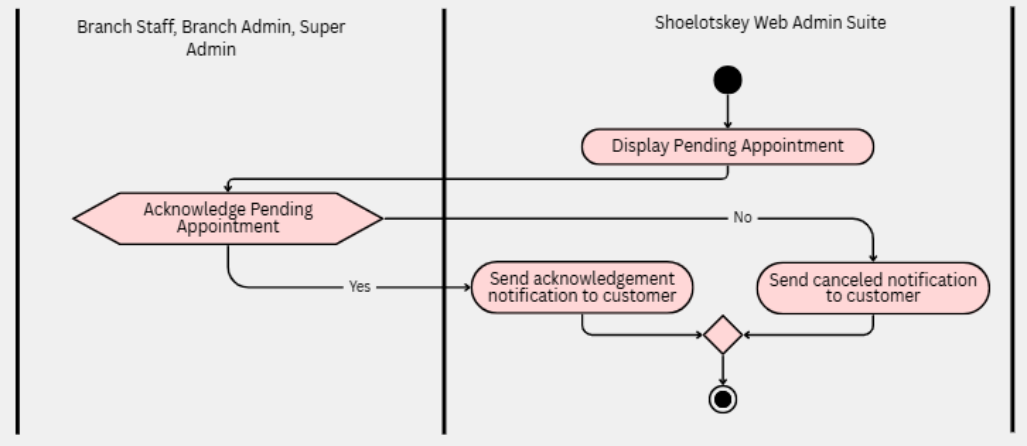
Admins can visualize performance metrics using this flow. Upon accessing the analytics module, the system loads and filters data based on selected parameters. It displays charts for trends like income and service types. Errors from empty filters or data issues are handled smoothly.

Figure 23. Activity Diagram for Manage Users and Schedules



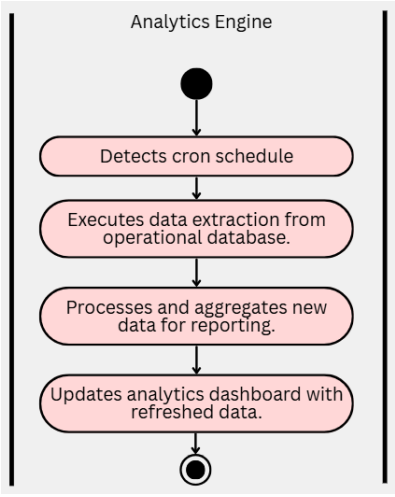
This activity diagram shows how branch staff manage schedules and announcements. Staff can edit appointment availability and publish announcements which sync to the customer app. The flow ensures changes are recorded and shared publicly. Error prompts occur for schedule conflicts.

Figure 24. Activity Diagram for Send Customer Notifications (Admin Side Notifications)



This process manages the acknowledgment of customer appointments within the Shoelotskey Web Admin Suite. When a customer books an appointment, the system displays it under “Pending Appointments” for branch or admin users to review. Staff can then acknowledge or cancel the booking.

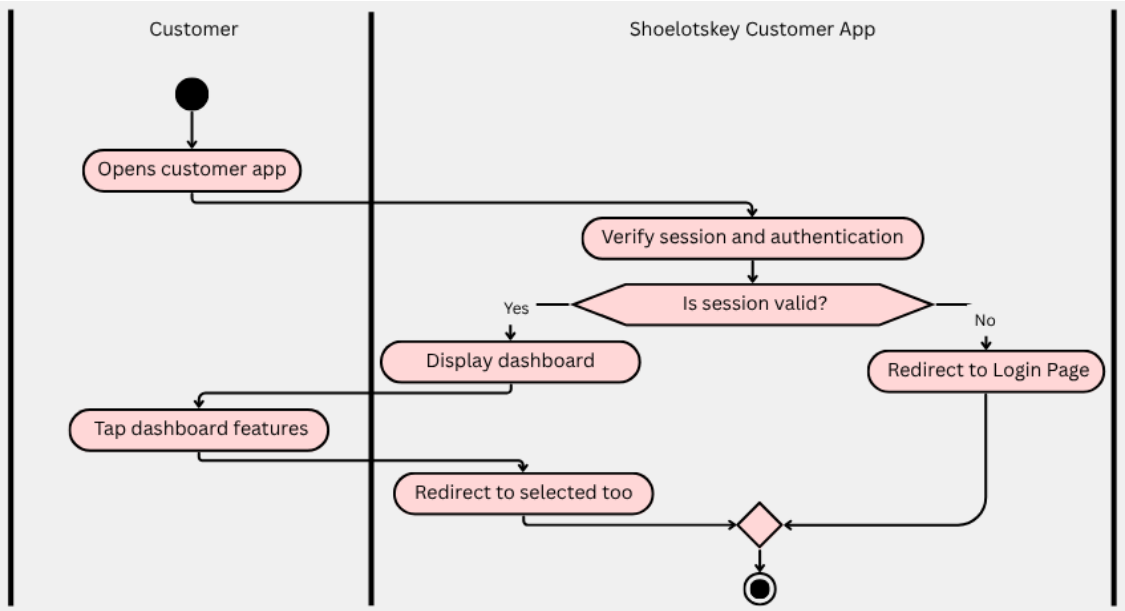
Figure 25. Activity Diagram for Automated System Processes



This diagram illustrates the automated analytics workflow handled by the system’s cron job. On a scheduled interval, the analytics engine detects the cron trigger, extracts updated data from the operational database, processes and

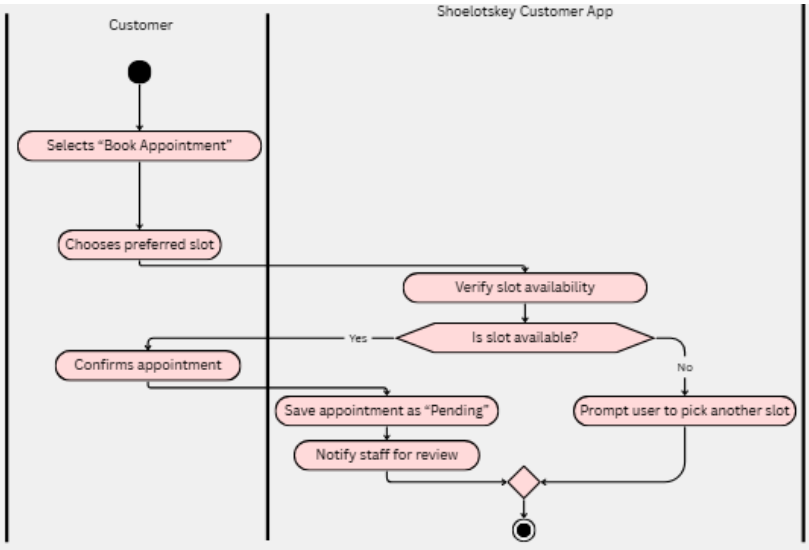
aggregates it for reporting, and refreshes the analytics dashboard. These automated updates ensure that performance metrics and operational insights remain current without requiring manual intervention. Any failures in the process are logged and queued for retry to maintain data consistency while ensuring transparency.

Figure 26. Activity Diagram for Access Customer Dashboard



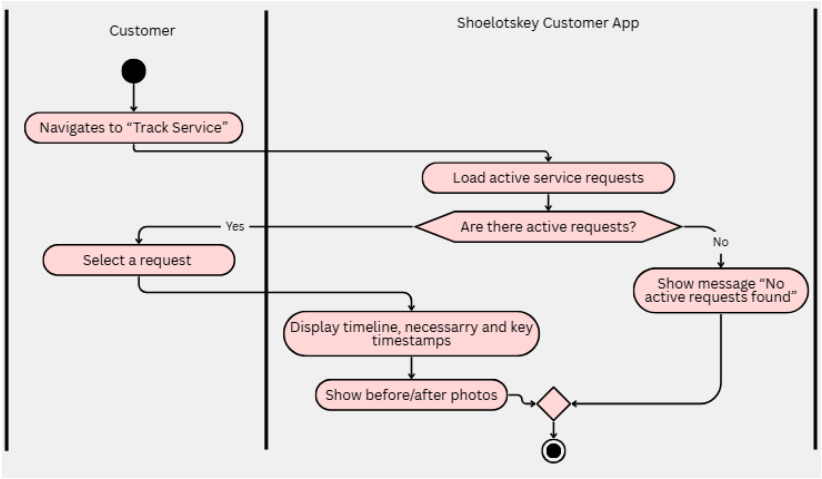
Customers gain access to personalized information after logging in. The dashboard displays ongoing services, promos, and notifications. Tapping features redirects to relevant service tools. If sessions expire or data fails to load, fallback actions are triggered.

Figure 27. Activity Diagram for Book Appointment Slots



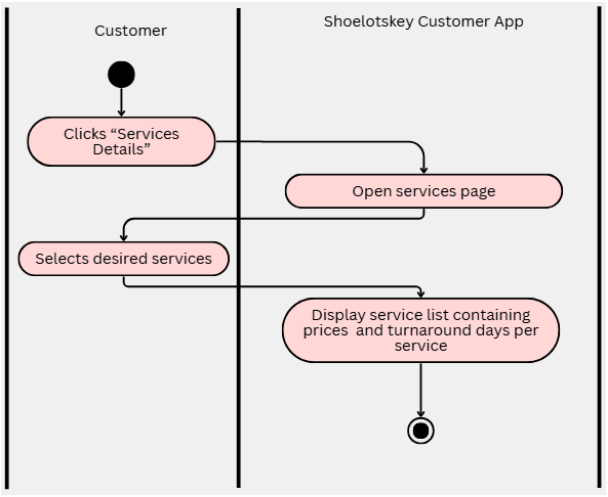
This diagram shows the customer's booking workflow. After selecting a slot, the system checks availability and confirms the request. If slots are unavailable, the customer is prompted to choose again.

Figure 28. Activity Diagram for Track Service Status



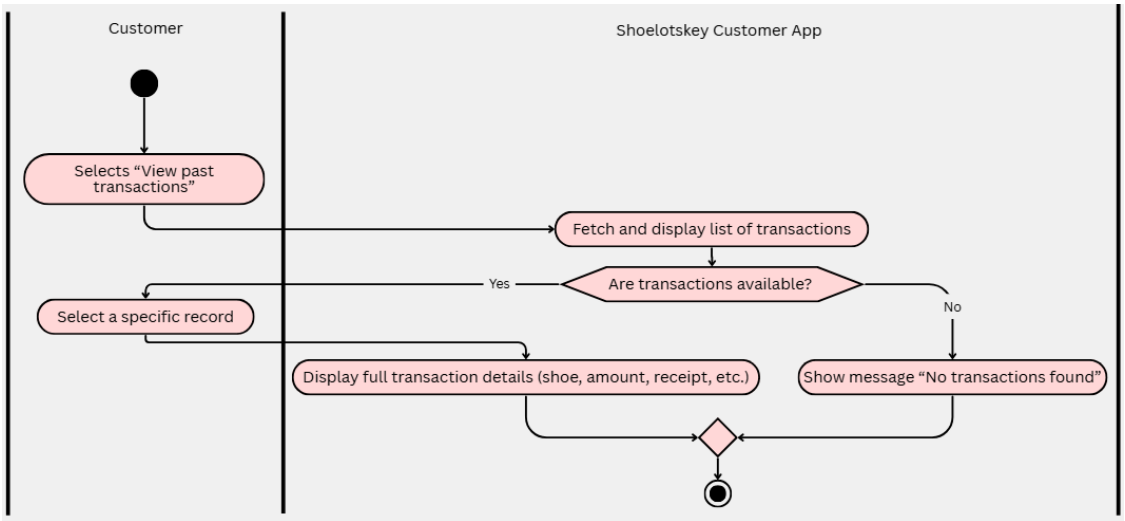
Customers can monitor their items in real-time through this flow. After selecting a request, they are shown a timeline with key updates and photos. If no active services exist, a message is shown. System errors offer retry options to maintain usability.

Figure 29. Activity Diagram for Estimating Service Cost and Turnaround Time



This tool allows customers to see service costs before booking. The customer selects desired services and views costs and possible days it takes per service.

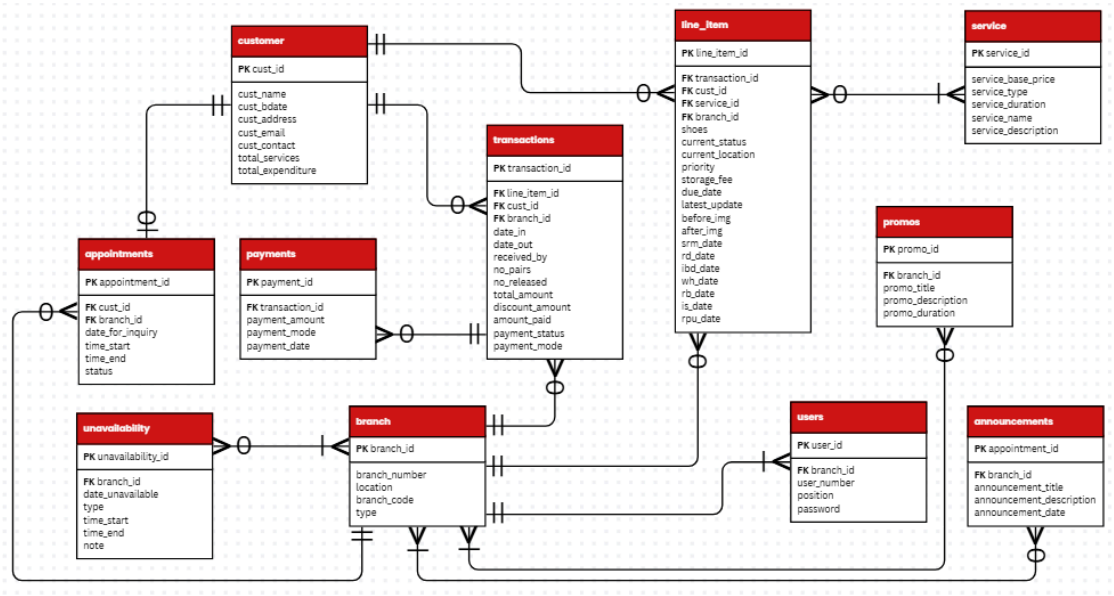
Figure 30. Activity Diagram for View Transaction History



Customers can review past services via this flow. After choosing "Transaction History," they can view detailed records of each service, including receipts and dates. If no history exists, the system displays a message. Failures in loading also show prompts for retry.

Database Schema

Figure 31. Entity Relationship Diagram



Data Dictionary

Table 22

Line Item Data Dictionary

Key	Data Name	Data Type	Constraints	Description
PK	line_item_id	VARCHAR(20)	PRIMARY KEY, NOT NULL	Entry Header - Receipt ID Entry Detail - Identifies each item/order under that receipt. Third Section - Branch it came from Sample: 00001-001-BRANCH 00001-002-BRANCH 00002-001-BRANCH

Continuation of **Table 22**

Key	Data Name	Data Type	Constraints	Description
FK	transaction_id	VARCHAR(20)	FOREIGN KEY REFERENCES transactions(transaction_id), NOT NULL	YYYY - 4-digit year MM - 2-digit month 00001 - increments each receipt. Resets every new month/beginning of month BRANCH - branch code
FK	cust_id	VARCHAR(15)	FOREIGN KEY REFERENCES customers(cust_id), NOT NULL	Customer ID. Format: CUST-01-00001 CUST - just to indicate this is a customer 01 - account creation identifier(base on branch ID, 00 if mobile) 00001 - auto increment
FK	service_id	VARCHAR(10)	NOT NULL	Service or additional service. Format: SRV-01 or ADD-01 SRV / ADD - service or additional 01 - service or additional number Note: Can have multiple service_id
FK	branch_id	VARCHAR(15)	NOT NULL	Branch ID. Format: BRNCH-T-REG BRNCH - Branch code T - type(H/B), hub or branch REG - Region Sample: VAL-B-NCR, SMV-B-NCR
	priority	ENUM('Rush','Normal')	NOT NULL, DEFAULT 'Normal'	Indicates if this item is rush or normal

Continuation for **Table 22**

Key	Data Name	Data Type	Constraints	Description
	storage_fee	DECIMAL(10,2)	DEFAULT 0.00	Fee per day from ready-to-pickup date 100 peso per day from when updated to ready to pickup
	shoes	VARCHAR(50)	NOT NULL	Shoe Model Name
	current_location	ENUM('Hub','Branch')	NOT NULL	Current location where the shoes are. Based on status.
	current_status	VARCHAR(30)	NOT NULL	Service status
	due_date	DATE	NULL	Estimated completion date of service
	latest_update	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Date of last service status update. will base on dates table
	before_img	VARCHAR(255)	NULL	Cloudinary link for before image
	after_img	VARCHAR(255)	NULL	Cloudinary link for after image
	before_img	VARCHAR / BLOB	NULLABLE	Image of shoes before service
	after_img	VARCHAR / BLOB	NULLABLE	Image of shoes after service
	srm_date	DATETIME	NULL	Service Request Queued Date: the moment a transaction starts in Service Request Management and appears in the Ops Service Queue
	rd_date	DATETIME	NULL	Ready for Delivery Date: when an item is moved from Service Queue to Ready for Delivery

Continuation for **Table 22**

Key	Data Name	Data Type	Constraints	Description
	ibd_date	DATETIME	NULL	Incoming Branch Delivery Date: when an item is moved from Ready for Delivery to Incoming Branch Delivery
	wh_date	DATETIME	NULL	Warehouse Date: when an item is moved from Incoming Branch Delivery to Warehouse In Processing
	rb_date	DATETIME	NULL	Return to Branch Date: when an item is moved from Warehouse to Return to Branch
	is_date	DATETIME	NULL	In Store Date: when an item is moved from Return to Branch to In Store
	rpu_date	DATETIME	NULL	Ready for Pickup Date: when an item is moved from In Store to Ready for Pickup

Table 23

Transaction Data Dictionary

Key	Data Name	Data Type	Constraints	Description
PK	transaction_id	VARCHAR(20)	PRIMARY KEY, NOT NULL	YYYY - 4-digit year MM - 2-digit month 00001 - increments each receipt. Resets every new month/beginning of the month BRANCH - branch code Sample: 2025-09-00001-SMVAL 2025-09-00002-SMVAL 2025-09-00001-VALEN

Continuation for **Table 23**

Key	Data Name	Data Type	Constraints	Description
FK	line_item_id	VARCHAR(20)	NOT NULL, FOREIGN KEY → line_items(line_item_id)	Entry Header - Receipt ID Entry Detail - Identifies each item/order under that receipt. Third Section - Branch it came from Sample: 00001-001-SMVAL 00001-002-SMVAL 00002-001-VALEN
FK	branch_id	VARCHAR(30)	Foreign Key, Not Null	Branch ID. Format: BRNCH-T-REG BRNCH - Branch code T - type(H/B), hub or branch REG - Region Sample: VAL-B-NCR, SMV-B-NCR, SMGRA-B-NCR
	date_in	DATETIME	NOT NULL	Date the transaction/item was received
	received_by	VARCHAR(50)	NOT NULL	Staff who received the transaction/item
	date_out	DATETIME	NULL	Date the transaction/item was completed/released

Continuation for **Table 23**

Key	Data Name	Data Type	Constraints	Description
FK	cust_id	VARCHAR(15)	Foreign Key, Not Null	Customer ID. Format: CUST-01-00001 CUST - just to indicate this is a customer 01 - account creation identifier(base on branch ID, 00 if mobile) (00 - Mobile 01 - Admin 02 - SM Valenzuela 03 - Valenzuela 04 - SM Grand) 00001 - auto increment
	no_pairs	INT	DEFAULT 0	Number of pairs in the transaction
	no_released	INT	DEFAULT 0	Number of pairs released to customer
	total_amount	DECIMAL(10,2)	DEFAULT 0.00	Total amount for the transaction
	discount_amount	DECIMAL(10,2)	DEFAULT 0.00	Discount amount for the transaction
	amount_paid	DECIMAL(10,2)	DEFAULT 0.00	Amount paid by customer
	payment_status	ENUM('NP','PAID','PARTIAL')	NOT NULL, DEFAULT 'NP'	Payment status of the transaction
	payment_mode	ENUM('Cash','Card','GCash','Other')	NOT NULL	Mode of payment used

Table 24

Customer Data Dictionary

Key	Data Name	Data Type	Constraints	Description
PK	cust_id	VARCHAR(15)	PRIMARY KEY, NOT NULL	Unique Customer ID, e.g., CUST-01-00001
	cust_name	VARCHAR(100)	NOT NULL	Full name of the customer
	cust_bdate	DATE	NULL	Customer's birth date
	cust_address	VARCHAR(255)	NULL	Customer's address
	cust_email	VARCHAR(100)	UNIQUE, NULL	Customer's email
	cust_contact	VARCHAR(20)	NULL	Customer's contact number
	total_services	INT	DEFAULT 0	Total number of services availed
	total_expenditure	DECIMAL(10,2)	DEFAULT 0.00	Total money spent by the customer

Table 25

Service Data Dictionary

Key	Data Name	Data Type	Constraints	Description
PK	service_id	VARCHAR(10)	PRIMARY KEY, NOT NULL	Unique Service ID, e.g., SRV-01 or ADD-01
	service_base_price	DECIMAL(10,2)	NOT NULL	Base price of the service
	service_type	ENUM('Service',' Additional')	NOT NULL	Type of service: main service or additional
	service_duration	INT	NOT NULL	Duration of service in days
	service_id	VARCHAR(10)	PRIMARY KEY, NOT NULL	Unique Service ID, e.g., SRV-01 or ADD-01
	service_base_price	DECIMAL(10,2)	NOT NULL	Base price of the service

Table 26

Branch Data Dictionary

Key	Data Name	Data Type	Constraints	Description
PK	branch_id	VARCHAR(30)	Branch ID. Format: BRNCH-T-REG BRNCH - Branch code T - type(H/B), hub or branch REG - Region Sample: VAL-B-NCR, SMV-B-NCR	Primary Key, Unique, Not Null
	branch_number	INT	Auto-increment numeric ID starting at 1 (e.g., 1, 2, 3 …).	Auto Increment, Unique, Not Null
	location	VARCHAR(100)	Full location name of the branch (e.g., Valenzuela, NCR).	Not Null
	branch_code	VARCHAR(20)	Short branch code for easy reference (e.g., VAL, SMV).	Unique, Not Null
	type	VARCHAR(1)	Type of branch: H = Hub, B = Branch.	Not Null

Table 27

User Data Dictionary

Key	Data Name	Data Type	Constraints	Description
PK	user_id	VARCHAR(30)	Unique ID for each user (e.g., NCR-VAL-B). Format: REG-<BRANCHCODE> -<TYPE>.	Primary Key, Unique, Not Null
	user_number	INT	Auto-increment numeric ID starting from 1.	Auto Increment, Unique, Not Null

Continuation of **Table 27**

Key	Data Name	Data Type	Constraints	Description
	branch_id	VARCHAR(30)	The branch where the user is assigned. References branch.branch_id.	Foreign Key, Not Null
	position	VARCHAR(50)	User's role or position (e.g., Admin, Staff, Manager).	Not Null

Table 28

Promos Data Dictionary

Key	Data Name	Data Type	Constraints	Description
PK	promo_id	VARCHAR(20)	Unique ID for each promo (e.g., PROMO-001). Prefix PROMO- + zero-padded auto-increment.	Primary Key, Unique, Not Null
FK	branch_id	VARCHAR(30)	The branch where the user is assigned. References branch.branch_id.	Foreign Key, Not Null
	promo_title	VARCHAR(100)	Title or headline of the promo.	Not Null
	promo_description	TEXT	Detailed description of the promo.	Nullable
	promo_duration	VARCHAR(255)	Duration of the promo written as text (e.g., Jan 1-3 2025, Jan 6-10 2025).	Not Null

Table 29

Appointment Data Dictionary

Key	Data Name	Data Type	Constraints	Description
PK	appointment_id	VARCHAR(20)	Unique ID for each pending appointment (e.g., PEND-001). Prefix PEND- + zero-padded auto-increment.	Primary Key, Unique, Not Null

Continuation of **Table 29**

Key	Data Name	Data Type	Constraints	Description
FK	cust_id	VARCHAR(20)	References the customer who made the inquiry.	Foreign Key → customers(customer_id), Not Null
FK	branch_id	VARCHAR(30)	The branch where the user is assigned. References branch.branch_id.	Foreign Key, Not Null
	date_for_inquiry	DATE	Date the customer is requesting for an appointment.	Not Null
	time_start	TIME	Specific start time for appointment	Not Null
	time_end	TIME	Specific end time for appointment	Not Null
	status	VARCHAR(20)	Status of the pending appointment (e.g., Pending, Canceled, Approved).	Default = Pending, Not Null

Table 30

Unavailability Data Dictionary

Key	Data Name	Data Type	Constraints	Description
PK	unavailability_id	VARCHAR(20)	Unique ID for each unavailability record (e.g., UNAV-001). Prefix UNAV- + zero-padded auto-increment.	Primary Key, Unique, Not Null
FK	branch_id	VARCHAR(30)	The branch where the user is assigned. References branch.branch_id.	Foreign Key, Not Null
	date_unavailable	DATE	Date when the user/staff is unavailable.	Not Null
	type	VARCHAR(20)	Type of unavailability: Full Day or Partial Day.	Not Null
	time_start	TIME	Specific start time for partial day unavailability	Nullable
	time_end	TIME	Specific end time for partial day unavailability .	Nullable
	note	VARCHAR(255)	Title or short note describing the reason (e.g., “Meeting”).	Nullable

Table 31

Announcement Data Dictionary

Key	Data Name	Data Type	Constraints	Description
PK	announcement_id	ANN-001	Unique ID for each announcement. Prefix ANN- + zero-padded auto-increment.	Primary Key, Unique, Not Null
FK	branch_id	VARCHAR(30)	The branch where the user is assigned. References branch.branch_id.	Foreign Key, Not Null
	announcement_title	Text (e.g., VARCHAR 100)	Title or headline of the announcement.	Not Null
	announcement_description	Text (e.g., TEXT)	Full details/content of the announcement.	Not Null
	announcement_date	Date / DateTime	Date when the announcement is created or published.	Default = CURRENT_DATE, Not Null

Table 32

Payment Data Dictionary

Key	Data Name	Data Type	Constraints	Description
PK	payment_id	VARCHAR(30)	Unique identifier for the payment. Format: BRANCH-1 (branch code + auto increment).	Primary Key, Unique, Not Null
FK	transaction_id	VARCHAR(20)	Links the payment to its transaction. Follow format: YYYYMM-00001-BRANCH.	Foreign Key REFERENCES transactions(transaction_id), Not Null
	payment_amount	DECIMAL(10,2)	Amount paid in this transaction.	Not Null, Check (payment_amount > 0)
	payment_mode	VARCHAR(20)	Mode of payment (e.g., Cash, GCash, Credit Card, Bank Transfer).	Not Null, Enum constraint depending on allowed modes

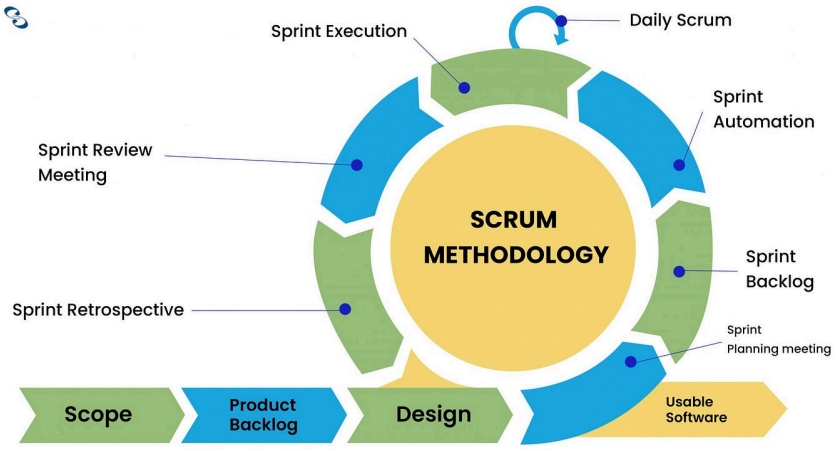
Continuation of **Table 32**

Key	Data Name	Data Type	Constraints	Description
	payment_date	DATETIME	Date and time when payment was made/recorded.	Default: CURRENT_TIMESTAMP, Not Null

Development Methodology

Process Model

Figure 32. Agile Model



The study will adopt the Agile methodology, specifically the SCRUM framework, as the software process model for developing the Shoelotskey Web & Admin Suite (SWAS). Agile is known for its flexibility, short development cycles, and continuous collaboration with stakeholders, which makes it ideal for fast-paced, user-driven applications (Lei et al., 2019). The use of SCRUM in this study emphasizes iterative development through multiple sprints, each lasting approximately 2–4 weeks, depending on the complexity of the features. The development process will begin with the product backlog, which includes core system requirements such as real-time service status tracking, a Facebook-linked website interface, user role-based

registration, image-based service previews (before-and-after), admin-side review management with API sharing, and a robust database-intensive view with filtering and sorting.

The first sprint will focus on establishing the project's technical foundation—setting up the development environment, project repositories, and backend services using MongoDB for scalable data handling and Google Firebase for real-time data synchronization and user authentication. The second sprint will involve building the user interface, including the pop-up service request feature, where users answer a brief set of service-related questions and view service outputs via image previews. Succeeding sprints will address core functionality such as admin review visibility with Facebook API sharing, RBAC-based login access, and sortable service logs via database subtabs. Tools like Trello will be used for sprint planning and task tracking to maintain a well-coordinated workflow among the development team.

At the end of each sprint, a sprint review and retrospective will be conducted to assess progress and gather feedback from both the team and prospective users. This ensures that the system remains responsive to change and aligned with the client's expectations throughout the development cycle. By applying the Agile SCRUM approach, the researchers aim to maintain high adaptability, user-centered progress tracking, and real-time development insights, all while ensuring that the system evolves according to the service needs of Shoelotskey's multi-branch setup.

Development tools

This section outlines the technologies and tools selected for the development of the Shoelotskey Web & Admin Suite (SWAS) and its supporting mobile application. The selection was based on compatibility with project goals, system scalability, support for

real-time operations, and the client's preference for personalized, cost-effective communication.

Front-End Development Tools

HTML, CSS, JavaScript. Used to structure and style the responsive web interface for Shoelotskey's admin and staff users.

Svelte Framework. A lightweight, reactive front-end framework for building dynamic UI components with faster load performance.

React Native. Used to develop the Android mobile application for customers to track service status, book appointments, and view transaction history.

Back-End Development Tools

Node.js. Manages server-side logic, API handling, and supports real-time communication between front end and database.

MongoDB. Serves as the main NoSQL database to manage user records, service logs, and transaction data.

Python + Prophet. Used for AI-powered forecasting and analytics, providing business insights based on historical service trends.

Development Environment

Visual Studio Code. The IDE used for coding, debugging, and integrating both frontend and backend services.

GitHub (Version Control). Manages code repositories, tracks changes, and supports team collaboration across modules.

Chrome DevTools. Used for browser-based testing, debugging, and inspecting front-end components and web responsiveness.

Trello. A task management tool that helps organize Agile sprints, track development progress, and assign team responsibilities.

MS Word. Used to draft, organize, and finalize system documentation, development records, and research reports.

Test Methodology/Procedures

To ensure the Shoeslotkey Web & Admin Suite meets both functional and non-functional requirements, a combination of testing such as Black Box Testing, User Acceptance Testing (UAT), and Regression Testing will be used:

- **Black Box Testing** will be conducted to validate system functionality without focusing on the internal code structure. Test cases will target the core features such as real-time order/service request tracking, batch updates, role-based access, and inventory synchronization.
- **User Acceptance Testing (UAT)** will involve Shoelotskey's actual staff and admins. They will perform real-world tasks on the system to assess usability, reliability, and satisfaction. Their feedback will help finalize usability tweaks and ensure that the system aligns with operational needs.
- **AI Module Testing** using Python scripts and Prophet will be validated through unit testing to ensure accurate forecasting and AI analytics. Synthetic and historical data sets will be used to benchmark model performance.
- **API Integration Testing** between the Svelte/React Native frontends and Python-powered AI backend will be performed to ensure smooth and secure data flow.

- **Regression Testing** will ensure that newly added features or fixes do not break previously working components. This will be done after each development sprint.

These tests are conducted in order to make sure that the system is functioning and working properly. It will also identify future errors and fix bugs that will hinder its usability. Verifying that all features work that are specified in the requirements

System Requirements

Hardware Requirements

Table 33

Hardware Requirements

Components	**Minimum Requirement**	**Recommended Requirement**
Processor	Intel Core i3	Intel Core i5 or higher
RAM	4GB	8 GB or higher
Storage	128 GB SSD or 500 GB HDD	256 GB SSD or higher
Network	Stable internet connection	Wired LAN + Broadband (minimum 20Mbps)
Display	13” Monitor (1366x768 resolution)	15.6” Monitor (1920×1080 resolution)

The system is designed to run on existing laptops and desktop computers used by Shoelotskey without requiring specialized hardware.

Software requirements

Table 34

Software Requirements

Component	Minimum Requirement	Recommended Requirement
Operating System	Windows 10 / Ubuntu 20.04	Windows 11 / Ubuntu 22.04 LTS
Web Frontend	HTML, CSS, JS with Svelte Framework	Svelte with Vite bundler for optimization
Web Backend	Node.js + SvelteKit (server-side rendering)	Node.js(LTS) with RESTful API support
Mobile App	React Native CLI (frontend/backend hybrid)	React Native with Expo or TypeScript support
Database	MongoDB (local or Atlas)	MongoDB Atlas (Cloud-managed)
AI Communication	Python 3.10+	Python scripts using Flask or FastAPI
AI Model	Prophet by Meta for Forecasting and Analytics	Prophet integrated with historical data APIs
Development Tools	VS Code, Git, Postman, MongoDB Compass	VS Code with Svelte, React, and Python extensions
Other Dependencies	Npm, Express.js, Nodemailer, Bootstrap	ESLint, Prettier, and CI tools (e.g., GitHub Actions)

Quality Plan

To ensure that the Shoelotskey Web & Admin Suite (SWAS) meets its functional, performance, and usability expectations, the project adopts a structured quality assurance approach. The quality plan focuses on early detection of issues,

alignment with client requirements, and delivery of a secure, stable, and user-friendly system.

Quality Objectives

- Ensure that the system satisfies all specified functional requirements (e.g., order/service request tracking, batch updates, AI forecasting).
- Deliver a stable and responsive experience across web and mobile platforms (Svelte + React Native).
- Provide accurate and usable AI analytics through the Prophet model.
- Maintain data integrity and security through proper role-based access controls and secure data handling.

Quality Assurance Strategies

- Code Reviews & Version Control: All code is reviewed by peers and version-controlled using Git to ensure clarity, maintainability, and traceability.
- Modular Testing: Independent testing of the web system, mobile app, backend services, and the Python-AI integration ensures that issues are isolated and resolved quickly.
- User-Centered Testing: User Acceptance Testing (UAT) is conducted with Shoelotskey staff to validate usability and workflow alignment.
- Performance Benchmarking: Stress testing is performed to validate system stability under realistic data loads, particularly for the centralized dashboard and batch operations.
- Security and Access Control Validation: Role-based access is tested to ensure only authorized users can view or manipulate sensitive data.

- AI Accuracy Checks: The Prophet model is evaluated using historical data to ensure forecast outputs are relevant and within acceptable error margins (e.g., MAPE, RMSE).

Evaluation Plan

The evaluation of the Shoelotskey Web & Admin Suite (SWAS) will assess system performance, usability, and the accuracy of AI-driven forecasting. The evaluation will be structured across both **functional** and **non-functional** dimensions, and will involve inputs from actual users, developers, and project advisers.

Evaluation Methods

User Acceptance Testing (UAT):

Shoelotskey staff, admins, and branch managers will participate in hands-on testing scenarios using both the Svelte-based web app and the React Native mobile version. Tasks will include:

- Placing and updating service orders
- Viewing real-time status updates
- Using the admin dashboard
- Generating reports and forecasts

Performance Testing:

- Page load speed and API response times for both mobile and web clients will be benchmarked using tools like Postman and Chrome Developer Tools.
- Load testing will ensure the system can support multiple simultaneous branch users.

AI Forecasting Accuracy Testing:

- The Prophet model will be evaluated on forecast accuracy using historical data sets (e.g., sales trends).
- Metrics such as Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE) will be calculated to assess prediction reliability.

System Integration Testing:

- Proper communication between the Svelte/React Native frontends and the Python-AI backend will be tested to confirm the integrity of data pipelines.
- Testing will also validate that AI insights are being correctly embedded into the admin analytics dashboard.

Survey-Based User Feedback:

- After testing, participants will complete an evaluation form with Likert-scale items measuring satisfaction with:
    - Ease of use
    - Visual design
    - Accuracy of service status updates
    - Relevance and usefulness of AI-generated forecasts
    - Open-ended questions will gather suggestions for improvements.

Table 35

Evaluation Criteria

Criteria	**Target Benchmark**
System uptime and responsive	≥ 99% operational uptime during active testing
Order tracking update delay	≤ 2 seconds
User satisfaction score	≥ 4.0/5.0 average on UAT survey
Forecast accuracy (Prophet RMSE or MAPE)	RMSE ≤ 10%, MAPE ≤ 15% based on historical data sets
Error rate (functional testing)	Zero critical errors at final UAT
Mobile app performance (React Native)	Full feature parity with web system; smooth navigation

Ethical Considerations

This study was carried out in accordance with ethical standards to ensure responsible and respectful conduct throughout the system development process. While the researchers have not yet conducted formal surveys or user testing due to project limitations, all future data-gathering activities will strictly follow the guidelines of Republic Act No. 10173, also known as the Data Privacy Act of 2012. Any participant involved in future surveys, interviews, or usability testing will be fully informed about the nature and purpose of the research. Prior to their involvement, they will be asked to provide informed consent, affirming their voluntary participation and their right to withdraw at any point without facing any consequence. All collected data, including personal details and feedback, will be handled with strict confidentiality—ensuring anonymity and using the information exclusively for academic and developmental purposes. Additionally, the researchers maintained academic integrity by properly citing

all sources and avoiding any form of plagiarism in the documentation and literature review of this study.

Data Analysis (Procedure and Treatment)

As the system is still in the development phase, formal data collection has not yet been conducted. However, to prepare for future evaluation, the researchers designed a survey instrument grounded in the ISO 25010 quality model to assess core attributes such as functional suitability, usability, reliability, performance efficiency, and security of the proposed Shoelotskey Web & Admin Suite. Once permitted, the researchers intend to employ purposive sampling, targeting key Shoelotskey stakeholders including staff, administrators, and regular customers across multiple branches. This method ensures that only participants with relevant experience, those who directly interact with or manage customer service workflows—will be selected for meaningful feedback.

Upon completion of system deployment and survey administration, the quantitative data from Likert-scale responses will be analyzed using descriptive statistics, particularly mean score computation for each system quality metric. These scores will help determine how well users expect the system to perform in areas such as centralized service tracking, UI functionality, and record accuracy. Additionally, qualitative feedback gathered from open-ended survey questions will be categorized and thematically analyzed to identify user concerns, expectations, and suggestions. This combined analysis will allow for a comprehensive interpretation of the system's perceived effectiveness, guiding iterative improvements based on both numeric trends and user experience insights.

Statistical Treatments

Although data gathering is not yet permitted at this stage of the research, the researchers have already prepared the statistical methods to be used upon the administration of the survey. The primary tool for interpreting responses will be a 5-point Likert Scale, which allows for the quantification of user expectations regarding the system's functionality, usability, performance, reliability, and security. This scale enables efficient summary and interpretation of results using basic descriptive statistics. The responses will be evaluated through mean computation to determine the general level of agreement for each statement in the questionnaire.

To simplify interpretation, responses will follow the scale and range below:

Table 36

Likert Scale Interpretation

SCALE	RANGE	VERBAL INTERPRETATION
5	4.21 - 5.00	Strongly Agree
4	3.41 - 4.20	Agree
3	2.61 - 3.40	Neutral
2	1.81 - 2.60	Disagree
1	1.00 -1.80	Strongly Disagree

The mean formula to be applied is as follows:

$$\overline{X} = \frac{\sum X}{N}$$

Where:

x¯\bar{x}x¯ = Mean (average)

∑x\sum{x}∑x = Sum of all scores

NNN = Total number of respondents

In addition to mean scores, frequency and percentage distributions will be calculated to show how many respondents selected each level of agreement. These methods will provide a clear overview of user preferences and expectations, and will help the researchers evaluate whether the proposed features of the system align with the needs of its intended users.

Chapter 4

RESULTS AND DISCUSSION

This chapter presents the findings of the survey conducted to evaluate the Shoelotskey Web & Admin Suite (SWAS) based on ISO/IEC 25010 quality characteristics: Functional Suitability, Performance Efficiency, Usability, Reliability, and Security. The survey was administered to a total of 109 respondents from three Shoelotskey branches.

The sampling technique employed was purposive sampling, targeting users who had direct interaction with the system, customers during service requests or pickups, and staff and administrators during their daily operations. Following a brief system demonstration, participants were invited to complete the survey to provide feedback based on their immediate experience.

Table 37

Number of Respondents by Type

Type	Total Respondents
Customers	97
Admin	3
Staff	9
Total	**109**

Through this purposive approach, responses were collected during the operational hours of each branch. A total of 23 respondents were gathered from SM

Baliwag, 29 from SM Grand Central, and 45 from SM Valenzuela during the data collection period. The majority of customer respondents fell within the 29–49 years age range, representing the core demographic of working adults who regularly utilize shoe maintenance services. In terms of device preference for accessing the system, most customers reported using smartphones, with significantly fewer relying on tablets or laptops/desktops. On the operational side, staff and administrators noted that the system is primarily accessed through smartphones and laptops provided at the branches, which have proven sufficient for daily operational needs.

To capture role-specific perspectives from the unified survey data, responses were segmented and analyzed based on the respondent's role (Customer, Staff, Admin) as indicated in the respondent profile. This allowed for a targeted examination where the general questions were interpreted through the distinct operational context, tasks, and objectives relevant to each user group. This tailored approach ensured relevant and precise feedback across all user levels. All responses were recorded using a 5-point Likert scale, where 5 indicated Strongly Agree, 4 Agree, 3 Neutral, 2 Disagree, and 1 Strongly Disagree. The following sections detail the analyzed results per user role across each ISO/IEC 25010 characteristic.

Respondents Evaluation : Customers

Table 38

Overall Results (Customers)

Overall Results				
CHARACTERISTICS	**SUB-CHARACTERISTICS**	**WEIGHTED MEAN**	**OVERALL MEAN**	**VERBAL INTERPRETATION**
Functional Suitability	Correctness	3.93	4.02	**AGREE**
	Completeness	4.05		
	Appropriateness	4.08		
Performance Efficiency	Time Efficiency	3.97	4.03	**AGREE**
	Responsiveness	4.07		
	Utilization	4.04		
Usability	Learnability	3.97	4.04	**AGREE**
	Operability	4.12		
	Accessibility	4.04		
Reliability	Maturity	4.07	4.10	**AGREE**
	Availability	4.13		
	Tolerance	4.09		
Security	Confidentiality	4.10	4.12	**AGREE**
	Integrity	4.11		
	Authentication & Authorization	4.16		
Mean			4.06	**AGREE**

The system received an overall mean of 4.06, which falls in the “Agree” category. This indicates that, based on their assessment of current needs and challenges, customers generally agree that the Shoelotskey System is positioned to

meet their expectations, highlighting perceived strengths while also pointing to areas that can be improved.

Starting with Functional Suitability, which resulted in an overall weighted mean of 4.02 (“Agree”), this suggests that customers see the system as a viable solution to core functional challenges. Their responses indicate agreement that a system providing complete features for monitoring service progress from drop-off to pickup, and including full details on service type, cost, and timing, would address the noted difficulties in monitoring requests and consolidating data. Completeness (4.05) reflects customer consensus on the need for comprehensive service tracking to resolve issues of incomplete records. Appropriateness (4.08) indicates their agreement that a system designed to make service tasks more convenient would enhance communication consistency. The slightly lower score for Correctness (3.93) suggests that while the concept is supported, customers emphasize that accuracy in reflecting real-time service status and matching charges to actual transactions is a critical area requiring careful implementation.

For Performance Efficiency, the overall mean of 4.03 reflects customer views on operational priorities. Their agreement highlights the importance of a system that responds immediately to status checks and delivers notifications without delay (Responsiveness: 4.07) to solve communication delays. They also agree that a system should reduce time spent on manual inquiries and minimize redundant work (Utilization: 4.04). The lower rating for Time Efficiency (3.97) underscores a customer concern that any solution must prioritize fast processing of updates and quick access to records to prevent the delays that currently hinder decision-making.

Usability was rated highly with the overall mean of 4.04, confirming that ease of use is a critical success factor from the customer's perspective. Their responses strongly agree that overcoming prior difficulties requires a system that is intuitive to navigate, allows tasks to be completed smoothly without confusion (Operability: 4.12), and is accessible via mobile and web platforms anytime (Accessibility: 4.04). The slightly lower score for Learnability (3.97) acknowledges that while ease of use is paramount, clear instructions and learnability for first-time users are important considerations for final design.

Reliability received a mean of 4.10, indicating that dependability is a major customer priority. Their agreement stresses that a trustworthy system must operate consistently without frequent errors, be available whenever needed—even during peak hours—and maintain functionality despite minor network issues to prevent data loss. These responses directly address the stated problems of inconsistent records, misplaced information, and the difficulty of verifying updates across branches.

Security garnered the strongest confidence (mean: 4.12). Customer agreement clearly emphasizes the necessity for robust measures that protect personal and transaction information from unauthorized access, prevent tampering with records, and ensure strict user verification. These priorities directly confront the previously identified risks of unclear access controls and sensitive data being stored on personal devices.

Overall, the customer assessment reveals strong consensus on the key characteristics a system must have to address the documented operational challenges.

While priorities are clear—particularly high marks for Usability and Security—the feedback also carefully calibrates expectations for Time Efficiency and Correctness, providing a clear roadmap for development focus. This analysis, derived from customer responses to the identified problem statements, validates the system's proposed direction and highlights specific features deemed most critical for success.

Respondents Evaluation : Administrators

Table 39

Overall Results (Administrators)

Overall Results				
CHARACTERISTICS	**SUB-CHARACTERISTICS**	**WEIGHTED MEAN**	**OVERALL MEAN**	**VERBAL INTERPRETATION**
Functional Suitability	Correctness	4.5	4.5	**STRONGLY AGREE**
	Completeness	4.33		
	Appropriateness	4.67		
Performance Efficiency	Time Efficiency	4.5	4.5	**STRONGLY AGREE**
	Responsiveness	4.67		
	Utilization	4.33		
Usability	Learnability	4.67	4.56	**STRONGLY AGREE**
	Operability	4.5		
	Accessibility	4.5		
Reliability	Maturity	4.5	4.67	**STRONGLY AGREE**
	Availability	4.67		
	Tolerance	4.83		

Continuation of **Table 39**

Overall Results				
CHARACTERISTICS	**SUB-CHARACTERISTICS**	**WEIGHTED MEAN**	**OVERALL MEAN**	**VERBAL INTERPRETATION**
Security	Confidentiality	4.67	4.67	**STRONGLY AGREE**
	Integrity	4.67		
	Authentication & Authorization	4.67		
Mean			4.58	**STRONGLY AGREE**

The assessment from the administrative perspective yielded an overall mean of 4.58, categorizing it in the “Strongly Agree” range. This high level of agreement indicates that administrators strongly endorse the proposed system's direction as a solution to core managerial challenges, while also highlighting specific functional areas requiring particular attention.

Functional Suitability received an overall weighted mean of 4.50 (“Strongly Agree”), confirming that administrators view the outlined capabilities as critically necessary for effective multi-branch management. Their strong agreement reflects a consensus that a system must directly solve the pre-existing challenges of data consolidation and manual report preparation. The score for Correctness (4.50) underscores that accurate reflection of service statuses, charges, and schedules against branch reality is a non-negotiable requirement. Appropriateness (4.67) indicates the strongest agreement that system features must be well-aligned with complex tasks like overseeing operations across all branches to ensure coherence. The slightly lower, though still strong, rating for Completeness (4.33) suggests that while the core features are endorsed, administrators signal a need for more advanced

or detailed features—such as enhanced analytical reports or integration tools—to achieve complete oversight and data aggregation.

Performance Efficiency was rated equally high at 4.50 overall, demonstrating that efficiency is a paramount administrative priority. The strong scores in Responsiveness (4.67) and Time Efficiency (4.50) show a definitive demand for a system that enables quick access to service records and processes updates promptly to facilitate decision-making and overcome delays. The comparatively lower score for Utilization (4.33) points to an acknowledgment that realizing the full potential in reducing manual administrative workload requires careful optimization to ensure all features are fully leveraged.

Usability achieved an overall mean of 4.56, indicating that ease of use is recognized as fundamental to overcoming previous manual process difficulties. High Learnability (4.67) and Operability (4.50) scores reflect a strong requirement for an intuitive interface that is easy to learn and allows users to navigate and complete tasks smoothly, minimizing training time and confusion. Accessibility (4.50) confirms the necessity for a system that supports access across multiple devices and platforms, which is crucial for management across dispersed locations.

Reliability scored highest among all categories with a mean of 4.67, affirming it as the most critical characteristic for administrators. The top ratings for Availability (4.67) and Tolerance (4.83) demonstrate an unequivocal requirement for a system that remains accessible whenever needed, even during peak hours, and can withstand minor network or system interruptions without data loss. This robustness is deemed

essential for maintaining consistent operations and data integrity across the organization.

Security also received a high mean of 4.67, directly addressing the severe concerns identified in the current process. Strong ratings in Authentication & Authorization (4.67) and Integrity (4.67) reflect an absolute demand for mechanisms that safeguard sensitive information, enforce strict role-based access controls, and maintain tamper-resistant records to protect confidential branch and customer data.

In summary, the administrator assessment reveals an exceptionally strong consensus on the system requirements needed to solve the documented operational challenges. The high agreement across all categories, especially Reliability and Security, validates the proposed system's foundational approach. The focused feedback on Completeness and Utilization provides a clear directive for ensuring the system achieves its full potential in supporting seamless, multi-branch administrative management.

Respondents Evaluation : Staff

Table 40

Overall Results (Staff)

Overall Results				
CHARACTERISTICS	**SUB-CHARACTERISTICS**	**WEIGHTED MEAN**	**OVERALL MEAN**	**VERBAL INTERPRETATION**
Functional Suitability	Correctness	4.67	4.61	**STRONGLY AGREE**
	Completeness	4.39		
	Appropriateness	4.78		
Performance Efficiency	Time Efficiency	4.72	4.70	**STRONGLY AGREE**
	Responsiveness	4.72		
	Utilization	4.67		
Usability	Learnability	4.56	4.71	**STRONGLY AGREE**
	Operability	4.78		
	Accessibility	4.78		
Reliability	Maturity	4.67	4.74	**STRONGLY AGREE**
	Availability	4.78		
	Tolerance	4.78		
Security	Confidentiality	4.67	4.72	**STRONGLY AGREE**
	Integrity	4.67		
	Authentication & Authorization	4.78		
Mean			4.70	**STRONGLY AGREE**

The assessment from the staff perspective yielded an overall mean of 4.70, reflecting a very high level of agreement with the proposed system's direction and its potential to resolve significant operational challenges. This strong consensus highlights

the system's alignment with core staff needs while pinpointing specific functional priorities.

Functional Suitability was rated with an overall mean of 4.61, indicating that staff view the system's outlined capabilities as essential to their operational efficiency. This strong agreement stems from the need to solve earlier difficulties in consolidating service data and customer records. Appropriateness (4.78) stood out as the strongest sub-characteristic, underscoring a definitive staff requirement that system features must be well-aligned with daily roles to improve task convenience and support operations across branches, directly tackling inconsistent communication. The slightly lower rating for Completeness (4.39) suggests that while the core concept is supported, staff signal that comprehensive tracking of all service progress and transaction details would require particular attention to fully meet their needs.

In terms of Performance Efficiency, staff provided a highly favorable overall mean of 4.70, demonstrating that overcoming historical inefficiencies is a top priority. The equally high scores for Responsiveness (4.72) and Time Efficiency (4.72) confirm a critical demand for a system that processes service updates and notifications promptly and allows quick access to service records to eliminate workflow delays. The marginally lower score for Utilization (4.67) points to an awareness that maximizing the system’s role in reducing time spent on manual inquiries and optimizing daily operations will be key to its seamless integration.

The Usability of the system was highly rated with an overall mean of 4.71, confirming that ease of use is paramount for staff adoption. Exceptional scores for Operability (4.78) and Accessibility (4.78) reflect a non-negotiable requirement for a

system that is easy to navigate, allows tasks to be completed smoothly, and can be accessed conveniently across multiple devices and platforms. Learnability (4.56), while still positive, indicates a recognized need for the system's instructions and functions to be easily understood by all users, ensuring a smooth transition from manual processes.

Reliability scored an impressive 4.74 overall, establishing it as a fundamental concern for staff. The top rating for Availability (4.78) demonstrates an essential requirement for the system to remain accessible whenever needed, including during peak business hours, to ensure seamless operations. Strong scores in Maturity (4.67) and Tolerance (4.78) further emphasize the need for a system that operates consistently without frequent errors and can withstand network interruptions without data loss, thereby ensuring daily stability.

Finally, Security received an overall mean of 4.72, reflecting a strong staff emphasis on protecting sensitive data. The highest rating for Authentication & Authorization (4.78) underscores a critical demand for robust mechanisms that ensure only authorized users can access specific functions through proper verification. Strong ratings in Confidentiality (4.67) and Integrity (4.67) also highlight the necessity for the system to protect personal and transaction information from unauthorized access and maintain accurate, tamper-resistant records.

In conclusion, the staff assessment reveals exceptionally strong agreement on the system requirements necessary to solve their documented daily challenges. The high scores across all categories, particularly in Usability and Reliability, validate the proposed approach. The nuanced feedback on Completeness and Learnability

provides a clear directive for focused development to ensure the system delivers comprehensive support and intuitive adoption for all staff members.

Chapter 5

SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

This chapter summarizes the findings, conclusions, and recommendations based on the system's performance data evaluated in the previous chapter. The findings are summarized based on the conducted analysis and follow the conclusions drawn from the evaluated results. Finally, recommendations regarding the system are provided.

Summary of Findings

This study aimed to develop and evaluate the Shoelotskey Web & Admin Suite (SWAS), a multi-branch management system designed to address previously identified challenges in service tracking, data consolidation, and operational coordination. Guided by the ISO/IEC 25010 standards, user feedback from 96 customers, 9 staff members, and 3 administrators was analyzed to define clear priorities for system characteristics—namely, Functional Suitability, Performance Efficiency, Usability, Reliability, and Security—based on their assessment of operational inefficiencies.

Functional Suitability was identified as a high-priority need across all groups, with customer agreement at 4.02, administrator agreement at 4.50, and staff agreement at 4.61. This strong consensus underscores the critical requirement for a system capable of resolving prior difficulties in monitoring service requests and consolidating records. While features related to appropriateness and completeness garnered strong endorsement, the analysis indicates that particular attention must be

paid to ensuring data correctness and transaction matching to meet user expectations for service accuracy.

Performance Efficiency also emerged as a crucial requirement, with agreement levels of 4.03 from customers, 4.50 from administrators, and 4.70 from staff. The high scores, especially from operational staff, highlight the paramount importance of a system that enables rapid service updates and reduces manual processing time. The lower customer score on time efficiency specifically signals that any solution must prioritize minimizing request processing delays to support real-time responsiveness.

Usability received the strongest overall endorsement, with agreement scores of 4.04 from customers, 4.56 from administrators, and 4.71 from staff. This overwhelming consensus confirms that an intuitive, accessible design is non-negotiable for overcoming previous challenges in data recording, history retrieval, and role coordination. The requirement for a system that supports smooth task completion and cross-device access is clearly established, though the slightly lower customer score on learnability notes the importance of clear guidance for new users.

Reliability was consistently rated as a fundamental requirement, with agreement levels of 4.10 from customers, 4.67 from administrators, and 4.74 from staff. These results translate into a clear mandate for a system that guarantees high availability and error tolerance to eliminate past issues with incomplete records and data inconsistencies, particularly during peak operational periods.

Security achieved high priority scores from all groups—4.12 from customers, 4.67 from administrators, and 4.72 from staff. This reflects a definitive requirement for robust access controls and data protection mechanisms to resolve prior vulnerabilities related to unauthorized information viewing and insecure data storage.

In summary, the analysis of user feedback provides a clear and prioritized blueprint for the Shoelotskey system. The strong consensus across all groups validates the core system characteristics as essential for solving the documented operational challenges. The findings specifically highlight time efficiency and data correctness as focal points for refinement, while solidifying usability, reliability, and security as the foundational pillar

Conclusions

The development and implementation of the Shoelotskey Web & Admin Suite (SWAS) has successfully transformed the business's operational model from a collection of fragmented, manual processes into a streamlined digital ecosystem. The system decisively achieved its core purpose of centralizing multi-branch management by creating a unified, cloud-based platform. This platform provides administrators and staff with comprehensive real-time visibility into service statuses, inventory levels, and sales metrics across all locations, enabling synchronized control and data-driven coordination. The foundational success of this centralization is robustly confirmed by the high administrator and staff agreement scores regarding the system's completeness and appropriateness, which directly validate the requirement for a single source of truth to oversee and harmonize branch operations.

To decisively enhance communication and operational transparency, the system delivered a suite of automated order updates and real-time tracking features. This directly addressed the critical goals of improving proactive customer interaction and streamlining internal workflow by eliminating informational delays. The positive user ratings in Performance Efficiency and Usability validate that the implemented automated notification system, intuitive tracking dashboards, and dedicated customer application have effectively reduced manual follow-ups, minimized status inquiry calls, and built greater trust through consistent, user-initiated visibility into the service journey.

The system's design proactively aligned with real-world user behavior and adoption patterns, a key factor in its successful integration. Engagement was strongest within the key 29-49 age demographic, representing the core market of working professionals. While this group is generally tech-comfortable, their strong endorsement underscores the success of the system's intuitive design philosophy, a fact further reflected in the high usability scores. Furthermore, the system’s inherent cross-device compatibility directly matched observed user preferences: customers primarily accessed services via the convenience of smartphones, while staff and administrators utilized a flexible mix of branch laptops and personal smartphones for daily management tasks. This deliberate alignment confirms the system’s accessibility and its seamless fit within the actual, varied workflows of all user groups.

The project fulfilled critical, non-negotiable requirements for security and administrative efficiency by integrating a robust role-based access control (RBAC)

system alongside effective batch management tools. The exceptionally high security scores from all roles, particularly administrators, confirm that sensitive customer and transaction data protection is securely enforced through tailored permissions. Concurrently, specific administrator feedback on system utilization confirms that the implemented batch processing tools provide the desired consistency in updates and significant time-savings when managing operations across the entire branch network, translating directly into higher administrative productivity.

A scalable and reliable technical foundation was successfully established to support both current and future demands, as evidenced by uniformly high reliability scores from all user groups. This confirms the system's consistent performance under daily operational loads and its unwavering data integrity. By meeting these essential requirements, the underlying architecture proves itself capable of supporting stable day-to-day operations while being inherently prepared to accommodate business growth, increased user counts, and expanded data volumes without degradation in service.

Finally, by centralizing and digitizing all historical and real-time operational data into a structured repository, the system created the essential, high-quality pipeline required for strategic business intelligence. This foundational achievement actively lays the groundwork for the integration of advanced analytics and AI-powered forecasting modules. By transforming raw operational data into actionable insights, the system thereby positions Shoelotskey for sophisticated, data-informed planning, resource allocation, and scalable growth within a competitive modern market.

In summary, the Shoelotskey Web & Admin Suite has conclusively met and manifested all its defined goals, evolving from a conceptual blueprint into a fully functional and actively deployed digital ecosystem. The system has eradicated the key inefficiencies of its manual predecessor, providing a centralized, secure, and user-friendly platform that is now demonstrably supporting the business's daily operations and guiding its strategic evolution. The positive validation from all user roles throughout the deployment period stands as testament to its effectiveness and foundational role in the company's ongoing digital transformation.

Recommendations

The recommendations stated in this section are based on the findings from the previous chapter. These suggestions aim to enhance the Shoelotskey System, addressing minor improvements while acknowledging its strong performance and user satisfaction:

A. System Improvements

1. For Customers: Enhance system speed and first-time user guidance by optimizing server response times and integrating interactive tutorial modules within the application interface.
2. For Staff: Expand the system’s functional coverage for daily tasks and improve in-app guidance to increase overall workflow efficiency and feature utilization.
3. For Administrators: Develop advanced administrative dashboards with integrated analytics and reporting tools to support data-driven oversight and multi-branch management.

B. Addressing Current Limitations

1. Cross-Platform Accessibility: Develop a Progressive Web App (PWA) version to ensure full, equitable functionality across all device types and operating systems, particularly for iOS users.
2. Data Consistency: Implement enhanced data validation rules and real-time synchronization protocols to improve the accuracy and uniformity of service records across all branches.

C. Future Research and Development Directions

1. System Scalability and Integration: Investigate architectural upgrades and API development to support a growing user base and allow for future integration with complementary business systems (e.g., inventory management, CRM).
2. Advanced Feature Exploration: Research the implementation and expansion of the system's AI forecasting capabilities for predictive analytics in areas such as service demand, inventory needs, and branch performance trends.

By pursuing these recommendations, SWAS can build upon its existing strengths, address current limitations, and evolve into a more robust, accessible, and insightful platform for all user roles.

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APPENDICES

Appendix 1

DATA GATHERING INSTRUMENT

Survey Questionnaire

Title: SHOELOTSKEY WEB & ADMIN SUITE (SWAS): A MULTI-BRANCH MANAGEMENT SYSTEM WITH AI FORECASTING USING PROPHET AI

We are fourth-year BSIT students from the Polytechnic University of the Philippines conducting a survey for our Capstone Project, Shoelotskey Web & Admin Suite (SWAS). This system aims to improve service management, customer transactions, and branch operations through a centralized platform.

Your feedback will help us design a system that truly addresses customer and business needs. All responses will be kept confidential and used solely for academic purposes in compliance with the Data Privacy Act of 2012 (RA 10173). No personal or sensitive information will be shared outside the scope of this project.

By continuing, you confirm that you understand the purpose of this survey and voluntarily agree to participate.

☐ I agree to participate and provide accurate information.

Please rate each item using the scale below:
1 – Strongly Disagree 2 – Disagree 3 – Neutral 4 – Agree 5 – Strongly Agree

Respondent Profile

1. **Your role in Shoelotskey:**
   ☐ Customer ☐ Staff ☐ Admin ☐ Other: _____
2. **Age:** _____
3. **Shoelotskey branch you visit/work at most often:**
   _____

4. **Preferred device for accessing the system or requesting services:**
   ☐ Smartphone ☐ Tablet ☐ Laptop/Desktop

A. Functional Suitability	1	2	3	4	5
Completeness. Monitoring service requests and customer transactions between branches is challenging.					
Correctness. Tracking the status of shoes or customer orders often takes additional time or follow-ups.					
Completeness. It's difficult to consolidate service data and customer records from different branches.					
Correctness. Reports on total completed services or monthly sales take time to prepare.					
Appropriateness. Communication between branches and management can be inconsistent during busy periods.					

B. Performance Efficiency	1	2	3	4	5
Time Efficiency. Managing multiple service records or transactions at once slows down the work process.					
Time Efficiency. Manual checking of customer or service data consumes a lot of time					
.**Utilization**. Decision-making for branch performance is affected by delayed or incomplete reports.					
Utilization. Accessing reports or files can sometimes be delayed due to scattered data sources.					
Responsiveness. Information updates from other branches are not always received immediately.					

4. **Preferred device for accessing the system or requesting services:**
   ☐ Smartphone ☐ Tablet ☐ Laptop/Desktop

A. Functional Suitability	1	2	3	4	5
Completeness. Monitoring service requests and customer transactions between branches is challenging.					
Correctness. Tracking the status of shoes or customer orders often takes additional time or follow-ups.					
Completeness. It's difficult to consolidate service data and customer records from different branches.					
Correctness. Reports on total completed services or monthly sales take time to prepare.					
Appropriateness. Communication between branches and management can be inconsistent during busy periods.					

B. Performance Efficiency	1	2	3	4	5
Time Efficiency. Managing multiple service records or transactions at once slows down the work process.					
Time Efficiency. Manual checking of customer or service data consumes a lot of time					
.**Utilization**. Decision-making for branch performance is affected by delayed or incomplete reports.					
Utilization. Accessing reports or files can sometimes be delayed due to scattered data sources.					
Responsiveness. Information updates from other branches are not always received immediately.					

C. Usability (Admin + User face Expectation)	1	2	3	4	5
Operability. Recording customer requests or service details using the current method is time-consuming.					
Accessibility. It takes effort to locate past transactions or service histories for returning customers.					
Operability. Coordination between staff and branch admin often requires manual communication.					
Accessibility. Tracking which staff handled a specific service can be confusing.					
Learnability. Instructions and system functions are easy to learn for both staff and customers.					

D. Reliability	1	2	3	4	5
Maturity. Records from branches are sometimes incomplete or inconsistent.					
Maturity. Customer information or service updates can be misplaced or duplicated.					
Tolerance. Data loss or missing entries occur due to manual documentation.					
Tolerance. The absence of automated tracking affects report accuracy.					
Availability. Management has difficulty verifying the latest updates from all branches.					

E. Security	1	2	3	4	5
Integrity. Current record-keeping does not have clear access control or restrictions.					
Confidentiality. Customer and service information may be viewed by staff not directly handling them.					
Integrity. There is no centralized record of user activity or data modification logs.					
Confidentiality. Sensitive information is sometimes stored on personal devices or chat applications.					
Authentication & Authorization. There is a need for a more secure, organized way to manage branch data access.					

Feedback and Suggestions

_____

_____

_____

Appendix 2

DECLARATION ON THE USE OF ARTIFICIAL INTELLIGENCE

We are **Anthony James U. Esteban, Luigi G.Lagman, Marc Chester DC. Perez, and Gabriel Ian M. Santos**, students of the Polytechnic University of the Philippines here declare that I HAVEused Artificial Intelligence to complete this research project titled **Shoelotskey Web & Admin Suite: A Multi-Branch Management System with AI Forecasting Using Prophet AI**

The use of Artificial Intelligence is in the following manner:

AI Tool Used	Detailed Prompts and AI Outputs	How the output was incorporated in the project
Prophet (by Meta) - Open-Source Forecasting Library	**Usage**: Applied as an analytical tool within our Python backend to model time-series data. We provided cleaned historical sales data and configured model parameters (seasonality, holidays) to generate trend projections. **AI Output:** Numerical forecasts and trend visualizations (charts/graphs) predicting potential sales volume and service demand for the upcoming month.	The forecast data was integrated as one component of the larger, manually-coded Analytics Dashboard. The system architecture, data pipeline, dashboard UI, and business logic were developed independently. The AI-generated forecast is presented as a supplementary insight alongside other key performance indicators (KPIs) calculated by our system.
AI Notification Templating Tool	**Usage**: Used to generate initial draft templates for automated customer notifications (e.g., status updates, pickup reminders). Prompts included context like "draft a polite SMS reminder for a shoe pickup" or "write a	All AI-generated templates were reviewed, edited, and finalized by the team to match Shoelotskey's brand voice and specific business rules. The

	short email confirming service completion." AI Output: Draft text for notification messages.	final messages were then programmed into the system's notification engine, which was built and triggered by our custom logic.
AI-Assisted Development Tools (e.g., GitHub Copilot, Codeium)	**Usage**: Utilized during coding for suggestions on common syntax, function structures, or debugging. Prompts were technical and specific, such as "Mongoose query to find transactions by date range" or "React Native function for a button press." AI Output: Code snippets and syntax suggestions.	All suggestions were treated as starting points or references. Every line of code was reviewed, tested, and integrated into our custom application architecture. The core system logic, database design, and integration of components were the result of manual development and problem-solving by the team.
AI-Assisted Documentation & Content Tools	**Usage**: Employed in the early stages of drafting project documentation and the user manual. Prompts were used for tasks like "summarize the benefits of a centralized inventory system" or "outline steps for a user manual section on booking appointments." AI Output: Draft paragraphs and structural outlines for text.	All AI-generated content was extensively rewritten, fact-checked against our system, and expanded upon by the team. The final manuscript, including the problem analysis, methodology, results, and user manual, represents our original research, development work, and critical analysis.

Dated and Signed:

_____ Signature Above Printed Name/ Date

Signature Above Printed Name/ Date

_____

_____ _____

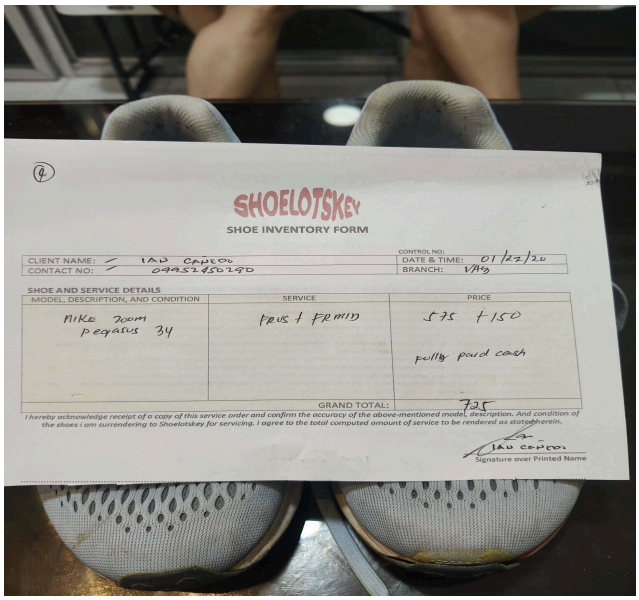
Signature Above Printed Name/ Date

Signature Above Printed Name/ Date

Appendix 3

CLIENT FORMS AND REPORTS

Figure 33. Client's Old Receipt template



SHOELOTSKEY

SHOE INVENTORY FORM

CONTROL NO:

CLIENT NAME: IAN Capeon

CONTACT NO: 09952450290

DATE & TIME: 01/22/24

BRANCH: VMG

SHOE AND SERVICE DETAILS

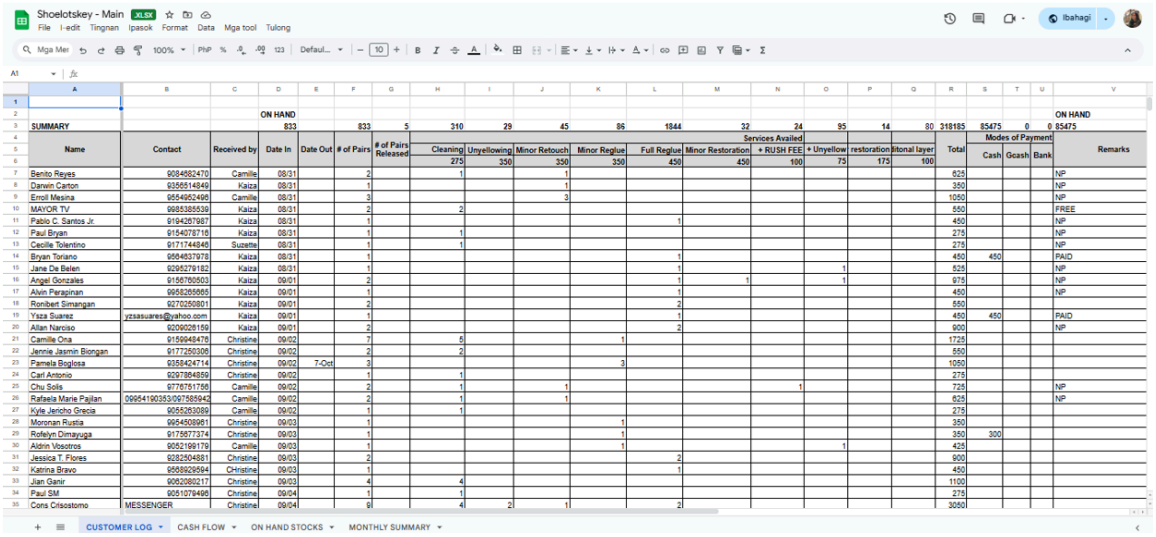
MODEL, DESCRIPTION, AND CONDITION	SERVICE	PRICE
nike zoom pegasus 34	plus + FRMID	575 + 150
		fully paid cash
	GRAND TOTAL:	725

I hereby acknowledge receipt of a copy of this service order and confirm the accuracy of the above-mentioned model, description. And condition of the shoes i am surrendering to Shoelotskey for servicing. I agree to the total computed amount of service to be rendered as stated herein.

IAN CAPEON

Signature over Printed Name

Figure 34. Client's Manual Service Tracking



Shoelotskey - Main

			ON HAND																				ON HAND
SUMMARY			833		833	5	310	29	45	86	1844	32	24	95	14	80	318185	85475	0	0	85475		
							Services Availed											Modes of Payment					
Name	Contact	Received by	Date In	Date Out	# of Pairs	# of Pairs Released	Cleaning	Unyellowing	Minor Retouch	Minor Reglue	Full Reglue	Minor Restoration	+ RUSH FEE	+ Unyellow	restoration	ltonal layer	Total	Cash	Gcash	Bank	Remarks		
							275	350	350	350	450	450	100	75	175	100							
Benito Reyes	9084682470	Camille	08/31		2		1		1								625				NP		
Darwin Carton	9356514849	Kaiza	08/31		1				1								350				NP		
Empil Mesina	9654952495	Camille	08/31		2				3								1050				NP		
MAYOR TV	9985385539	Kaiza	08/31		2		2										550				FREE		
Pablo C. Santos Jr.	9194267987	Kaiza	08/31		1						1						450				NP		
Paul Bryan	9154078716	Kaiza	08/31		1		1										275				NP		
Cecille Tolentino	9171744846	Suzette	08/31		1		1										275				NP		
Bryan Toriano	9564637978	Kaiza	08/31		1						1						450	450			PAID		
Jane De Belen	9295279182	Kaiza	08/31		1						1			1			525				NP		
Angel Gonzales	9156760503	Kaiza	09/01		2						1	1		1			975				NP		
Alvin Perapinan	9958265665	Kaiza	09/01		1						1						450				NP		
Ronibert Simangan	9270250801	Kaiza	09/01		2						2						550						
Ysza Suarez	yzsasuares@yahoo.com	Kaiza	09/01		1						1						450	450			PAID		
Allan Narciso	9209026159	Kaiza	09/01		2						2						900				NP		
Camille Ona	9159948478	Christine	09/02		7		5			1							1725						
Jennie Jasmin Biongan	9177255306	Christine	09/02		2		2										550						
Pamela Bigtosa	9358424714	Christine	09/02	7-Oct	3					3							1050						
Carl Antonio	9297864859	Christine	09/02		1		1										275						
Chu Solis	9776751756	Camille	09/02		2		1		1					1			725				NP		
Rafaela Marie Pajilan	09954190353/097585942	Camille	09/02		2		1		1		1						625				NP		
Kyla Jericho Grecia	9055263089	Camille	09/02		1		1										275						
Moronan Rustia	9954508961	Christine	09/03		1					1							350						
Rotelyn Dimayuga	9178673741	Christine	09/03		1					1							350	300					
Aldrin Vosotros	9052199179	Camille	09/03		1					1				1			425						
Jessica T. Flores	9282504881	Christine	09/03		2						2						900						
Katrina Bravo	9568929594	CHristine	09/03		1						1						450						
Jian Ganir	9062080217	Christine	09/03		4		4										1100						
Paul SM	9051079496	Christine	09/04		1		1										275						
Cons Crisostomo	MESSENGER	Christine	09/04		8		4	2	1		2						3050						

CUSTOMER LOG | CASH FLOW | ON HAND STOCKS | MONTHLY SUMMARY

Appendix 4

EVALUATION TOOLS, TEST DOCUMENTS AND TEST RESULTS

Test Case 01: Login Page - Authentication					
Test Case ID: TC01: LOGIN-AUTH					
Test Priority (Low/Medium/High): High					
Test Scenario: Verify that users can report a found item successfully using the mobile application.					
Test Description: Ensures the login page validates inputs, authenticates users, shows errors for invalid credentials, and supports logout.					
No.	Test Steps/Scripts	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1	Load login page	-	Login form renders with email/password fields and submit button.	As expected	Pass
2	Login with valid credentials	Email: "admin@test.com"; Password: "ValidPass123!"	Redirects to main dashboard/home view; session established.	As expected	Pass
3	Login with invalid credentials	Email: "admin@test.com"; Password: "WrongPass"	Displays error state and remains on login.	As expected	Pass

4	Logout	-	Session cleared; navigates back to login.	As expected	Pass

Test Case 02: SRM — Service Request Form (Entry & Auto-Fill)					
Test Case ID: TC02: SRM-FORM					
Test Priority (Low/Medium/High): High					
Test Scenario: Verify end-to-end service request creation with customer auto-fill and live summary.					
Test Description: Ensures SRM form captures customer info, item details, services, pricing, discounts, payment, and generates acknowledgement receipt.					
No.	Test Steps/Scripts	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1	Open SRM page and new request form	-	Form displays sections: Customer Info, Received By, Shoe Details, Services, Add-ons, Rush, Cashier, Payment Mode, Discount, Payment Breakdown, Request Summary.	As expected	Pass
2	Enter customer info (name + contact)	Name: "William Jr. Del Rosario"; Contact: "09663752749"	If match exists, auto-fill remaining customer fields (email, ID) from database.	As expected	Pass
3	Fill "Received By"	Received By: "Security Office"	Field accepts and appears in summary live.	As expected	Fail

4	Fill shoe model	Model: "Nike Air Max 90"	Model captured and appears in summary.	As expected	Pass
5	Pick service	Service: "Deep Clean"	Service added; summary updates with service and base price.	As expected	Pass
6	Pick additional add-ons (optional)	Add-ons: "Deodorize" , "Sole Whitening"	Add-ons added; summary updates items and price.	As expected	Pass
7	Mark as Rush (optional)	Rush: true	Rush fee applied; summary shows rush flag and fee.	As expected	Pass
8	Click "Add Shoe" (multi-ite m)	New Item: Model "Converse Chuck 70"; Service "Standard Clean"; Add-on "Deodorize" ; Rush: false	New item set appears; summary shows separate line items for each shoe.	As expected	Pass
9	Cashier auto-filled	-	Cashier defaults to logged-in user’s name.	As expected	Pass
10	Pick mode of payment	Mode: "Cash"	Payment mode reflected in summary/payment section.	As expected	Pass
11	Apply discount (percent)	Discount: 10% off	Discount applied to subtotal; summary totals adjust accordingly.	As expected	Pass
12	Apply discount (fixed)	Discount: PHP 100.00	Fixed discount applied correctly; totals update.	As expected	Pass
13	Pick payment	Breakdown: "Full"	Total sales equals full subtotal after	As expected	Pass

	breakdown - Full		discounts/fees; amount due defaults to full.		
14	Pick payment breakdown - 50%	Breakdown: "50%"	Total sales auto-calculates 50% due now; summary reflects remaining balance.	As expected	Pass
15	Pick payment breakdown - Custom	Breakdown: "Custom"; Custom %: 30%	Total sales due now calculates 30%; remaining shown in summary.	As expected	Pass
16	Fill amount due now	Amount Due Now: PHP 750.00	Field accepts; summary shows due now and remaining correctly.	As expected	Pass
17	Fill customer paid; auto calculate change	Paid: PHP 800.00	System computes change = PHP 50.00; displays in summary.	As expected	Pass
18	Verify request summary updates live	-	All chosen fields (customer, items, services, fees, discounts, breakdown, amounts) appear in real-time.	As expected	Pass
19	Confirm Service Request	Click "Confirm Service Request"	System validates, saves to DB, and generates downloadable acknowledgement receipt including the request summary.	As expected	Pass

Test Case 03: Operations - Service Queue to Ready for Pickup (Batch Pipeline)					
Test Case ID: TC03: OPS-PIPELINE					
Test Priority (Low/Medium/High): High					
Test Scenario: Verify batch status transitions across operations pipeline with image uploads and notifications.					
Test Description: Ensures items progress through Service Queue → Ready for Delivery → Incoming Branch Delivery → In Warehouse → Return to Branch → In Store → Ready for Pickup with required actions and validations.					
No.	Test Steps/Scripts	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1	Search and filter to locate line items	Keyword: "ORD-1001"; Status: "Service Queue"	Table shows matching items in Service Queue.	As expected	Pass
2	Batch select and Mark as Ready	Action: "Mark as Ready"	Selected items move to "Ready for Delivery" list.	As expected	Pass
3	Navigate to Ready for Delivery and Move to Branch	Action: "Move to Branch"	Selected items move to "Incoming Branch Delivery" list.	As expected	Pass
4	Navigate to Incoming Branch Delivery; upload Before image and Mark In Process	Before Image: before.jpg (original > required size)	Image is resized to required dimensions via image processing; items move to "In Warehouse".	As expected	Pass
5	Navigate to In	After Image: after.jpg	After image uploads; items	As expected	Pass

	Warehouse; upload After image and Return to Branch		move to "Return to Branch" list.		
6	Navigate to Return to Branch; Mark as Received	Action: "Mark as Received"	Items move to "In Store" list.	As expected	Pass
7	From In Store; Mark as Ready for Pickup	Action: "Mark as Ready for Pickup"	Items move to "Ready for Pickup"; customer mobile app receives notification.	As expected	Pass
8	Navigate to Ready for Pickup; verify overdue days	Notification date: 2025-01-01; Today: 2025-01-05	Pickup notice shows 4 days overdue since notification.	As expected	Pass

Test Case 04: Operations - Service Queue to Ready for Pickup (Batch Pipeline)					
Test Case ID: TC04: OPS-PIPELINE					
Test Priority (Low/Medium/High): High					
Test Scenario: Verify batch status transitions across operations pipeline with image uploads and notifications.					
Test Description: Ensures items progress through Service Queue → Ready for Delivery → Incoming Branch Delivery → In Warehouse → Return to Branch → In Store → Ready for Pickup with required actions and validations.					
No.	Test Steps/Scrip	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

	ts				
1	Search and filter to locate line items	Keyword: "ORD-1001"; Status: "Service Queue"	Table shows matching items in Service Queue.	As expected	Pass
2	Batch select and Mark as Ready	Action: "Mark as Ready"	Selected items move to "Ready for Delivery" list.	As expected	Pass
3	Navigate to Ready for Delivery and Move to Branch	Action: "Move to Branch"	Selected items move to "Incoming Branch Delivery" list.	As expected	Pass
4	Navigate to Incoming Branch Delivery; upload Before image and Mark In Process	Before Image: before.jpg (original > required size)	Image is resized to required dimensions via image processing; items move to "In Warehouse".	As expected	Pass
5	Navigate to In Warehouse; upload After image and Return to Branch	After Image: after.jpg	After image uploads; items move to "Return to Branch" list.	As expected	Pass
6	Navigate to Return to Branch; Mark as Received	Action: "Mark as Received"	Items move to "In Store" list.	As expected	Pass
7	From In Store; Mark as Ready for Pickup	Action: "Mark as Ready for Pickup"	Items move to "Ready for Pickup"; customer mobile app receives notification.	As expected	Pass

8	Navigate to Ready for Pickup; verify overdue days	Notification date: 2025-01-01; Today: 2025-01-05	Pickup notice shows 4 days overdue since notification.	As expected	Pass

Test Case 05: Operations — Payment & Pickup (Full Flow)					
Test Case ID: TC05: OPS-PAY-PICK					
Test Priority (Low/Medium/High): High					
Test Scenario: Verify search/filter, request summary review, cashier auto-fill, payment processing, overdue storage fee deduction, and pickup confirmation with receipt.					
Test Description: Ensures end-to-end payment & pickup workflow: find line item, review summary, auto-fill cashier, select payment mode, compute remaining balance including overdue storage fees, enter due now and customer paid, calculate change, update balance and status (PAID/NP/PARTIAL), save & mark as pickup to update DB and download acknowledgement receipt with updated payment history.					
No.	Test Steps/Scripts	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1	Search and filter line item	Keyword: "ORD-1001" ; Status: "Ready for Pickup"	Table shows matching order in Ready for Pickup.	As expected	Pass
2	Click Select button to open request summary	-	The request summary displays all details (items, services, discounts, history) correctly.	As expected	Pass
3	Verify cashier auto-fill	Logged-in user: "Grace S. Felix"	Cashier field auto-fills with "Grace S. Felix".	As expected	Pass

4	Pick mode of payment	Mode: "Cash"	Payment mode reflected in payment section.	As expected	Pass
5	Compute remaining balance with overdue storage fee	Overdue days: 4; Storage fee per day: PHP 20.00	Remaining balance increases by PHP 80.00 and displays updated remaining.	As expected	Pass
6	Fill Due Now	Due Now: PHP 750.00	Field accepts; remaining and status preview update.	As expected	Pass
7	Fill Customer Paid; auto-calculate change	Paid: PHP 800.00	Change auto-calculated = PHP 50.00; displayed in summary.	As expected	Pass
8	Show updated balance and status	-	Status reflects payment outcome: PAID if remaining = 0; PARTIAL if remaining > 0; NP if no payment.	As expected	Pass
9	Click Save and Mark as Pickup	Action: "Save & Mark as Pickup"	System updates DB, moves item to picked-up state, and downloads acknowledgement receipt including new payment history and updated balance.		

Test Case 06: Database View - Central View (Tables/Filters)					
Test Case ID: TC06: DB-CENTRAL					
Test Priority (Low/Medium/High): High					
Test Scenario: Verify central view tables render and support filters/sorting.					
Test Description: Ensures central tables (e.g., receipts/users/branches overview) display data with common controls.					
No.	Test Steps/Scripts	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1	Open Central View	-	Tables render with columns and toolbar actions.	As expected	Pass
2	Apply filter	Filter: "Last 7 days"	Tables update to filtered dataset.	As expected	Pass
3	Sort by date	Sort: "Date desc"	Rows reorder accordingly.	As expected	Pass

Test Case 07: Database View - Customer Information (CRUD)					
Test Case ID: TC07: DB-CUSTOMERS					
Test Priority (Low/Medium/High): High					
Test Scenario: Verify listing, creating, editing, and deleting customers.					
Test Description: Ensures `CustomerTable.tsx`, `AddCustomer/EditCustomerDialog` flows validate fields and persist changes.					
No.	Test Steps/Scripts	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1	List customers	-	Table displays customers with pagination and search.	As expected	Pass

2	Add customer	Name: "Jane Doe"; Email: "jane@example.com"; Phone: "09123456789"	Creates record and updates table.	As expected	Pass
3	Edit customer	Phone: "09999999999"	Saves and shows updated value.	As expected	Pass
4	Delete customer	Customer: "Jane Doe"	Confirms removal and refreshes list.	As expected	Pass

Test Case 08: Database View - Branches (Create/Edit)					
Test Case ID: TC08: DB-BRANCHES					
Test Priority (Low/Medium/High): Medium					
Test Scenario: Verify branch add/edit, including storage location.					
Test Description: Ensures `AddBranchDialog.tsx` and `EditBranchDialog.tsx` persist and display storage location.					
No.	Test Steps/Scripts	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1	Add branch	Name: "North Campus"; Storage: "Cabinet A"	Creates branch and shows in list.	As expected	Pass
2	Edit storage location	Storage: "Cabinet 8"	Updates and displays across the central view.	As expected	Pass

Test Case 09: Analytics - Date Range, View Toggle, Export					
Test Case ID: TC09: ANALYTICS					
Test Priority (Low/Medium/High): Medium					
Test Scenario: Verify analytics charts update by date range and support exporting.					
Test Description: Ensures `DateRangePicker.tsx`, `ViewToggle.tsx`, and `ExportButton.tsx` operate correctly with dummy datasets.					
No.	Test Steps/Scripts	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1	Open analytics page	-	Charts render with default range.	As expected	Pass
2	Select range	Range: "Last 30 days"	Charts update datasets.	As expected	Pass
3	Toggle view	-	Switches between chart and table view.	As expected	Pass
4	Export data	-	Downloads CSV for current view.	As expected	Pass

Test Case 10: User Management - Appointments (List/Schedule/Update)					
Test Case ID: TC10: UM-APPT					
Test Priority (Low/Medium/High): High					
Test Scenario: Verify listing appointments and scheduling/updating status.					
Test Description: Ensures appointments table supports create/update and validations.					
No.	Test Steps/Scripts	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1	List	-	The table shows	As	Pass

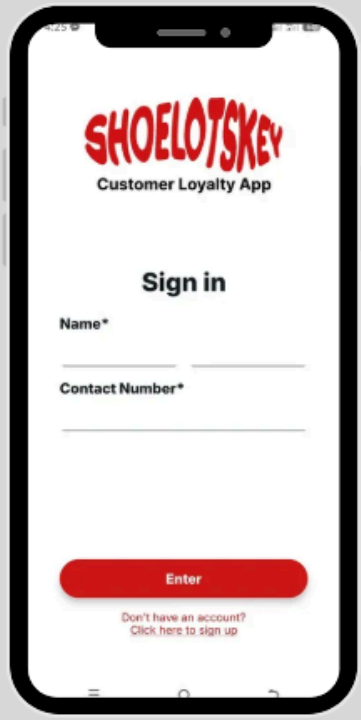
	appointments		appointments with date/time and status.	expected	
2	Schedule appointment	Date: "2025-01-10 09:00"	Creates schedule and displays in list.	As expected	Pass
3	Update status	Status: "Completed"	Row reflects updated status.	As expected	Pass

Test Case 11: User Management - Announcements (Create/Edit/Delete)					
Test Case ID: TC11: UM-ANN					
Test Priority (Low/Medium/High): Medium					
Test Scenario: Verify creating, editing, and deleting announcements.					
Test Description: Ensures announcement form validates and updates the announcements list.					
No.	Test Steps/Scripts	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1	Create announcement	Title: "Holiday Hours"; Body: "Office closed on Jan 1"	Adds announcement to list.	As expected	Pass
2	Edit announcement	Body: "Office closed on Jan 1-2"	Saves changes and updates list.	As expected	Pass
3	Delete announcement	Title: "Holiday Hours"	Removes from list after confirmation.	As expected	Pass

Test Case 12: Notifications Sheet - Display and Actions					
Test Case ID: TC12: NOTIF-SHEET					
Test Priority (Low/Medium/High): Medium					
Test Scenario: Verify notifications sheet opens/closes and supports item actions.					
Test Description: Ensures `NotifSheet.tsx` renders a sheet with notifications and basic actions (mark read, open related view).					
No.	Test Steps/Scripts	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1	Open notifications sheet from navbar/icon	-	The sheet opens displaying notifications.	As expected	Pass
2	Mark notification as read	Item: "New Claim Submitted" Item state changes to read; count decreases.	System generates and downloads the report in a PDF format	As expected	Pass
3	Open related view from notification	Action: "View"	Navigates to corresponding page (e.g., claim details).	As expected	Pass

Appendix 5

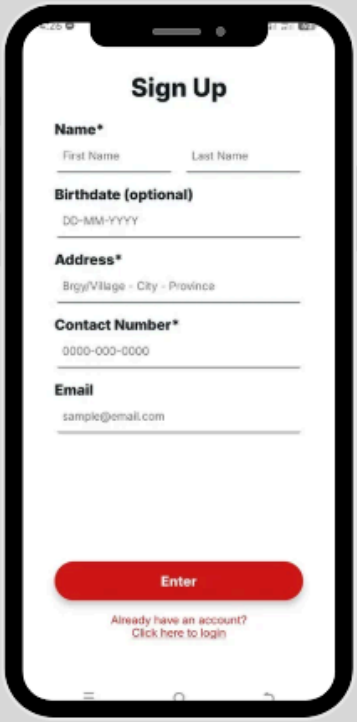
USER'S MANUAL



Sign In Screen

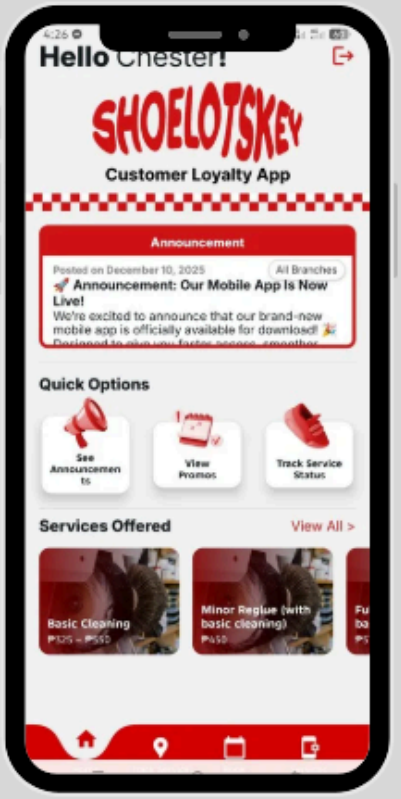
This screen allows users to log in using their Name and Contact Number. After entering the required fields, tap Enter to access the app. A link at the bottom allows users without an account to proceed to the Sign Up page.

Contact number was used as password in order to strengthen customer's password uniqueness



Sign Up Screen

New users create an account. Required information includes Name, Address, and Contact Number, while Birthdate is optional. After completing the form, click Enter to register. A link is provided for users who want to return to the Login page.

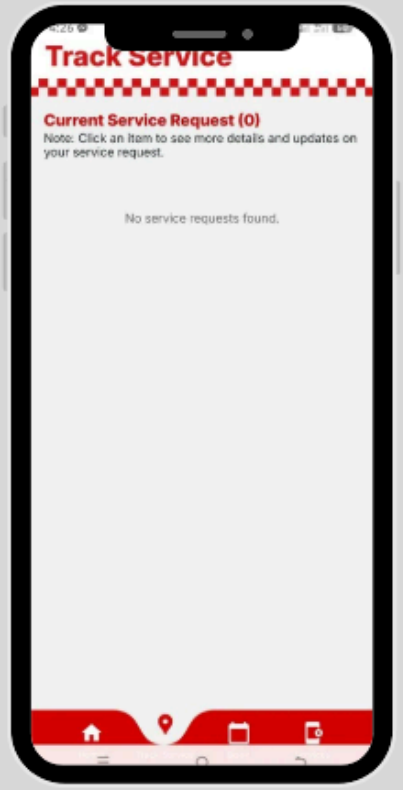


Home Screen

Main dashboard of the Shoelotskey Customer Loyalty App. It displays a welcome message, the app banner, and the Announcements section for updates. Users can access quick actions such as:

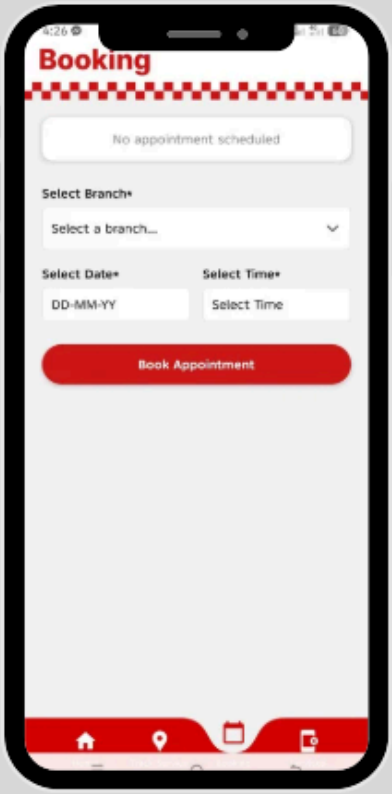
- See Announcements
- View Promos
- Track Service Status

Below the quick options, users can view featured Services Offered, each showing a service card with price and basic details.



Track Service Screen

Track Services displays the user's current service requests.
If there are no active services, the screen shows "No service requests found." Users can tap any listed service (when available) to view its details and progress.

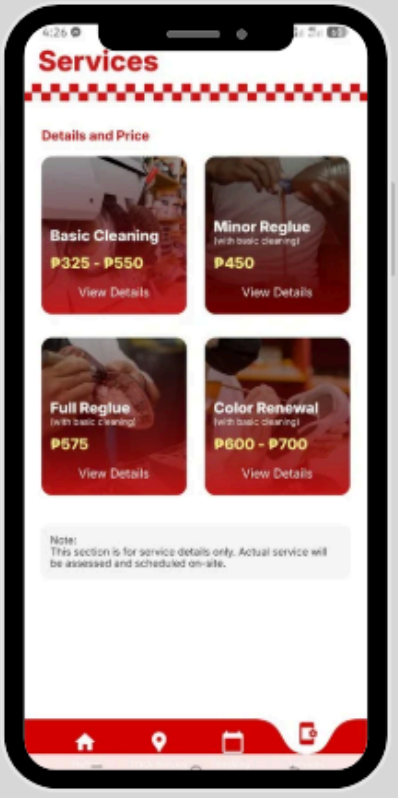


Booking Screen

This screen allows users to book a service appointment.
Users need to provide:

- Branch
- Date
- Time

After selecting the required information, tap the Book Appointment button to submit an appointment request. The top portion of the screen also indicates if the user has any scheduled appointments.



Services Screen

This screen lists all available services along with their corresponding prices.
Each service is displayed in a card format with an image, title, and price range:

- Basic Cleaning
- Minor Reglue
- Full Reglue
- Color Renewal

Users can tap View Details on each service to learn more.
A short note indicates that all services shown are for reference only and final assessment will be done on-site.

Login and Sign-Up

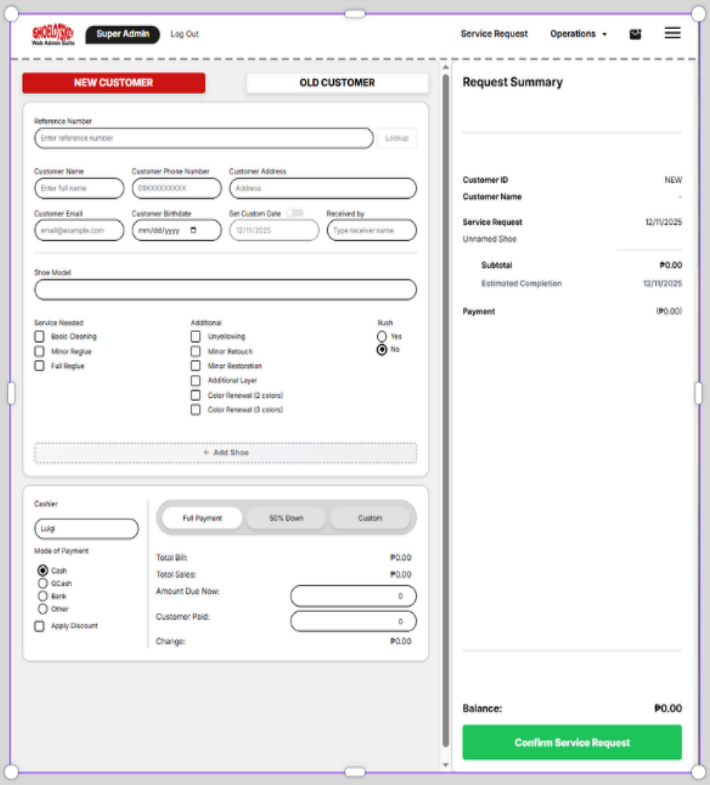
This page allows admins and staff to log in using their User ID and password. Each branch of Shoelotskey have their own User ID that separates each branches staff and make the inventory management easy.

Home Page - Customer Request Tab

This is the Home Page tab of the Shoelostkey Web Admin Suite. In this tab admins and staff take orders, sort them and list all customer in. The home page has many uses, staffs can access quick actions such as:

- Searching new and old customers
- Take customer service request
- Automatic billing procedures
- Receipt

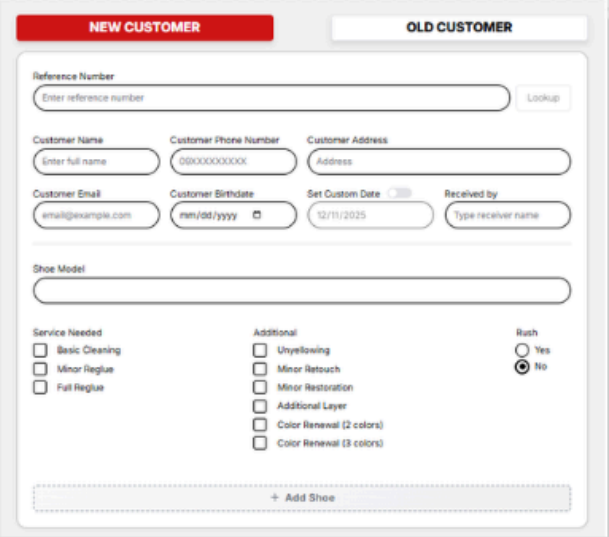
This makes it more easier to navigate customer who have had previous service request with Shoelotskey.



Home Page - Customer Search Panel

Customer search panel uses reference number of the customer who makes the reservation in the app and pull up their credentials for faster operations.

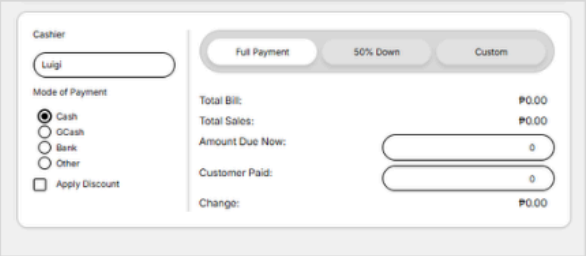
After this, staffs can now proceed in getting customer services request by listing their shoe model and the services they need.



Home Page - Billing Procedures

Billing procedures uses the staffs name for the cashier who takes the order in order to reduce fraud. Their names cannot be changed until the account that is logged-in is sign out.

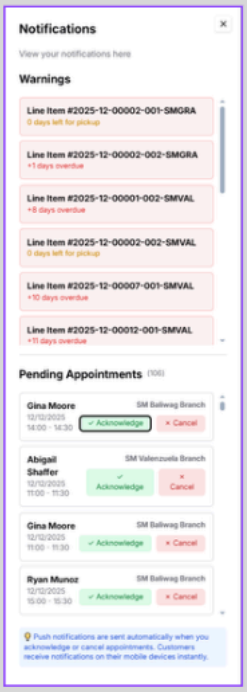
Billing procedures processes customers mode of payment as well as calculate their total bill. Customers may pay the whole amount or half. After making the request this will automatically create a receipt that will be sent to the customers email and will be generated on the staff side as well.



Notifications

Notifications give staffs the visibility of item items that are near over due and pending appointments that are not yet approved or declined.

In pending appointments, once a staff has acknowledge an appointment, it will automatically send a notification to the customers mobile app and will give them a reference number that staff can use in order to pull up the customers credentials.



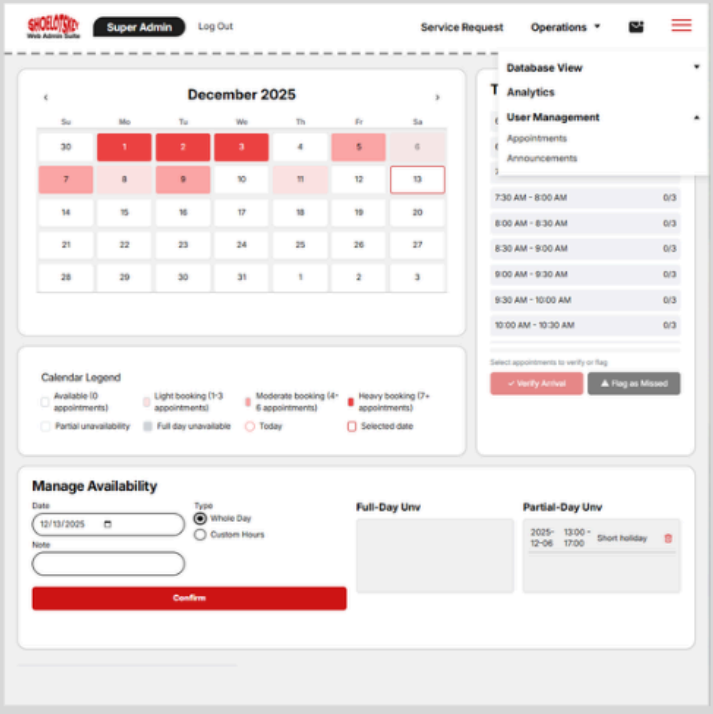
← Warning

Notify staff that some customer service request hasn't been completed yet or their item has been overdue after the cleaning process was finished.

← Pending Appointments

Staff are notified that customers who have the app has made an appointment and staff can either acknowledge or decline depending on date that the branch is open.

User Management Tab - Appointments

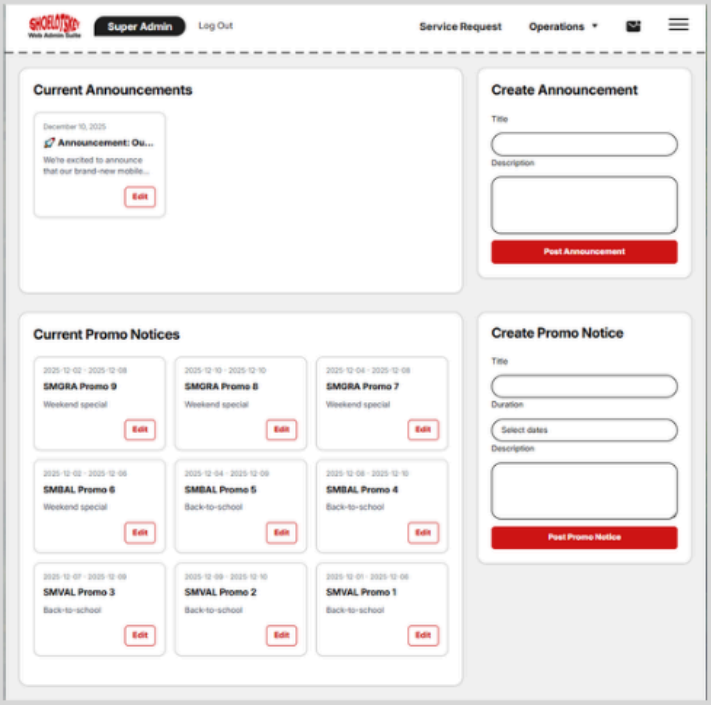


In **User Management Tab** that is for appointments, employees and staff can see here the date where customers have booked a date for an appointment.

Through the **calendar legend** provided. Staffs can easily know which dates have light, moderate and heavy booking.

They can also manage the branches availability through the **Manage Availability** section to make an announcement whether the branch is open or close for a day or for a specific hour

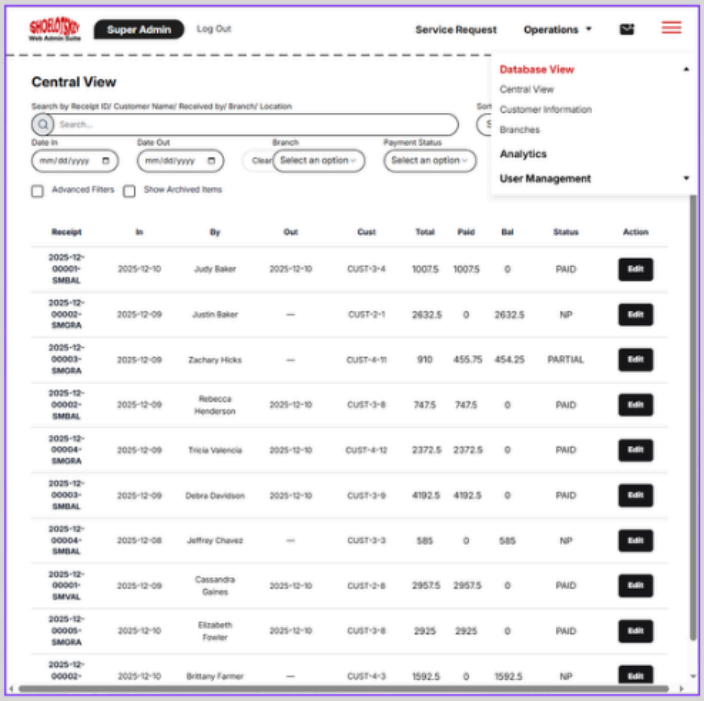
User Management Tab – Announcements



In **User Management Tab** that is for announcement, employees and staff can manually make an announcement that customers can see through the mobile app's home page.

They can also create promo notice that customers may avail once specific conditions are met.

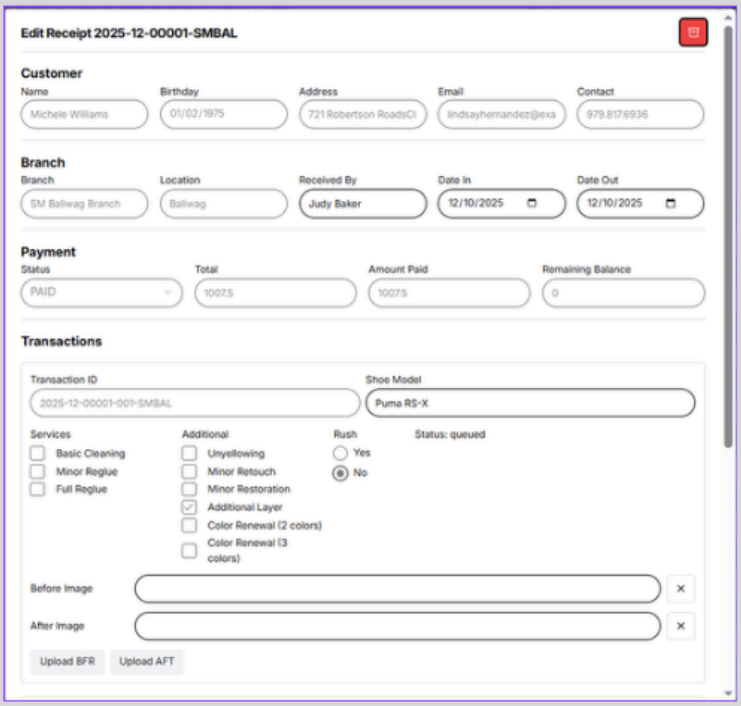
Database View – Central View



In this page, employees can track customers service request using their receipt, customer name, who received their order, or branch location.

This also shows how much customer paid for their service request as well as their remaining balance.

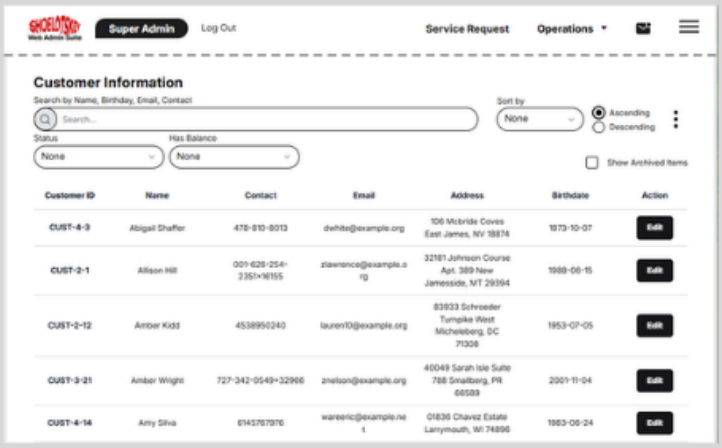
Database View – Central View



Using the **edit action**. Staffs can manually change if there was a human error on their end and fix the problem quickly.

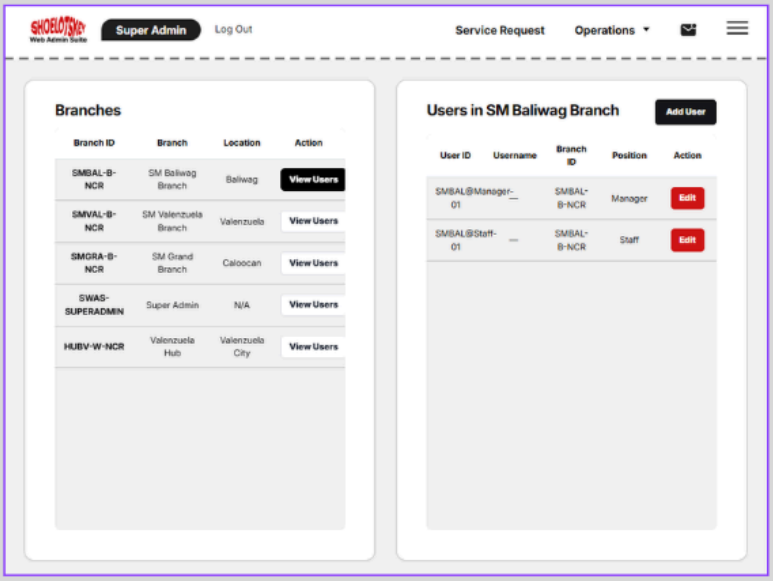
The only items that are available to be edited are the dates, who received the item as well as the name or the shoe model.

Database View – Customer Information



The Customer information tab lets the staffs or employees see and change(only by a customer request) a customer credentials that they have inputted when they created their account through the mobile app.

Database View – Branches

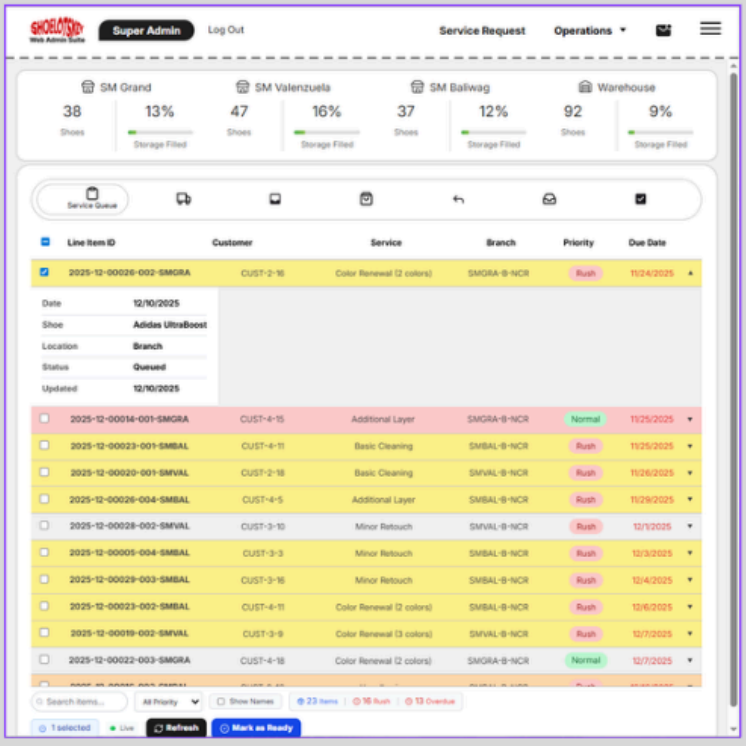


In **branches tab** of the Database View, only the super admin can see this tab as they are the only one who have the highest access level of control from all the other branches.

In this tab, the super admin can add or edit a staffs username and password only when requested.

The owners of the business or the super admins can track how many employees they have in any of their branches

Operations tab – Service queue



Operations tab give staff the access to complete the process of a shoes services need per step.

Operations tab have **7 steps**. Services queue, ready for delivery, incoming branch delivery, In warehouse, return to branch, In store, and ready for pick-up.

All of this steps consist of the line item ID, customer, service, branch, priority and due date.

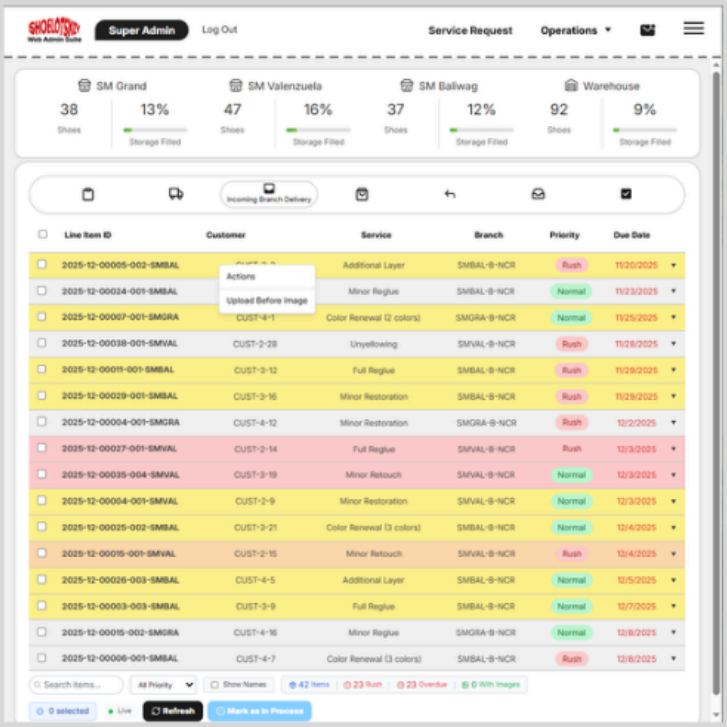
Once a step is done, it automatically notifies the customer where their shoe is through the mobile app real time.

Operations tab - Incoming Branch Delivery

This step indicates that the shoes have already been picked-up and is on the way to the warehouse of the service request.

The staff from the branch may also take a picture of the shoes before the cleaning process which the customer may view using the Shoelotskey's mobile app.

Operations tab - In Warehouse

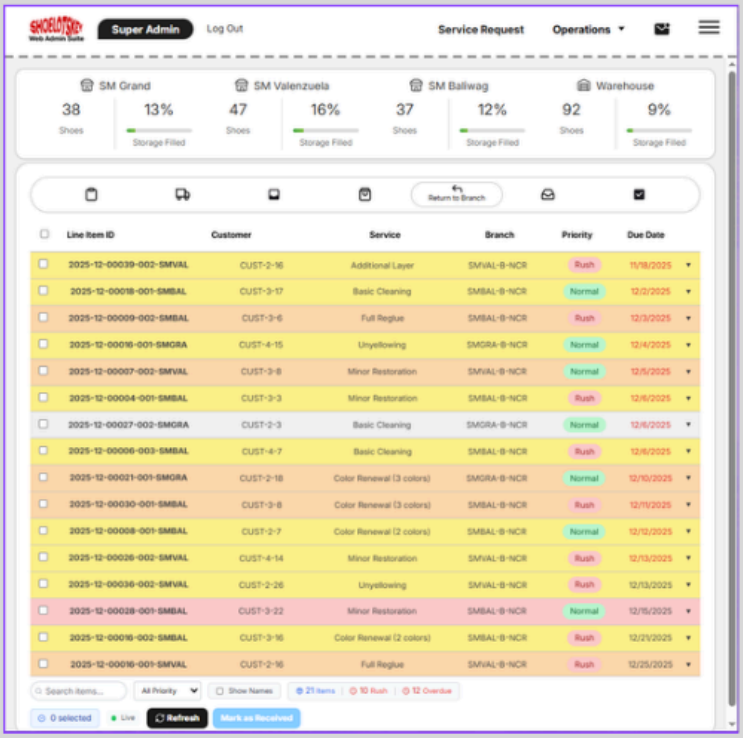


The In Warehouse step or process makes sure that all shoes that have been picked-up from each branch have been accounted for and listed.

This is also when the shoes service request processed and will take days until it is ready for delivery back to the branch.

Staffs may also upload pictures of the after image of the shoes after the cleaning process which the customer may see using the mobile app.

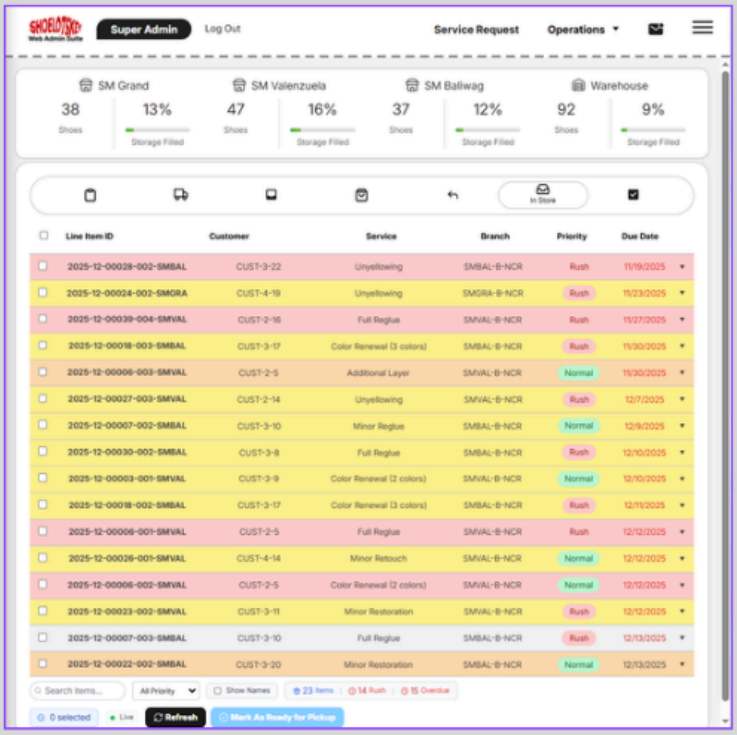
Operations tab – Return to Branch



Return to Branch indicates that the cleaning process was already finished and the staffs from the warehouse have already inspect the shoes and the service request as done.

This step also includes the transportation of the shoes as well back to the branch where the shoes was requested from.

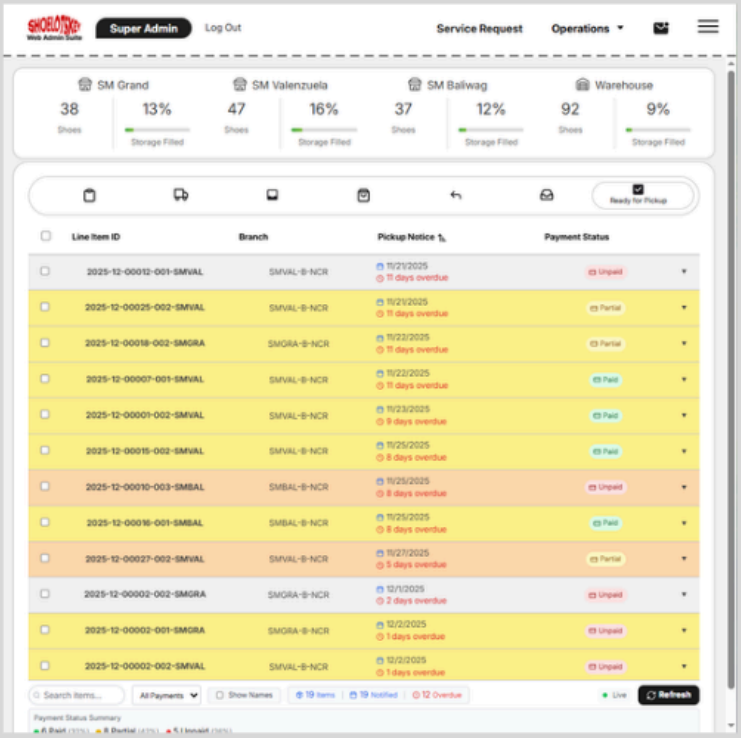
Operations tab – In Store



This step indicates that the shoes have already been received by the branch and its inspection is already finished.

Once in this step, it will notify the customer that their shoes is already at the branch and is ready for pick-up

Operations tab – Ready for Pickup

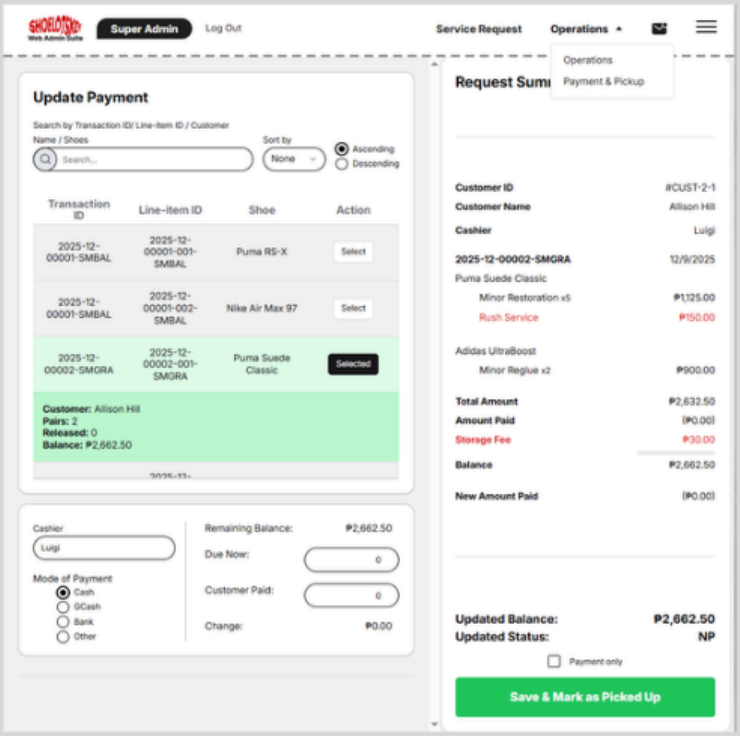


Ready for pick-up is the last step of the operations process where the customer can pick up their shoes.

In this step once a service was marked as ready for pick-up it will start the count down wherein customer must pick-up their shoes within the time period or a storage fee of ₱30.00 is applied per day and will be added to the exact bill.

The item that are in this tab will not be archived as long as the customer does not pay in full for their service request.

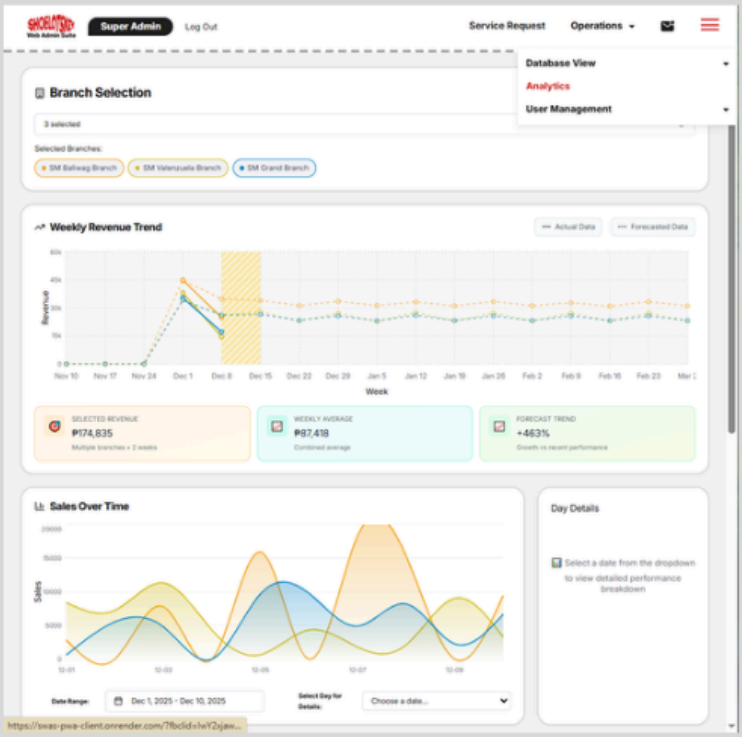
Operations tab – Payment and Pick-up



This is the payment and pick-up section of the operations where the customer completes the payment for their service request.

Once done, customer's may take their shoes home and the item that are within the ready for pick-up tab will be automatically archived in order to keep the list clean.

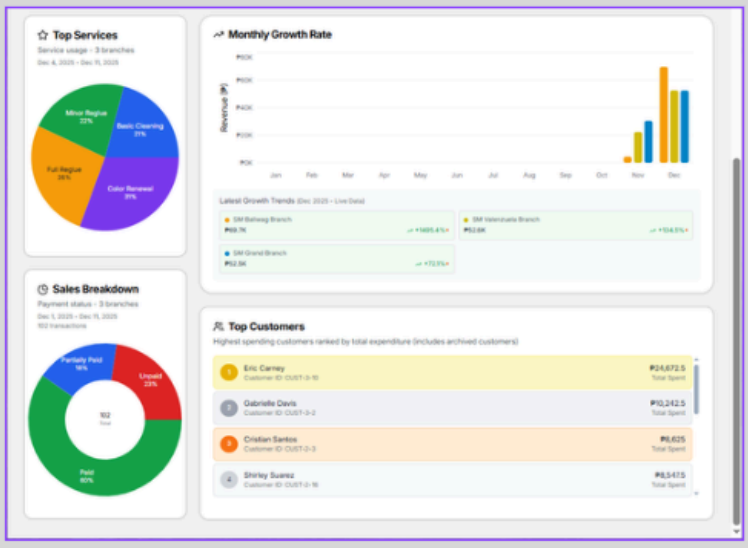
Analytics Tab



This section serves as the central analytics and reporting dashboard of the Shoelotskey Web Admin Suite. It provides the Super Admin with a consolidated view of business performance across selected branches, allowing real-time monitoring of sales, revenue, service demand, and customer activity.

Within this tab, administrators can analyze weekly revenue trends by comparing actual and forecasted data, track sales performance over time, and evaluate monthly growth rates to measure business expansion.

Analytics Tab



The dashboard also highlights top-performing services, displays a sales breakdown based on payment status, and identifies top customers based on total spending.
By using the branch selection feature, admins can filter analytics per branch or view combined performance across multiple locations.

Overall, the Analytics tab supports data-driven decision-making, helping management assess operational efficiency, optimize services, monitor financial health, and plan future strategies effectively.

Appendix 6

SAMPLE GENERATED OUTPUTS

SHOELOTSKEY
SHOE CLEANING SERVICES
SWAS-SUPERADMIN

ACKNOWLEDGEMENT RECEIPT

2025-08-00008-SMBAL

PAY-1-SWAS-SA

Date: 11/10/2025

Customer ID: CUST-3-34
Customer Name: Jessica Valdez
Address: 15336 Guerrero Hills
North Brandon, OK 72546
Date Dropped off: 08/07/2025
Date Pick-up: -

Nike Cortez

Minor Restoration x3	675.00
Rush Service	150.00
Subtotal	825.00

Nike Blazer Mid

Color Renewal (2 colors) x4	1,100.00
Subtotal	1,100.00

Total Bill:	1,775.00
Less: PAY-7-SMBAL	1,775.00
Add: Storage Fee	130.00
Less: PAY-1-SWAS-SA	130.00
Balance:	PHP 0.00

lo

Cashier/Authorized Signature

SWAS ANALYTICS REPORT

Generated: November 5th, 2025

Data Period: 90 days

Report Period: 2025-08-06 to 2025-11-04

EXECUTIVE SUMMARY

Total Revenue
PHP 601,528

Daily Average
PHP 6,684

BRANCH PERFORMANCE

Branch	Revenue	Percentage	Performance
SM Baliwag Branch	PHP 185,308	30.8%	Average
SM Valenzuela Branch	PHP 225,160	37.4%	Above Average
SM Grand Branch	PHP 191,060	31.8%	Average

SALES OVER TIME ANALYSIS (Last 31 days)

Date	SM Baliwag Branch	SM Valenzuela Branch	SM Grand Branch	Total	vs Avg %	Performance
2025-10-04	PHP 3,225	PHP 1,325	PHP 6,500	PHP 11,050	65.3%	Above Avg
2025-10-05	PHP 3,950	PHP 7,300	PHP 0	PHP 11,250	68.3%	Above Avg
2025-10-06	PHP 4,625	PHP 0	PHP 6,525	PHP 11,150	66.8%	Above Avg
2025-10-07	PHP 3,425	PHP 0	PHP 4,000	PHP 7,425	11.1%	Above Avg

Date	SM Baliwag Branch	SM Valenzuela Branch	SM Grand Branch	Total	vs Avg %	Performance
2025-10-08	PHP 0	PHP 1,950	PHP 3,850	PHP 5,800	-13.2%	Below Avg
2025-10-10	PHP 1,525	PHP 0	PHP 4,225	PHP 5,750	-14.0%	Below Avg
2025-10-11	PHP 0	PHP 2,375	PHP 5,650	PHP 8,025	20.1%	Above Avg
2025-10-12	PHP 0	PHP 3,875	PHP 0	PHP 3,875	-42.0%	Below Avg
2025-10-13	PHP 0	PHP 8,975	PHP 4,050	PHP 13,025	94.9%	Above Avg
2025-10-14	PHP 650	PHP 1,918	PHP 0	PHP 2,568	-61.6%	Below Avg
2025-10-15	PHP 700	PHP 10,108	PHP 2,400	PHP 13,208	97.6%	Above Avg
2025-10-16	PHP 2,775	PHP 8,840	PHP 5,275	PHP 16,890	152.7%	Above Avg
2025-10-17	PHP 1,175	PHP 8,515	PHP 2,100	PHP 11,790	76.4%	Above Avg
2025-10-18	PHP 0	PHP 4,850	PHP 1,025	PHP 5,875	-12.1%	Below Avg
2025-10-19	PHP 2,600	PHP 4,075	PHP 0	PHP 6,675	-0.1%	Average
2025-10-20	PHP 6,200	PHP 0	PHP 1,725	PHP 7,925	18.6%	Above Avg
2025-10-21	PHP 4,160	PHP 6,750	PHP 0	PHP 10,910	63.2%	Above Avg
2025-10-22	PHP 1,250	PHP 0	PHP 2,075	PHP 3,325	-50.3%	Below Avg
2025-10-23	PHP 2,175	PHP 0	PHP 2,275	PHP 4,450	-33.4%	Below Avg
2025-10-24	PHP 0	PHP 3,525	PHP 5,475	PHP 9,000	34.7%	Above Avg
2025-10-25	PHP 0	PHP 3,175	PHP 2,550	PHP 5,725	-14.3%	Below Avg
2025-10-26	PHP 2,200	PHP 2,250	PHP 2,225	PHP 6,675	-0.1%	Average
2025-10-27	PHP 0	PHP 975	PHP 675	PHP 1,650	-75.3%	Below Avg
2025-10-28	PHP 5,675	PHP 0	PHP 0	PHP 5,675	-15.1%	Below Avg
2025-10-29	PHP 650	PHP 2,350	PHP 3,500	PHP 6,500	-2.7%	Average
2025-10-30	PHP 5,500	PHP 4,550	PHP 0	PHP 10,050	50.4%	Above Avg
2025-10-31	PHP 9,550	PHP 0	PHP 0	PHP 9,550	42.9%	Above Avg
2025-11-01	PHP 0	PHP 4,225	PHP 0	PHP 4,225	-36.8%	Below Avg
2025-11-02	PHP 1,625	PHP 1,500	PHP 750	PHP 3,875	-42.0%	Below Avg

Date	SM Baliwag Branch	SM Valenzuela Branch	SM Grand Branch	Total	vs Avg %	Performance
2025-11-03	PHP 0	PHP 2,025	PHP 2,100	PHP 4,125	-38.3%	Below Avg
2025-11-04	PHP 250	PHP 3,900	PHP 3,000	PHP 7,150	7.0%	Average

MONTHLY GROWTH ANALYSIS

Month	SM Baliwag Branch	SM Valenzuela Branch	SM Grand Branch	Total	Growth %
2025-01	PHP 0	PHP 85,421	PHP 75,632	PHP 249,695	N/A
2025-02	PHP 0	PHP 95,214	PHP 56,324	PHP 253,101	1.4%
2025-03	PHP 0	PHP 78,776	PHP 78,547	PHP 209,446	-17.2%
2025-04	PHP 0	PHP 98,963	PHP 88,852	PHP 230,171	9.9%
2025-05	PHP 0	PHP 45,652	PHP 89,652	PHP 179,428	-22.0%
2025-06	PHP 0	PHP 98,438	PHP 98,689	PHP 249,811	39.2%
2025-07	PHP 0	PHP 75,689	PHP 78,542	PHP 250,552	0.3%
2025-08	PHP 0	PHP 57,896	PHP 90,763	PHP 234,301	-6.5%
2025-09	PHP 0	PHP 64,408	PHP 87,403	PHP 213,955	-8.7%
2025-10	PHP 0	PHP 26,075	PHP 35,588	PHP 77,263	-63.9%
2025-11	PHP 1,875	PHP 11,650	PHP 5,850	PHP 19,375	-74.9%

FORECAST ANALYSIS

SM Baliwag Branch

Week Start	Actual	Forecast	Variance %	Accuracy %
2025-10-06	PHP 9,575	PHP 14,605	-34.4%	65.6%
2025-10-13	PHP 7,900	PHP 14,137	-44.1%	55.9%
2025-10-20	PHP 15,985	PHP 13,098	22.0%	78.0%
2025-10-27	PHP 23,000	PHP 13,082	75.8%	24.2%
2025-11-03	PHP 250	PHP 13,753	-98.2%	1.8%
2025-11-10	N/A	PHP 9,540	N/A	N/A
2025-11-17	N/A	PHP 14,275	N/A	N/A
2025-11-24	N/A	PHP 16,012	N/A	N/A
2025-12-01	N/A	PHP 10,343	N/A	N/A
2025-12-08	N/A	PHP 10,425	N/A	N/A
2025-12-15	N/A	PHP 14,265	N/A	N/A
2025-12-22	N/A	PHP 16,197	N/A	N/A
2025-12-29	N/A	PHP 10,836	N/A	N/A
2026-01-05	N/A	PHP 10,110	N/A	N/A
2026-01-12	N/A	PHP 14,205	N/A	N/A
2026-01-19	N/A	PHP 16,223	N/A	N/A
2026-01-26	N/A	PHP 10,865	N/A	N/A

SM Valenzuela Branch

Week Start	Actual	Forecast	Variance %	Accuracy %
2025-10-06	PHP 8,200	PHP 13,822	-40.7%	59.3%
2025-10-13	PHP 47,280	PHP 13,586	248.0%	-148.0%
2025-10-20	PHP 15,700	PHP 15,088	4.1%	95.9%
2025-10-27	PHP 13,600	PHP 13,763	-1.2%	98.8%
2025-11-03	PHP 5,925	PHP 14,093	-58.0%	42.0%
2025-11-10	N/A	PHP 18,293	N/A	N/A
2025-11-17	N/A	PHP 14,505	N/A	N/A
2025-11-24	N/A	PHP 13,862	N/A	N/A
2025-12-01	N/A	PHP 9,874	N/A	N/A
2025-12-08	N/A	PHP 18,293	N/A	N/A
2025-12-15	N/A	PHP 14,860	N/A	N/A
2025-12-22	N/A	PHP 14,061	N/A	N/A
2025-12-29	N/A	PHP 9,961	N/A	N/A
2026-01-05	N/A	PHP 18,293	N/A	N/A
2026-01-12	N/A	PHP 14,922	N/A	N/A
2026-01-19	N/A	PHP 14,084	N/A	N/A
2026-01-26	N/A	PHP 9,965	N/A	N/A

SM Grand Branch

Week Start	Actual	Forecast	Variance %	Accuracy %
2025-10-06	PHP 24,250	PHP 13,880	74.7%	25.3%
2025-10-13	PHP 14,850	PHP 15,190	-2.2%	97.8%
2025-10-20	PHP 16,325	PHP 14,602	11.8%	88.2%
2025-10-27	PHP 4,925	PHP 14,216	-65.4%	34.6%
2025-11-03	PHP 5,100	PHP 13,991	-63.5%	36.5%
2025-11-10	N/A	PHP 11,778	N/A	N/A
2025-11-17	N/A	PHP 12,897	N/A	N/A
2025-11-24	N/A	PHP 9,679	N/A	N/A
2025-12-01	N/A	PHP 10,160	N/A	N/A
2025-12-08	N/A	PHP 12,542	N/A	N/A
2025-12-15	N/A	PHP 13,787	N/A	N/A
2025-12-22	N/A	PHP 10,649	N/A	N/A
2025-12-29	N/A	PHP 10,258	N/A	N/A
2026-01-05	N/A	PHP 12,424	N/A	N/A
2026-01-12	N/A	PHP 13,755	N/A	N/A
2026-01-19	N/A	PHP 10,614	N/A	N/A
2026-01-26	N/A	PHP 10,172	N/A	N/A

Appendix 7

CERTIFICATION OF SIMILARITY CHECK

REPUBLIC OF THE PHILIPPINES
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR ACADEMIC AFFAIRS
COLLEGE OF COMPUTER AND INFORMATION SCIENCES
DEPARTMENT OF INFORMATION TECHNOLOGY

BAGONG PILIPINAS

CERTIFICATION

This is to certify that the research paper of ***ANTHONY JAMES U. ESTEBAN, LUIGI G. LAGMAN, MARC CHESTER DC. PEREZ, GABRIEL IAN M. SANTOS*** with the title, ***SHOELOTSKEY WEB AND ADMIN SUITE (SWAS): A MULTI-BRANCH MANAGEMENT SYSTEMM WITH AI FORECASTING USING PROPHET AI*** submitted to the Polytechnic University of the Philippines, has passed the Originality Check with a detected text-matching similarity of **8%** using the University approved Plagiarism Detection System.

Certified true and correct:

ENGR. ARLENE B. CANLAS
Faculty in-Charge

Date: **02/03/2026**

Appendix 8

CERTIFICATION OF EDITING

CERTIFICATE

OF LANGUAGE EDITING/ GRAMMARIAN/PROOFREADING

This document certifies that the paper listed below was edited for English grammar, punctuation, spelling, correctness, clarity, engagement, delivery, and overall English language usage by **Mr. MARVIN DOMINIC B. BUENA**.

TITLE:

SHOELOTSKEY WEB & ADMIN SUITE: A MULTI-BRANCH MANAGEMENT SYSTEM WITH AI FORECASTING USING PROPHET AI

PROPONENT/S:

ESTEBAN, ANTHONY JAMES U.
LAGMAN, LUIGI G.
PEREZ, MARC CHESTER DC.
SANTOS, GABRIEL IAN M.

CTRL#2026-00158

2 FEBRUARY 2026

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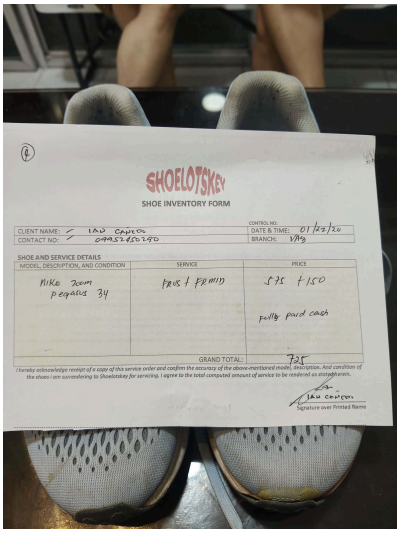
MB
M. Buena
Acad Research Consultant

191-384-584-820-1192-1466-1601-1670-1709-1770-1880-2050-138

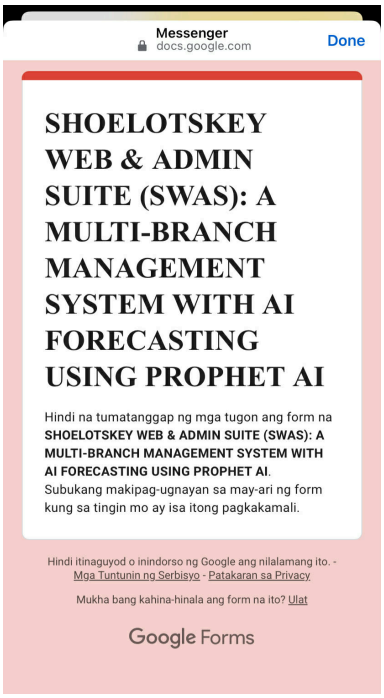
Appendix 9

IMPLEMENTATION REPORT

Following the development of the Shoelotskey Web & Admin Suite (SWAS), a comprehensive implementation and validation phase was conducted across the client's three branches: SM Baliwag, SM Grand Central, and SM Valenzuela. The primary objective was to transition the system from a theoretical model into a practical tool validated within the actual business environment. This process began with a formal presentation of the completed system to the Shoelotskey management team, including the Chief Operating Officer and branch administrators. The demonstration covered all core modules—real-time service tracking, the administrative dashboard, inventory management, and the customer mobile app—clearly illustrating how the system was designed to digitize their manual workflows, centralize multi-branch coordination, and enhance customer communication.



To gather critical operational data and validate the system's usability, we engaged directly with end-users at each branch location. This involved conducting walkthroughs and training sessions with staff members, including customer service representatives, who performed their daily tasks using the new platform. Simultaneously, select customers were invited to interact with the mobile application to test features like appointment booking, service status tracking, and transaction history viewing. Following these hands-on testing sessions, we administered structured surveys to all participants—spanning administrators, staff, and customers—to collect quantitative feedback and qualitative insights. The survey was designed to evaluate the system against key quality attributes such as functionality, usability, and reliability. The data gathered from this thorough on-site testing and feedback collection across all three branches were instrumental in confirming the system's alignment with operational needs and in identifying final refinements prior to its full deployment.



Appendix 10

CAPSTONE PROJECT REVISION MATRICES

REPUBLIC OF THE PHILIPPINES
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OFFICE OF THE VICE PRESIDENT FOR ACADEMIC AFFAIRS
COLLEGE OF COMPUTER AND INFORMATION SCIENCES
DEPARTMENT OF INFORMATION TECHNOLOGY

CAPSTONE PROJECT REVISION MATRIX

■Proposal ☐ Tool ☐ Final ☐ Re-defense

Title:	**SHOELOTSKEY WEB & ADMIN SUITE (SWAS): A MULTI-BRANCH MANAGEMENT SYSTEM WITH AI FORECASTING USING PROPHET AI**
Proponents:	1. Esteban, Anthony James U. 2. Lagman, Luigi G 3. Perez, Marc Chester DC. 4. Santos, Gabriel Ian M.

Comments / Suggestions / and Recommendations from the Panelists	Action Taken	Signature of the Panelist
Panelist 1 Name: Montaigne G. Molejon		
1. **Revise the whole Fishbone diagram.** The current Fishbone diagram needs to be revised entirely to better align with the content and flow of the paper. 2. **Remove the technicality.** Please remove the overly technical elements in the diagram or explanation, as they may not be appropriate for the intended audience or the scope of the paper. 3. **The topic was not discussed in the paper—far from Fishbone.** This particular part was not discussed within the body of the paper, making it seem unrelated or distant from the purpose of the Fishbone diagram. 4. **Prone to human error.** The existing process or system appears to be susceptible to human error, which should be addressed or acknowledged for improvement. 5. **Improve the dashboard.** The dashboard component needs to be improved in terms of design, usability, or clarity to better support the users. 6. **Should have a trigger notification so everyone will be aware that it is done.**	1. The Fishbone Diagram was reconstructed to reflect actual operational problems (e.g., customer miscommunication, unclaimed shoes, inconsistent intake) without referencing system features.**(page 8)**. 2. All technical references (e.g., "no database" or "no system") were removed from the Fishbone Diagram. The updated version focuses solely on operational pain points .**(page 8)**. 3. A detailed explanation of the revised Fishbone Diagram is now included in **Section 1.3.1** with clear linkages between causes and effects.**(page 8)**. 4. The researchers added a "last date modified" display and sorting logic in the Coordinate Branch and Warehouse Operations Use Case Diagram to allow staff to easily monitor the length of time each service request has remained in its current status. This feature helps prioritize rush items and older pending transactions.**(page 29)**	6/30/2025

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It is recommended to add a trigger notification feature so that all relevant parties are immediately informed once a task or process is completed. 7. **Label for forecasting—short interpretation (narrative).** The forecasting label should include a brief interpretation or narrative to help readers quickly understand the meaning and implication of the data.	5. The researchers added clearly labeled data visuals in the Generate and View Business Analytics Use Case Diagram and Use Case Report to improve readability and user comprehension of trends and metrics displayed in the analytics tab.**(page 31)** 6. The researchers added an automated notification trigger in the Send Admin Notifications Use Case Diagram that notifies warehouse staff when a service request is nearing its estimated completion time and also alerts admins and staff if a transaction record has not been updated for a number of days, helping ensure timeliness, accountability, and proactive handling of delayed service progress.**(page 33)** 7. The researchers added a narrative reporting feature in the Generate and View Business Analytics Use Case Diagram to generate descriptive summaries of forecast outputs, compensating for the absence of built-in interpretive functionality in the forecasting tool.**(page 31)**	
Panelist 2 Name: Maria Espereanza Hogar-Reyes		
1. **Title too long—remove irrelevant words.** The current title is overly long and includes unnecessary terms. Please revise it to be more concise by removing words that do not directly contribute to the core message. **Grammatically incorrect:** "The warehouse is used for	1. Title was shortened to: *Shoelotskey Web & Admin Suite (SWAS): A Multi-Branch Management System with AI Forecasting Using Prophet AI.* **(Title page)** 2. Revised to: The laptop and printer are used to encode the customer records using Excel, print	ghmg. rgf 06.30.2025

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encoding customer record" and "due to past or manual processes. 2. Change "status of their **shoes" to "status of their service."** Please modify the phrase "status of their shoes" to "status of their service" to better reflect the intended meaning and context of the statement. 3.	receipts, while performing limited data handling tasks.functionalities.**(page 5)** 3. All instances of "status of their shoes" were revised for clarity and accuracy. **(page 11)**.	mgsarmiento 06.30.2025
Panelist 3 Name: Marigyn Sarmiento		
1. **Do not mention the database in the title.** Please remove any mention of the specific database from the title to maintain a more general and professional tone. 2. **Change the word "Automate" to "system shall provide."** It is recommended to replace the word "Automate" with the phrase "The system shall provide" to align with standard requirement phrasing and ensure consistency throughout the document.	1. "MongoDB" was removed from the title for a more generalized and professional wording. **(Title page)**. 2. Requirement phrasing was standardized to "The system shall provide..." throughout the Problem–Requirement Matrix.**(Pages 9–10)**	

Noted by:

Arlene B. Canlas
Faculty in-Charge

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CAPSTONE PROJECT REVISION MATRIX

☐ Proposal ☑ Tool ☐ Final ☐ Re-defense

Title:	Shoelotskey Web & Admin Suite (SWAS): A Multi-Branch Management System with AI Forecasting Using Prophet AI
Proponents:	1.Esteban, Anthony James U. 2. Lagman, Luigi G. 3. Perez, Marc Chester DC. 4. Santos, Gabriel Ian M.

Comments / Suggestions / and Recommendations from the Panelists	Action Taken	Signature of the Panelist
Panelist 1 Name: Montaigne Molejon		
1. Customer info still recorded manually; add formatted reference number. 2. Add user roles for better tracking and preventing errors or fraud. 3. Improve charts and avoid cluttered categories.	1. Added auto-generated formatted reference numbers and auto-fill for returning customers. 2. Implemented new RBAC role layers and audit logs to track staff actions per transaction. 3. Revised pie chart design, moved labels inside, removed overloaded categories.	12/10/2025
Panelist 2 Name: Maria Esperanza Hogar-Reyes		
1. Possible issue: records may show a cashier's name who is not on duty.	1. Implemented staff-on-duty tagging and automatic user stamping per transaction to prevent mismatches.	12.15.2025

Noted by:

Engr. Arlene Canlas
Faculty in-Charge

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Comments / Suggestions / and Recommendations from the Panelists	Action Taken	Signature of the Panelist
Panelist 1 Name: Montaigne Molejon		
1. Ensure all collected data was directly reflected in the conclusions, or remove irrelevant questions. 2. Survey questions should be tailored to specific user roles to improve data analysis. 3. Design role-specific questionnaires that clearly map to objectives. 4. Provide deeper, role-specific interpretations of results instead of generalized conclusions.	1. All conclusions in Chapters 4 and 5 are now explicitly and directly derived from the collected survey responses. The analysis ensures every key finding is traceable to specific data points, strengthening the validity of the results. 2. To extract role-specific insights from the unified survey instrument, the dataset was rigorously segmented by respondent role (Customer, Staff, Admin) as per the profile question. This allowed for a targeted analysis where the same general questions were interpreted through the distinct operational context and priorities of each user group. 3. A precise cross-walk was developed to connect each general survey question to the specific objectives and pain points relevant to each role. This mapping is clearly demonstrated in the revised Chapter 4, where each role's	

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	evaluation details how their responses to questions about, for instance, "Completeness" or "Responsiveness," directly reflect their unique goals (e.g., customer tracking vs. admin oversight). 4. Strengthened the interpretive link between objectives, data, and conclusions: The conclusion was revised to directly and explicitly connect the role-specific findings back to each project objective. Instead of presenting generalized outcomes, the results for each user group (Customer, Staff, Admin) are now clearly discussed in the context of how they validate and fulfill the specific goals of centralization, automation, security, and scalability outlined at the project's outset.	

Noted by:

Engr. Arlene Canlas
Faculty in-Charge

Appendix 11

ETHICS CLEARANCE AND TERMINAL REPORT

Republic of the Philippines
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
Sector
Office
UNIVERSITY RESEARCH ETHICS COMMITTEE

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Date: 11/21/2025

TO/FOR: **Gabriel Ian M. Santos et al.**

Subject: Ethical Clearance

From: **RIA A. SAGUM, MCS,LPT**
Panel Chair, CLREC-CCIS

Recommendation From:
Aileen Camille B. Dimatatac, MC (sgd)
Chief, Research Ethics Section

This is to inform you that the documentary requirements you submitted for your research project titled **"Shoelotskey Web & Admin Suite (SWAS): A Multi-Branch Management System with AI Forecasting Using Prophet AI"** passed the evaluation of the PUP Research Ethics Committee (REC) through the **CLREC-CCIS Research Ethics Committee** under the requirements set by the Philippine Health Research Ethics Board (PHREB).

UREC Code	CL-BSIT-2025-72
Type of Review	**Approved on Nov 21, 2025**
Approval Date	11/20/2025
Expiry Date	11/19/2026
PUP-UREC Decision	**Approved**

The standard conditions of this approval are as follows:

1. Conduct the project following the submitted and approved research protocol and other documentary requirements.
2. If changes are to be made in the research project/study that will affect the research participants, an amendment of the research protocol must be submitted to **clrec-ccis@pup.edu.ph** before implementing such changes.
3. When ethical clearance is about to expire, the researcher must apply to resubmit the research protocol.
4. A final report/terminal report must be submitted when the research project/study is complete.
5. Researchers must advise the **clrec-ccis@pup.edu.ph** in writing if the research project/study has been discontinued.
6. You may now commence your research project/study. Good luck.

Appendix 12

BIOGRAPHICAL STATEMENTS

Anthony James U. Esteban is a motivated Information Technology student at the Polytechnic University of the Philippines and a DOST-SEI JLSS Scholar. He has hands-on experience in full-stack web and mobile development, building scalable, user-centered systems using React, Svelte, React Native, and Python. His projects showcase expertise in API integration, database management, AI-powered solutions, and real-time system design, blending technical skills with practical problem-solving to deliver impactful digital solutions.

Luigi G. Lagman is an Information Technology student from the Polytechnic University of the Philippines with a strong focus on full-stack web and mobile development, UI/UX design, and AI-integrated systems. He has hands-on experience building scalable applications using technologies such as React, Svelte, React Native, and Python, including projects involving multi-branch systems, AI-powered tools, and location-based mobile apps. He is passionate about creating intuitive, user-centered digital solutions and continuously developing his technical and creative skills.

Marc Chester DC. Perez is an Information Technology professional with ten months of experience in customer service and technical support. He is currently a Real-Time Analyst Apprentice (RTAA), where he applies his logical mindset and passion for strategy to live operational monitoring. With a focused interest in becoming a Data Analyst, he is dedicated to utilizing his analytical skills and technical background to provide data-driven solutions in a professional and collaborative environment.

Gabriel Ian M. Santos is an Information Technology professional with experience in technical support and in-app assistance through his work with TaskUs. He also has a background as a freelance graphic and video editor, which strengthened his creative and visual communication skills. He has a strong interest in data analysis and project management and is eager to apply his technical, analytical, and creative skills in a professional and collaborative work environment.